THE INCOMPLETENESS OF OBSERVATION

The Equivalence of Quantum Mechanics and Embedded Observation

Observer geometry and physical carrier. SM §4.1 distinguishes two claims. First, finite equivariant projection is classical and exact: cubic symmetry is inherited by the projected Markov part and every memory kernel, forbidding quadratic anisotropy in the scalar symbol — so whatever nondegenerate quadratic structure survives is isotropic — while allowing cubic anisotropy at quartic order. The particular normalized nearest-neighbor operator $A/(2d)$ additionally requires an **observer-level** center-free kernel; microscopic center independence alone does not imply that after correlated hidden variables are marginalized. Second, the six signed simple-cubic links and their exact $3 \oplus 2 \oplus 1$ representation are unconditional geometry, but identifying that link space with the complete physical carrier is **H-link**; the single-copy clause gives $K = 6$, and the custodial condensate-stabilizer reading is **H-cust**. Accordingly, any downstream statement here that consumes $K = 6$, $N_f = 6$, the physical SM gauge carrier, or a fixed three-sector count inherits those named bridge conditions unless it is explicitly referring only to the six geometric links.

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ABSTRACT

An observer embedded in a finite deterministic system cannot access the complete state. This paper establishes an exact finite-horizon equivalence between three descriptions of what such an observer can record: finite stochastic laws (S), visible marginals of finite reversible deterministic systems (D), and finite-dimensional unitary systems with fixed-basis Born-rule readout (Q_{fb}). Thus

$$S \iff D \iff Q_{\mathrm{fb}}$$

at the finite observable-law level. The quantum representation is exact, not merely approximate; it is also universal, so representability alone does not distinguish the memory-bearing sector from an ordinary Markov process. The Born dictionary $p = |U|^2$ is part of the definition of Q_{fb}: the equivalence exhibits the Born form and does not select the quadratic exponent, which is named as a frontier item (§3.1). The discriminating content is history readback. When (C4) holds, every faithful deterministic completion contains readout-relevant hidden predictive memory, and for one fixed finite reversible representative the full rooted visible process is indivisible somewhere in its recurrence cycle: finite recurrence returns the visible map to the identity, while a C4 readback gap forces

a non-permutation rooted map, which cannot possess a stochastic inverse. Conditions (C1) and (C3) are then necessary diagnostics of the hidden mediation and its capacity; (C2-structural) is the persistence needed for the stored record to survive until readback, with the slow-bath inequality one sufficient physical mechanism.

For spatial OI systems, the coupling graph defined by one-step dynamical dependence supplies exact causal cones: influence propagates at most one graph edge per update. Bell accounting is correspondingly sharp: a deterministic completion that keeps the pointwise cone and a setting-independent pre-setting ensemble is a local hidden-variable model, so $|S_{\mathrm{CHSH}}| \leq 2$ and quantum correlations are unreachable. Reproducing them requires ontic parameter dependence — realized in the specific SM/GR branch through preparation-indexed state-dependent adjacency or an equivalent Bell-nonlocal response structure, with no operational signal — or else setting-dependent ontic ensembles, for which $S_{\mathrm{CHSH}} \leq 2 + 4\sqrt{2 \ln 2\, I_{\mathrm{ont}}}$ gives the model-independent necessary lower bound $I_{\mathrm{ont}} \geq 0.0309$ bits for the quantum value. Conversely, every finite stochastic law, and therefore every finite quantum record, admits a finite reversible realization; for fixed finite dimension, instrument menu, horizon, and accuracy ε, one finite OI construction reproduces the whole corresponding quantum experimental envelope to that accuracy. In this precise sense the stochastic description of embedded observation and its quantum representation are mathematically equivalent. Two logically separate completion questions remain beyond that finite-law result: whether the complete family of coherent interventions and composites is represented simultaneously by the standard quantum instrument theory (§3.4), and whether preparation-indexed Bell adjacency in the state-dependent cubic SM/GR graph preserves the metric/Ollivier–Ricci continuum limit needed by the gravitational reconstruction (H-Bell, [Substratum §3.3]). Accordingly, the exact equivalence claim is at the finite observable-law level; universal Hilbert-space representability is not promoted into a uniqueness theorem for operational quantum mechanics entire. The specific nearest-neighbor cubic representative used for the Standard-Model sector is not by itself a Bell-violating completion under measurement independence; compatibility of preparation-indexed Bell adjacency with the continuum-curvature limit is the separate H-Bell hypothesis of [Substratum §3.3].

1. INTRODUCTION

1.1 The Problem of Embedded Observation

An observer embedded in a deterministic universe cannot access the complete state: degrees of freedom beyond causal reach are permanently hidden, and the observer's description is obtained by marginalizing over the hidden sector. Whether a description obtained this way can be considered complete is the question Einstein, Podolsky, and Rosen posed of quantum mechanics [46]; it returns here with the opposite valence — incompleteness is derived as a structural property of embedded observation, and quantum mechanics is identified as its form. The question is what this reduction imposes on the form of the observer's physical laws.

Prior work (QBism, relational QM, 't Hooft's cellular automaton [1]) takes observer-dependence as an interpretive starting point or derives it from specific microphysical models. This framework instead proves the finite observable-law correspondence directly: every finite stochastic law has both a finite reversible realization and an exact fixed-basis unitary/Born representation, and every finite reversible visible law lies in the same class (§3.4). History readback then identifies the nontrivial sector by forcing hidden predictive memory and, on a fixed finite recurrent representative, global indivisibility. The remaining distinction between this exact stochastic–quantum equivalence and *full operational quantum mechanics* is confined to the simultaneous coherent-instrument and composite structure stated explicitly in §3.4.

1.2 The Starting Point

The framework begins from a starting point usually stated in two words: *observation occurs*. An observer records distinguishable outcomes of interactions with a system not wholly under the observer's control. This is the cogito of Descartes made precise — not as philosophy, but as a mathematical constraint. As §1.2 makes explicit, that starting point is not a single indubitable fact but resolves into two axioms, one indubitable and one a substantive posit; the two-word phrase is the informal name for the pair.

The axiom has structural content beyond contingent occurrence. Substrata that do not admit embedded observers exist as mathematical objects with dynamics; they do not produce physics in the present framework's sense. The axiom thus commits to our universe being in the observer-admitting subset of substrata — substrata whose bijection structure satisfies the conditions C1–C4 below for some partition. This is a structural commitment about the framework's domain, not a contingent fact about whether someone happens to be looking.

The phrase "observation occurs" can be stated with more precision than a single undifferentiated axiom. Distilled, it is *two axioms*. **Axiom 1** is the evidential floor: tokened differentiation occurs — concrete differentiated content is *registered*, a this-not-that tokened from a locus against a remainder; equivalently, *embedded representation occurs*. It is indubitable, since to doubt it is itself to register a distinction from a locus — the cogito made precise, which is irreducibly perspectival: a doubting I, not a free-floating fact. Read this way, the primitive already contains three things that therefore need no separate derivation: a differentiation (the this-not-that), a locus that registers it (the observer), and the remainder it is registered against. The proper-part decomposition $V \subsetneq S$ — the observer embedded in a whole it does not exhaust — is accordingly constitutive of the primitive rather than a downstream posit: a registering that left no remainder would be the whole representing itself, not an observation. From Axiom 1, by analysis of the words "occurs" and "registered," the relational structure, the existence of a dynamics, determinism and finiteness per moment follow as lemmas; the bijective (reversible) representative is a reconstruction choice motivated by observed near-unitarity, and total finiteness of the substratum is a physical posit (status Remark, §1.4; ledger §4.5) — the package is not a set of independent assumptions, and it is not all theorems. One thing does not follow: that the differentiation *recurs* — that there is more than one moment for the dynamics to act across. This is **Axiom 2**: differentiation recurs, the temporal domain being the minimal such recurrence. Axiom 2 presupposes Axiom 1 (there is no recurrence of differentiation without differentiation) but is not derivable from it: a structure containing a single tokened differentiation and nothing further satisfies Axiom 1 while containing no recurrence, so no analysis of Axiom 1 alone can yield Axiom 2 — non-derivability by countermodel. The framework's foundation is therefore most honestly described not as a single axiom but as a *two-axiom observation base*: one indubitable claim, and one substantive posit that the first does not yield. The two invoke a single principle — tokened differentiation — and are thematically one; they are numerically two. The Definition and Lemmas below unpack this structure.

The perspectival reading of the primitive is a commitment, and it is worth stating what the alternative would cost. One could instead read the evidential floor *thinly* — as bare differentiated content with no locus occupying a side — in which case the embeddedness $V \subsetneq S$ does not follow and would have to be added as a third, independent axiom. The framework adopts the perspectival reading: the floor the cogito secures is already a registering, not a free-standing fact, so embeddedness belongs to the primitive rather than being posited beside it. This choice has a payoff beyond economy of axioms. On the perspectival reading the foundation's incompleteness is an *internal* consequence of the primitive, not an external import: an embedded representation cannot close over the whole that contains it, since a complete self-inclusive representation would have to represent its own representing without remainder, collapsing the locus into the whole. This is the same form of diagonal that limits formal systems (Gödel [49]) and embedded inference devices (Wolpert [50]); those results corroborate the internal statement rather than supply it. The choice between the perspectival primitive and the thin-plus-third-axiom formulation is presentational — it bears only on how the foundation's commitments are counted and changes no result downstream of the structure (S, φ, V) — but the perspectival statement is the more faithful to the indubitable floor the cogito secures, which is why it is adopted here.

We ask: what is the minimal mathematical structure consistent with this fact?

Definition. An *observation* is a triple (S, φ, V): a total system S, a dynamics $\varphi : S \to S$, and an observer $V \subsetneq S$ — a proper subsystem with finitely many distinguishable internal states, coupled to the complement $H = S \setminus V$ through φ.

This definition formalizes three features implicit in the concept of observation: there is a whole (S), the observer is not the whole ($V \subsetneq S$), and the observer registers changes (coupling through φ). From this definition, three structural properties follow. The dynamics φ itself is nothing exotic, and its form need not be posited: it follows from a single innocuous premise, that evolution *composes* — advancing the state by t and then by s equals advancing it by $t + s$. This composition (semigroup) law is Cauchy's functional equation, and it forces the evolution to be a one-parameter family generated by a fixed rule [45]; in the continuous case this is the *phase flow* foundational to classical mechanics [44], and in the discrete case relevant here it collapses to iteration of a single transition map, $x_t = \varphi^t(x_0)$. Thus the state-transition structure is not an assumption particular to this framework but a consequence of consistency of time evolution — the same structure standard physics rests on. What distinguishes the present setting is not the existence of such a map but that it acts on a *finite* S (Lemma 1); finiteness, not the dynamics, is the load-bearing commitment, and it is what renders energy and its conservation well defined downstream.

Lemma 1 (Finiteness). *The observer has finitely many distinguishable internal states, so the visible configuration space $\mathcal{C}_V$ is finite, with a discreteness scale ϵ providing a finite minimal cell volume.* Any observer bounded by a finite-area surface can couple to only finitely many modes across that boundary; independent support comes from holographic entropy bounds [2].

Lemma 2 (Causal partition). *The observer is a proper subsystem: $V \subsetneq S$. The complement $H = S \setminus V$ is the hidden sector.* This partition is not a modeling choice but the definition of embedded observation: an observer that could access all of S would have nothing external to observe. The product decomposition

$$\Gamma = \Gamma_V \times \Gamma_H, \qquad H_{\text{tot}} = H_V + H_H + H_{\text{int}}$$

follows, with cross-partition correlations entering only through H_{int}. The product decomposition is idealized; §4.5 addresses approximation quality.

Lemma 3 (Determinism, reversibility, and the selected measure). *φ is a function with a unique successor for every state (determinism) and, as the reversible representative of §1.2, a bijection: distinct states have distinct successors and predecessors (injectivity — reversibility). On each accessible orbit the invariant measure is unique (uniform on the cycle); globally, the counting measure is selected among invariant measures as the maximal-entropy one — a selection principle, not uniqueness from invariance alone.*

Remark (the continuum analogue). The same distinction appears in the continuum: Liouville measure is invariant but is not, in general, the unique absolutely continuous invariant measure.

Remark (the measure as a realization datum). Lemma 3 fixes the canonical measure — invariant, maximal-entropy — and the framework's baseline transition matrices (§2.3) and the physical realization use it. It does not preclude structured preparations: a realization in the sense of §3.4 carries its own fixed hidden prior μ_H as part of the realization datum, and the emergent law is a function of $(\varphi, \text{partition}, \mu_H)$ — the same bijection and partition under different priors generally yield different visible laws (trivially: the two-state swap under uniform versus point-mass hidden priors). Partition-relativity statements are therefore relative to the declared measure; the counting measure is the default, not a constraint on representations.

Remark (status of the early commitments). Of the structure above, only finite visible resolution is a theorem (Lemma 1). The bijective substratum is a representation choice — the minimal recurrent bijective representative of the hidden dynamics; total finiteness of that representative is a physical posit, carried in the posit ledger (§4.5); observed near-unitarity is the empirical input motivating the choice. The characterization theorem is accordingly conditional on finite reversible substratum dynamics — an empirically motivated reconstruction, not a pure derivation. Determinism (injectivity) holds by a two-pronged argument. For states differing in registered content, injectivity is the requirement that the dynamics preserve the observer's records — merging them would erase a distinction that has occurred. For states differing only in the hidden sector, a merge is either statistically observable or removable: a merge whose successor configurations differ visibly breaks the double stochasticity of the marginalized process at some step order (the marginal of a bijection under the uniform prior is doubly stochastic by Birkhoff–von Neumann, §3.2, and unitary emergent statistics are unistochastic), so it is excluded by the observed unitarity of quantum dynamics; a merge that leaves no statistical trace is removable, since any map on a finite set restricts to a bijection on its *eventual image*, where any recurrent registered history lives — the non-injective description is then equivalent, for all registered history, to a bijective one. Either horn delivers a bijective substratum for the dynamics governing registered history; the two prongs use different inputs — the first empirical (the observed unitarity of quantum dynamics), the second structural (finiteness and recurrence). On a finite set, injectivity implies surjectivity, so φ is a bijection. As for the measure: on the orbit of the actual microstate, the uniform measure is the unique φ-invariant measure; the selection among orbits is fixed by maximal entropy compatible with the observer's records — and the atypical short orbits of the idealized linear dynamics (fixed points, low-period cycles) are precisely observer-lacking substrata — an orbit of period p makes at most p states dynamically accessible, so for small p no partition of it can satisfy the high-capacity condition C3 — admitting no C1–C4 partition, hence outside the scope of every theorem below. The invariant counting measure, in the continuum limit, becomes the Liouville measure on Γ:

$$\frac{\partial \rho}{\partial t} = \{H_{\text{tot}}, \rho\} \equiv \mathcal{L}\rho$$

Remark (reversibility as the conservative assumption). The bijectivity of φ — equivalently, that the substratum conserves information — is not an exotic commitment peculiar to this framework but the orthodox assumption of mainstream physics, imported here rather than invented. Unitary evolution in quantum mechanics *is* information conservation: a pure state evolves from a unique pure state, so the dynamics is reversible and the past is in principle recoverable from the present [42]. The classical counterpart is Liouville's theorem, the conservation of phase-space volume recovered above. That the physics community treats information conservation as effectively non-negotiable is shown by the black-hole information paradox, in which the *apparent* violation of unitarity by Hawking evaporation is regarded as a paradox to be resolved rather than a discovery to be accepted [43, 42]; the no-hiding theorem [42] makes the underlying conservation statement rigorous. The framework therefore assumes no more than standard physics already does — and it is worth noting that the literature independently records the same information↔energy link that the companion Explainer draws via Noether's theorem: a violation of unitarity would also violate energy conservation. What is non-standard here is not the assumption of reversibility but the claim to *recover* it, below, from the two-pronged argument of Lemma 3; a reader unwilling to grant that recovery may simply take reversibility as the standard postulate it is elsewhere, with no cost to the emergence results.

These properties contain no quantum postulates. The claim is that quantum mechanics *emerges* under the conditions below.

1.3 Conditions on the Hidden Sector

Companions and independence. The four conditions below are diagnostics of a realization rather than hypotheses of the characterization: the equivalence of §3.4 quantifies over none of them (Remark, *the conditions are diagnostics, not hypotheses*), (C1) and (C3) follow from (C4) in any faithful realization, and (C2) is a physical-regime premise. They are stated and used because they are what one can check of a candidate partition — a cosmological horizon, a laboratory apparatus, an equilibrium fluctuation — where the bare existence claim cannot be checked directly. Every result uses only these, the two axioms, and the one external correspondence of §3.1. Their cosmological realization — the horizon as the hidden sector, with its capacity, its timescales, and its bidirectional read-write coupling — is developed in companion work and is nowhere consumed by a proof here; the realization clauses below are physical statements of that picture, not premises.

(C1) Non-zero coupling. $H_{\text{int}} \neq 0$: hidden degrees of freedom affect the visible evolution at some accessible step — the conditional law of some visible transition depends on the hidden state. A non-permutation one-step matrix T is the cheapest sufficient *witness* (§2.3), not an equivalent: influence can hide at step one and surface later (the delayed-revival family, realized by the §3.4 construction; `review4_probes.py`). (Status: doubly anchored. Necessity — observed non-Markovian dynamics force C1: with no hidden influence at any accessible step, every visible transition is a fixed function of the visible present and the process is Markov (§3.4); directly observed memory effects in open-system dynamics certify the antecedent. Bell violation supplies a separate spatial-composite anchor only on the adopted ontic-parameter-dependent branch of §3.3; it does **not** by itself imply temporal P-indivisibility. Realization — at the cosmological horizon the ADM boundary Hamiltonian and the constraint structure supply a *bidirectional* coupling stronger than $H_{\text{int}} \neq 0$. Bidirectionality is a property of the physical realization, not part of C1's defining content: a one-way influenced process can be P-indivisible — see the Remark at the non-permutation-witness lemma, §3.4. Dependency — C1 is *not independent of C4*: a readback gap requires hidden mediation, so any realization satisfying C4 satisfies C1, and zero-coupling realizations exhibit no gap at any order, certified by exhaustive enumeration at $(n_V, |\mathcal{C}_H|) = (2,2),(2,3),(3,2)$ in `primitive_probes.py`. C1 is retained as a separately checkable condition — the adopted Bell branch independently requires Bell-relevant cross-wing coupling, while C4 already forces hidden mediation — not as an independent postulate.)

(C2) Memory persistence. Stated in two named forms, because the corpus needs both and they are not equivalent. **(C2-structural):** a hidden record of visible history, once written, persists to the readback that uses it — the retention that clause (b) of the universal hidden-memory theorem shows every memory-bearing completion must contain (§3.4). **(C2-slow):** the timescale mechanism — the hidden sector's *coarse-grained mixing time* τ_B, the timescale on which the effective channel degrades a stored record of visible history, is long compared to the accessible window: $\tau_S \ll \tau_B$, the *inverse* of the Markovian regime. (C2-slow) implies (C2-structural) under the uniform-erosion estimate of §2.3 — a record eroding at rate $1/\tau_B$ survives windows $k\tau_S \ll \tau_B$ essentially intact — but not conversely: what is required is that records persist, not that the hidden sector evolve slowly. Slow evolution is one *sufficient mechanism* for persistence (a record that stays put survives), but fast autonomous hidden dynamics with conserved structure serve equally — a shift-register bath rotating five cells per visible step produces a maximal order-2 readback gap, because its circulation forgets nothing and routes the record back within the window, satisfying (C2-structural) while violating (C2-slow) (`fastbath_probes.py`). Unqualified "(C2)" in this paper means the persistence requirement, with the form named wherever the distinction bites. The same probe certifies the complementary boundary: persistence alone does not supply (C4) — with the return time pushed outside the accessible window the identical non-forgetting register yields an exactly i.i.d. visible process. (C4)'s operative demand on the hidden sector is thus persistence *together with in-window routing*; (C2) names the persistence half. (Status: **a physical-regime premise, not a structural condition** — and the distinction is load-bearing. C1, C3 and C4 are the structural conditions: the characterization's equivalence runs over those three (§3.4), and the existential realization satisfies C2 trivially with a static bath. What C2 supplies is physical: it is the mechanism by which a stored record survives to be read in realistic models, so that C4 holds rather than failing by erasure (§2.3); and in the other direction it is *derived* — observed conditional memory forces a lower bound on bath persistence, conditional on the hidden-sector mixing hypothesis, which is the physical-memory theorem of §3.4 rather than an assumption. Note also that C2 does not imply C4: a slow bath can store without reading, certified by a C1–C3 realization with identically zero backflow on arbitrarily long horizons (`review3_probes.py`), which is why C4 is a separate condition and not a consequence. ETH, C2's physical motivation — a well-supported conjecture rather than a theorem — is discussed at the remark in §3.4. Realization — $\tau_S/\tau_B \sim 10^{-32}$ at the horizon.)

(C3) Sufficient memory capacity. The hidden sector has enough distinguishable states to encode the observed conditional past–future information: $\log_2 |\mathcal{C}_H| \geq I^*$, the per-process bound of §3.4. (Status: necessity — the data-processing bound, $m \geq 2^{I^*}$, for any realization. Realization — at the horizon the margin is astronomical: $A/\epsilon^2 \sim 10^{122}$ hidden modes against $\sim 10^{80}$ baryons, a hierarchy $N_H \gg N_V$ with the bath imperturbable on laboratory timescales; the hierarchy and the imperturbability belong to this realization clause, not to the condition. Dependency — like C1, C3 follows from C4 within any faithful realization: the observed conditional information must pass through the hidden sector, so data processing gives $\log_2 |\mathcal{C}_H| \geq I^*$ automatically. Exhaustive enumeration confirms it and finds the bound SATURATED — realizations attaining $I^* = \log_2 |\mathcal{C}_H|$ exist at $(2, 2)$ and $(3, 2)$ (`primitive_probes.py`) — so C3 is tight rather than slack, and is retained as a separately checkable condition.)

(C4) History readback. History-sensitive hidden mediation: on accessible windows, hidden degrees of freedom carry information about the visible past into future visible conditionals — at some order, two visible histories with the same current state induce different next-step laws, mediated through the hidden state. The condition is operational; it does not by itself assert a causal write-then-read cycle — a pre-sampled hidden variable revealed by the history satisfies it (the response-table construction, §3.4). (Status: necessity — immediate given observed non-Markovianity, since any realization of a non-Markov process carries differing past-conditionals. Realization — in the cosmological realization the mediation *is* a causal read-write cycle: the interaction reads the same boundary degrees it writes, a property to be independently demonstrated there; this is what C1's redefinition relocated here. Separation — (C1)–(C3) alone do not imply (C4): a tape-and-ledger realization of the fair coin satisfies all three with identically zero backflow on arbitrarily long accessible horizons (`review3_probes.py`; §2.3).)

Theorem statement. The claims divide into three exact layers and one explicitly conditional extension. **(1) Representation equivalence:** on every finite accessible horizon, the stochastic laws S, their finite reversible deterministic realizations D, and fixed-basis unitary/Born representations Q_{fb} are the same class, $S \iff D \iff Q_{\text{fb}}$ (§3.4). **(2) Readback content:** C4 is not needed for the existence of a quantum representation; it is what makes the OI sector nontrivial. C4 forces conditional memory in the visible law, readout-relevant hidden predictive memory in every faithful completion, and — for a fixed finite reversible representative — global indivisibility somewhere in the recurrence cycle (§2.3). C1 and C3 follow as necessary diagnostics of the mediation and its capacity; C2-structural supplies persistence to readback, with C2-slow one physical mechanism. **(3) Reverse finite realization:** every finite stochastic law, including every finite quantum record, has a finite reversible OI realization; bounded quantum instrument families are uniformly realizable to arbitrary finite accuracy (§3.4). **(4) Full operational extension:** identifying one common *standard local quantum instrument/composite theory* for all coherent interventions requires the additional operational-lifting and composition hypotheses stated in §3.4. Bell is a separate constraint rather than one of those lifting hypotheses: for a response-complete deterministic completion, measurement independence plus ontic parameter independence would force $|S_{\mathrm{CHSH}}| \leq 2$, so the Bell-violating branch adopted in §3.3 retains measurement independence and relinquishes ontic parameter independence while preserving operational no-signaling.

1.4 Partition-Relativity

Lemma (partition-relativity under the canonical invariant measure). *Under the canonical measure of Lemma 3, the emergent description is uniquely determined by the partition; any parameters of the emergent theory depend only on the geometric and thermodynamic properties of the partition boundary. For the abstract representation theorems the prior is a third datum: canonically* $T = T(\varphi, \mathcal{P}, \mu_{inv})$, *in general* $T = T(\varphi, \mathcal{P}, \mu_H)$ *(§1.3 Remark).*

Proof. By Lemma 3, the Hamiltonian flow ϕ_t is unique. By Lemma 2, the partition is fixed. By Lemma 3, the observer uses the Liouville measure μ — the maximal-entropy invariant measure on the full phase space. (An embedded observer with incomplete knowledge of the total energy averages over energies with a smooth prior, recovering Liouville on a phase-space band — the continuum form of Lemma 3's maximal-entropy selection; singular invariant measures concentrated on atypical short orbits correspond to observer-lacking substrata and are outside scope by the same lemma.) Letting π_V denote projection onto the visible sector, the marginalized transition probabilities

$$T_{ij}(t_2, t_1) = \int_{\Gamma_H} \delta_{x_j}[\pi_V(\phi_{t_2 - t_1}(x_i, h))] \, d\mu(h)$$

are therefore uniquely determined by three inputs: dynamics (ϕ_t), partition (Γ_V, Γ_H), and measure (μ). Since ϕ_t and μ are fixed by the definition and lemmas, the stochastic process — and hence any emergent quantum description (§3.1) — depends only on the partition. □

1.5 The four layers

The results organize into four layers, and every claim should be read at its layer. **Layer 1 — reversible dilation** (§2.2, §3.4): exact mathematics, no quantum claim —

$$\text{finite-horizon non-Markovianity} \iff \exists\,\text{finite reversible hidden-state realization},$$

the paper's core. **Layer 2 — memory and capacity** (§2.3): (C4) forces positive conditional memory; the (C3) information bound; the physical-memory theorem ((C2), conditional on mixing). **Layer 3 — quantum representation** (§3.1, §3.4): the imported correspondence (Q1) and the in-house uncompressed form (Q2), scopes separated. **Layer 4 — physical and operational realization** (§3.2, §4, companions): interventions (classical comb closed, §3.4; quantum instruments open), coherent preparation, Bell and free-choice structure, H-spin' and H-χ', the cosmological realization — where the named hypotheses live. A Layer-1 result does not by itself establish any Layer-4 claim.

2. EMERGENT STOCHASTICITY AND P-INDIVISIBILITY

2.1 Emergent Stochasticity

The total system evolves deterministically, but the visible sector alone is stochastic. The observer knows x but not h; different hidden states h_k compatible with the same x send it to different futures x'_k. Transition probabilities are obtained by marginalizing over the Liouville measure (§1.4). This makes the framework a hidden variable theory; consistency with Bell's theorem is addressed in §3.3.

2.2 The Slow-Bath Regime

The Markovian limit requires $\tau_B \ll \tau_S$ [3]. Condition (C2) inverts this: $\tau_S \ll \tau_B$. A slow bath must be distinguished from a static field. By (C1), coupling is continuously active: each visible-sector transition imprints on the hidden sector. Because the hidden sector is slow (not static), imprints persist without thermal overwriting. On subsequent transitions, coupling reads back stored correlations, producing history-dependent transition probabilities — the non-Markovian regime [3].

2.3 P-Indivisibility

By Lemma 1 the visible sector is finite; under the total-finiteness posit (§1.2 Remark, §4.5 ledger) the hidden sector is also finite — configuration spaces $\mathcal{C}_V$ and $\mathcal{C}_H$ (the discrete counterparts of Γ_V and Γ_H). On these finite sets:

Theorem. *Let $\mathcal{C}_V$ and $\mathcal{C}_H$ be finite sets with $|\mathcal{C}_V| \geq 2$, and let $\varphi : \mathcal{C}_V \times \mathcal{C}_H \to \mathcal{C}_V \times \mathcal{C}_H$ be a bijection. Define:*

$$T_{ij} = \frac{|\{h \in \mathcal{C}_H : \pi_V(\varphi(x_i, h)) = x_j\}|}{|\mathcal{C}_H|}$$

and the k-step matrix $T_{ij}^{(k)}$ by applying φ^k. If T is not a permutation matrix, then the process is P-indivisible.

Proof. Uses total variation distance $d(p,q) = \frac{1}{2}\sum_k |p_k - q_k|$. P-divisibility $\implies$ $d(p(t), q(t))$ non-increasing for all initial p, q [4, 5] (a stochastic divisor is a total-variation contraction); the proof below uses only this direction, in contrapositive form. The converse is *false*: monotonicity constrains the divisor only on the image directions $(\delta_i - \delta_j)T(1,0)$, not on the extreme points of the zero-sum ball, and explicit stochastic pairs exist whose unique divisor $T(1,0)^{-1}T(2,0)$ carries a negative entry — no stochastic divisor, hence P-indivisible — while d is non-increasing for every p, q; one such pair is strictly contracting at $n = 2$ and one lies in the doubly stochastic class at $n = 3$ (`translation_probes.py`). Monotonicity is therefore a necessary, not sufficient, condition for P-divisibility, and is not used as a criterion anywhere below.

Step 1 (Recurrence). φ bijective on a finite set $\Rightarrow \exists N : \varphi^N = \mathrm{id}$. Thus $T^{(N)} = I$ and $d(\delta_i T^{(N)}, \delta_j T^{(N)}) = 1$ for $i \neq j$.

Step 2 (Strict contraction). T not a permutation $\Rightarrow \exists i, j, l$: $T_{il} > 0$ and $T_{jl} > 0$. Then:

$$d(\delta_i T, \delta_j T) = \frac{1}{2}\sum_k |T_{ik} - T_{jk}| < 1$$

Step 3. $d(\delta_i T^{(1)}, \delta_j T^{(1)}) < 1 = d(\delta_i T^{(N)}, \delta_j T^{(N)})$: non-monotonic, hence P-indivisible. □

The theorem requires only bijective dynamics (Lemma 3) and non-trivial coupling (C1). Lemma 1 gives the finite visible space; the hidden space is finite under the total-finiteness posit (§1.2, §4.5).

Lemma (stochastic inverse). *If a finite square stochastic matrix has a stochastic left or right inverse, then it is a permutation matrix.*

Proof. Suppose $AB = I$ with A and B stochastic. For $i \neq j$, $0 = (AB)_{ij} = \sum_k A_{ik}B_{kj}$ is a sum of nonnegative terms. Hence each k with $A_{ik} > 0$ has the kth row of B supported only on column i; stochastic normalization makes that row the basis vector e_i. Two positive entries in one row of A would therefore make two rows of B identical, contradicting invertibility. Thus every row of A has one unit entry, and invertibility makes the occupied columns distinct. The right-inverse case is symmetric. □

Theorem (history readback + finite recurrence implies indivisibility). *For a fixed finite reversible OI representative, genuine C4 history readback implies that the rooted visible stochastic process is indivisible somewhere in its full recurrence cycle.*

Proof. Because φ is a permutation of a finite set, it has finite order L, so the rooted visible map at L is $\Gamma_L = I$. If every earlier rooted visible map Γ_k were a permutation, the current visible configuration would be a bijective deterministic function of the initial visible configuration at every time. Two positive-probability histories with the same current endpoint would then identify the same initial visible state and could not have different next-step laws, contradicting C4. Hence some Γ_k is non-permutation. Divisibility from that time to the recurrence would require a stochastic divisor producing I from Γ_k, i.e. a stochastic left or right inverse, forbidden by the lemma. □

Scope. This is a global recurrence-cycle result. It does **not** say that C4 forces P-indivisibility on every accessible short-time window; the XOR control remains a counterexample to that stronger statement. What C4 buys exactly is hidden predictive readback and global indivisibility of the fixed finite recurrent law.

Genericity. The theorem's precondition — that T is not a permutation matrix — is a sufficient first-step *witness* of (C1) — neither necessary nor equivalent to the global condition (§1.3; the delayed-revival family hides its influence at step one, `review4_probes.py`) — and is generic among bijections on $\mathcal{C}_V \times \mathcal{C}_H$ in a quantifiable sense. A bijection φ on $\mathcal{C}_V \times \mathcal{C}_H$ produces T equal to a permutation matrix if and only if φ has the form $\varphi(x_i, h) = (\varphi_V(x_i), \sigma_{x_i}(h))$ — the V-dynamics is independent of h. Counting these: $|\mathcal{C}_V|!$ choices for φ_V times $(|\mathcal{C}_H|!)^{|\mathcal{C}_V|}$ choices for the family $\{\sigma_{x_i}\}$, giving a total of $|\mathcal{C}_V|!\,(|\mathcal{C}_H|!)^{|\mathcal{C}_V|}$ bijections with T a permutation, out of $(|\mathcal{C}_V|\,|\mathcal{C}_H|)!$ total. The fraction failing the precondition is therefore

$$P_{\text{perm}}(n_V, n_H) = \frac{n_V!\,(n_H!)^{n_V}}{(n_V n_H)!}, \quad n_V := |\mathcal{C}_V|,\ n_H := |\mathcal{C}_H|.$$

Stirling's approximation gives the asymptotic decay $\log P_{\text{perm}} \sim -n_V n_H \log n_V$ as $n_H \to \infty$ with n_V fixed — equivalently, $P_{\text{perm}} \sim n_V^{-n_V n_H}$. For any $n_V \geq 2$, the fraction of bijections failing C1 at the T-level decays exponentially in the hidden-sector capacity n_H. At modest physical scales — say $n_V = 10$, $n_H = 100$ — the fraction is $\sim 10^{-981}$; at the scales C3 invokes, it is astronomically smaller. The theorem is therefore non-vacuous on two independent grounds: the precondition excludes a structurally distinguished class of decoupled bijections (those whose V-dynamics have no h-dependence), and that class is a vanishing fraction of the configuration space as C3 is engaged. C3 (high-capacity hidden sector) and C1 (non-trivial coupling) are independent conditions in statement, but C3 is the *engine* of C1's genericity: enlarging n_H shrinks the measure of C1-violating bijections super-exponentially fast.

Relation to prior work. The P-indivisibility phenomenon is well-documented in the open-systems literature: reduced dynamics need not preserve positivity [6]; divisibility failure is equivalent to information backflow [4, 5]; process-tensor Markovianity provides an alternative characterization [7]; non-Markovian reduced dynamics arise in the slow-bath regime [8]. The Bylicka-Johansson-Acín equivalence between CP-divisibility and distinguishability monotonicity [9] narrows the gap between divisibility and non-Markovianity, but only for *invertible* reduced dynamics; the marginal $T(t)$ here is generally non-invertible, so the load-bearing property for the general argument remains P-indivisibility of the marginal process. Milz et al. [10] confirm CP-divisibility does not imply Markovianity in the process-tensor sense.

Continuous-time extension. The Hamiltonian flow on finite-dimensional phase space preserves Liouville measure on compact energy surfaces. $T_{ij}(t)$ is continuous with $T(0) = I$. By (C1), $T(t)$ departs from the permutation class for $t > 0$. By Poincaré recurrence, $\exists t_R$: the set of hidden states with $\pi_V(\varphi_{t_R}(x_i, h)) = x_i$ has measure $> 1 - \delta$ for any $\delta > 0$. For small δ, this gives non-monotonic trace distance, establishing P-indivisibility in continuous time.

The recurrence argument establishes P-indivisibility in principle. The following lemma shows that under (C2), information backflow occurs on observable timescales — not merely at Poincaré recurrence times.

Physical motivation (storage under C2). The coupling H_{int} writes visible-sector information into the hidden sector at rate $\sim 1/\tau_S$; between interactions the hidden sector's effective relaxation at rate $\sim 1/\tau_B$ (the coarse-grained channel's mixing; hypothesis (EM), §4.6) erodes it as $e^{-\tau_S/\tau_B}$ per step, so on windows $k\tau_S \ll \tau_B$ the stored history survives essentially intact. Storage, however, is not backflow: what is stored must also be *read*. The lemma below makes the reading quantitative, and the separation remark shows why it is a genuinely additional requirement.

Run backwards, the same estimate bounds what the failure of (C2) costs. Since the equivalence of §3.4 does not quantify over (C2), its failure carries no consequence except through that bound.

Lemma (fast bath erases readback). *Suppose the coarse-grained channel mixes fast relative to the visible step, $\tau_B \ll \tau_S$, and erodes the stored record uniformly, by $e^{-\tau_S/\tau_B}$ per step, across the history pairs at issue. Then for any $k \geq 2$ the order-k readback gap obeys*

$$\delta_k \;\leq\; \delta_0 \, e^{-(k-1)\tau_S/\tau_B}.$$

A fast, uniformly erasing bath therefore drives the readback gap to zero exponentially, and (C4) — which requires a gap at some order in the accessible window — fails.

Remark (what this lemma does not give). The backflow lemma above is a LOWER bound on $I(X_{<k}; X_{k+1} \mid X_k)$ given a gap; an upper bound on δ_k cannot be inserted into it to bound I from above. The corresponding upper bound on conditional mutual information is available, but by the continuity route — uniform total-variation closeness through the Fannes–Audenaert estimate — and it is proved where it is used, at the physical-memory theorem of §3.4. The statement above is confined to the gap.

Remark (why this matters for the exclusion arguments). Since (C2) sits outside the equivalence (§3.4), its failure excludes nothing by the characterization alone. It also matters which reading of (C2) is in play. Under the *evolution* reading — fast hidden dynamics — failure is compatible with (C4) holding at full strength:

a shift-register bath rotating five cells per visible step supports a maximal order-2 gap (`fastbath_probes.py`), which is why (C2) is stated as persistence rather than slowness (§1.3). It is the failure of PERSISTENCE, with the uniform erosion hypothesis of the lemma above, that carries (C4) down with it. The exclusion arguments below — the equilibrium phase, and the anti-Boltzmann-brain corollary of §4.6 — therefore run through this lemma: fast mixing erases the stored record before it can be read, (C4) fails, and it is (C4)'s failure that the characterization registers. (C2) is the mechanism; (C4) is the exclusion.

Lemma (accessible backflow from readback). *Under (C1)–(C3), suppose (C4) holds with a gap at order k in the accessible window: two visible histories $p \neq p'$ of length k with a common endpoint, each of probability at least p_0, induce next-step laws at total-variation distance at least δ. Then*

$$I(X_{<k};\, X_{k+1} \mid X_k) \;\geq\; \frac{p_0\,\delta^2}{\ln 2} \;>\; 0.$$

Proof. Condition on the common endpoint c and let w_a be the conditional weights of the length-k histories, P_a their next-step laws, and $\bar{P} = \sum_a w_a P_a$. The conditional mutual information given $X_k = c$ is the mixture divergence $\sum_a w_a\, D(P_a \| \bar{P})$, and Pinsker's inequality gives $D(P_a \| \bar{P}) \geq \frac{2}{\ln 2}\,\mathrm{TV}(P_a, \bar{P})^2$ in bits. By the triangle inequality $\mathrm{TV}(P, \bar{P}) + \mathrm{TV}(P', \bar{P}) \geq \delta$, so $\mathrm{TV}(P, \bar{P})^2 + \mathrm{TV}(P', \bar{P})^2 \geq \delta^2/2$. Multiplying by $\Pr[X_k = c]$ converts conditional weights to path probabilities, each at least p_0; hence $I(X_{<k}; X_{k+1} \mid X_k) \geq \frac{2}{\ln 2} \cdot p_0 \cdot \frac{\delta^2}{2} = p_0\delta^2/\ln 2$. □ (Certified exactly on gap families, the revival family, and a small-gap parametric family, with the coin silent, in `c4_backflow_probes.py`.)

Remark (why (C4) is a separate condition). (C1)–(C3) admit realizations with zero backflow throughout arbitrarily long accessible horizons: the fair coin realized by the §3.4 dilation construction — the tape influences, the ledger records, and the two never meet — has $T^{(1)} = \frac{1}{2}J$ (so (C1) holds), a static bath ($\tau_B = \infty$), vast capacity, and $I(X_{<t}; X_{>t} \mid X_t) = 0$ exactly at every horizon time (`review3_probes.py`). Influence plus storage plus capacity, without readback, is noise.

Role of (C2)–(C4). The Poincaré recurrence argument and the readback lemma are independent: the former establishes P-indivisibility from a non-permutation one-step witness and finiteness alone, at the recurrence time; the latter locates observable backflow inside accessible windows. (C2) keeps stored history alive between couplings; (C3) provides the room to store it; (C4) is the reading of it — and only the reading produces memory an observer can measure.

Weak-coupling and fast-bath regimes. Known P-divisible systems with non-zero coupling illustrate the necessity of (C2), not a failure of the theorem. The Jaynes-Cummings model with Lorentzian spectral density is P-divisible when dissipation dominates coupling — the fast-bath regime ($\tau_B \ll \tau_S$), violating (C2). The Davies-Merkli Markovian convergence theorem likewise applies in the Born-Markov (fast-bath) limit. The §3.4 necessity proof for (C2) establishes this formally.

2.4 Explicit Construction: A Minimal Model

Visible sector: $x \in \{0,1\}$. Hidden sector: $h \in \{1,\dots,6\}$. Dynamics: the permutation

$$\sigma = (0{,}1 \leftrightarrow 1{,}1)\ (0{,}2 \leftrightarrow 1{,}2)\ (0{,}3 \leftrightarrow 0{,}4)\ (0{,}5 \leftrightarrow 0{,}6)\ (1{,}3 \leftrightarrow 1{,}4)\ (1{,}5 \leftrightarrow 1{,}6)$$

satisfying (C1)–(C4) — the flip depends on hidden components that the dynamics itself conditions on the past, so readback is present. Since $\sigma^2 = \mathrm{id}$: at $t = 1$, two of six hidden states flip x, giving

$$T(1,0) = \begin{pmatrix} 2/3 & 1/3 \\ 1/3 & 2/3 \end{pmatrix}$$

At $t = 2$ ($\sigma^2 = \mathrm{id}$): $T(2,0) = I$. A Markov chain would predict continued mixing: $T(1,0)^2 = \left(\begin{smallmatrix} 5/9 & 4/9 \\ 4/9 & 5/9 \end{smallmatrix}\right)$. The actual dynamics show complete *un-mixing*. Computing the intermediate propagator:

$$\Lambda(2,1) = T(2,0) \cdot [T(1,0)]^{-1} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

Negative entries — no valid stochastic matrix exists. **P-indivisible.** □

Independent check (process-tensor non-Markovianity). The same model also fails the Pollock et al. [7] multi-time Markov condition without invoking $T(t)$-invertibility: $P(x_2 = 0 \mid x_1 = 1, x_0 = 0) = 1$ versus $P(x_2 = 0 \mid x_1 = 1) = 1/3$. Conditioning on x_0 changes the one-step prediction from 1/3 to 1 — a memory effect computed directly from multi-time conditional probabilities, valid regardless of $\det T(t)$. The non-Markovian character is therefore confirmed under both the divisibility-based criterion (Rivas et al. [4, 5]; Bylicka et al. [9] in the invertible case) and the process-tensor criterion (Pollock et al. [7] in the general case).

The mechanism. The hidden sector acts as a memory register. The transpositions $(0,h) \leftrightarrow (1,h)$ for $h \in \{1,2\}$ flip x while preserving h, so a state at $x = 1$ with $h \in \{1,2\}$ after step 1 carries the record "I was at $x = 0$." Step 2 reads this record and flips x back — information backflow impossible for a memoryless process. P-indivisibility is more extreme, not less, when memory is more persistent.

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3. THE EMERGENCE OF QUANTUM MECHANICS

3.1 The Stochastic-Quantum Correspondence

By Barandes' correspondence [11, 12], any finite-configuration member of the source's indivisible-stochastic-process class — its term of art for the tuple class, which includes Markov chains — embeds into a unitarily evolving quantum system on $\mathcal{H}$ of dimension $\leq n^3$; (T) is what admits the framework's processes to that class. The transition matrices themselves remain stochastic; what develops negative entries at fine time resolution is the candidate intermediate divisor Λ, which is what P-indivisibility asserts cannot be made stochastic; the Stinespring dilation theorem guarantees a dilated Hilbert space and unitary $\tilde{U}(t)$ whose unistochastic matrix $|\tilde{U}(t)|^2$ marginalizes over the ancilla index to $T_{ij}(t)$ — transition probabilities are exactly Born-rule probabilities. *The ancilla is not optional for this framework's own processes.* Marginalizing a bijection under the uniform hidden prior yields a doubly stochastic one-step matrix (§3.2), and for $n \geq 3$ doubly stochastic does not imply unistochastic: enumerating the construction's own realizations at $n_V = 3$, $|\mathcal{C}_H| = 2$ gives twenty-one distinct one-step matrices, all doubly stochastic, of which six — including $\begin{pmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 1/2 & 1/2 \end{pmatrix}$ — fail the row-orthogonality condition necessary for unistochasticity (`translation_probes.py`). An undilated $T = |U|^2$ representation is therefore unavailable on the framework's class, and the translation hypothesis (T) must be read against the ancilla-dilated statement — not against the bare one. What that costs is exactly one ancilla factor.

Lemma (one-step ancilla dilation). *For any stochastic T on n states, the map $W|i\rangle = \sum_j \sqrt{T_{ij}}\,|j\rangle \otimes |i\rangle$ is an isometry $\mathbb{C}^n \to \mathbb{C}^n \otimes \mathbb{C}^n$ whose ancilla-marginal is T: $\sum_\alpha |\langle j, \alpha|W|i\rangle|^2 = T_{ij}$. It extends to a unitary on $\mathbb{C}^n \otimes \mathbb{C}^n$.*

Proof. $\langle Wi|Wi'\rangle = \delta_{ii'} \sum_j T_{ij} = \delta_{ii'}$, so W is an isometry, and the marginal is $\sum_\alpha |\sqrt{T_{ij}}\,\delta_{\alpha i}|^2 = T_{ij}$; an isometry onto an n-dimensional subspace of an n^2-dimensional space extends to a unitary. □ (Certified exactly on all six non-unistochastic realizations, with the completion carried out explicitly for the displayed witness as a 9×9 unitary over the algebraic numbers, in `tdilate_probes.py`.)

So the obstruction above is priced rather than fatal: total dimension n^2 suffices at a single step, within the source's $\leq n^3$ bound for the full process family. Representability is not what (T) is about. **What (T) asserts is tuple instantiation**, and the source's membership conditions are the tuple's own — a finite configuration space, transition maps normalized and trivializing at the conditioning time, a prior, and the observable algebra — not a divisibility property. Divisibility failure is generic in the class rather than a condition of entry, so P-indivisibility is not what admits the framework's processes to it. What remains genuinely open is downstream: whether the ancilla-dilated representation can be made compatible with the passive $\hat{H}$ across horizons (§3.4). The proof requires a finite-dimensional configuration space (Lemma 1); Calvo [13] extends the correspondence to infinite dimensions, but this is not required here. Pimenta [14] distinguishes P-divisible and P-indivisible stochastic-quantum dynamics: the correspondence is non-trivial only in the P-indivisible regime, which is precisely the regime produced by §2.3. An independent derivation using only Stinespring's dilation theorem [15] and standard partial-trace properties [16] is given in §3.2; either route alone suffices for the representation of the transition data; the operational completion — physical preparation of coherent states, noncommuting interventions, the full instrument algebra — is a separate bridge, partially established by the one-step coherence theorem of §3.2 for the lattice realization and otherwise open.

Scope of the correspondence (what it does and does not deliver). Because the compressed-representation clause rests on this external result — the equivalence (ii) $\iff$ (iii) is proved in-house (§3.4), and the fixed-basis representation's existence is internal (the equivalence theorem), neither resting on it —, we state its scope precisely rather than treat it as a black-box equivalence. The correspondence is a published theorem (Barandes, *Philosophy of Physics* 3(1):8, 2025 [11]; proof in [12]) and is genuinely load-bearing here, but four scope conditions are carried throughout the framework and should be read as standing qualifications on every downstream statement that leans on the compressed representation. (i) *Directionality and non-uniqueness.* The map is many-to-one in both directions; the complex matrix Θ with $T_{ij} = |\Theta_{ij}|^2$ is guaranteed to exist but is not unique (an entrywise phase freedom analogous to a gauge potential), so the correspondence does not single out a unique quantum system from a stochastic process. (ii) *Scaffolding versus full QM.* The stochastic data directly supply only configuration-diagonal random variables; non-commuting observables enter as emergent patterns ("emergeables"), and spin enters by dilating the Hilbert space and imposing a rotation representation rather than being read off the stochastic side. Composite systems are on a different footing. For a spatial OI constitution the coupling graph is defined by one-step dependence, so its iterates give exact graph causal cones: before two cones meet, a remote intervention cannot dynamically alter the local substratum state. That kinematic locality does **not**, by itself, prove local tomography or that all visible composites and coherent interventions belong to one common tensor-product quantum instrument category. Those are the narrowly stated operational-lifting conditions of §3.4. Bell setting-independence is logically separate: the framework retains the measurement-independent branch and, for quantum Bell violation, gives up ontic parameter independence as stated in §3.3. The bare correspondence therefore reconstructs the fixed-basis/Born representation and quantum kinematic scaffolding, while the spatial architecture independently supplies causal propagation. Promotion to the full standard operational quantum theory is the coherent-instrument/composite bridge; compatibility of the specific local lattice realization with Bell-required ontic parameter dependence is the independent H-Bell bridge. Its unification with the staggered-to-Dirac spin structure of the lattice substratum — whose spinor decomposition is keyed to the lattice's bipartite partition — is the open step. One premise of that step is secured at the linear level: the bipartite grading itself survives observation exactly — the effective kernel obeys $\Gamma_V K_{\mathrm{eff}}(E)\,\Gamma_V = -K_{\mathrm{eff}}(-E)$ for any site partition (certified exactly in `chirality_grading_probes.py`), so the marginalization cannot generate a grading-breaking (staggered-mass) term and the structure the spinor decomposition is keyed to reaches the observer intact. (iii) *Restricted process class.* The representation $T_{ij} = |\Theta_{ij}|^2$ imposes substantive structure on the admissible processes; in particular Doukas (2026) shows the differentiable form forces the first-order rate matrix to vanish, excluding ordinary continuous-time Markov chains from the *differentiable* form — a restriction on the continuous-time reading, not on the discrete-time tuple class, whose membership conditions (above) include Markov chains — so "any stochastic process" is narrower than it sounds for that reading, and within the differentiable reading the qualifying regime is precisely the indivisible one §2.3 produces, not the generic one. The restriction also runs the other way. Doukas's accompanying bound — that marginalization over a *finite* environment produces no first-order dissipative term, secular leakage requiring an infinite reservoir — is, in continuous-time form, the same structural fact as the second-order estimate of §3.2 for the discrete substratum: a finite deterministic substratum observed by marginalization cannot generate first-order leakage, and its emergent dynamics is thereby confined to the near-unitary corner of process space, displaced from exact unitarity only by the residual dissipation $\mathcal{D}$. What appears as a restriction on the correspondence's process class is, under the present architecture, the mechanism's signature: the framework occupies the corner Doukas identifies not by selection but by structural necessity, conditional on its finiteness commitment (A1, argued at the axiom layer). This is explanatory unification rather than corroboration — near-unitarity is an empirical input elsewhere in the framework — but it converts scope condition (iii) from a qualification into a prediction: fundamental dynamics of continuous-time Markov type would require an actually infinite hidden sector, and the finite-ϵ displacement $\mathcal{D} \sim 10^{-64}$ is the corner-occupancy claim's falsifiable residue. (iv) *Reception.* The result is recent and located at the physics/philosophy boundary. Its mathematical core has since been independently reworked and generalized: the lift structure and its divisibility mechanics by Doukas [22], the countable-dimension extension by Calvo [13], the divisible/indivisible classification by Pimenta [14], and the Tsirelson-bound result in a mainstream physics venue [24, forthcoming in Phys. Rev. A] (a result for the causal-local indivisible construction of [24], not for generic P-indivisibility). The strongest interpretive readings (deflation of the wave function, of entanglement) remain contested philosophical overlays on the mathematical core, which is all the framework requires. None of (i)–(iv) undercuts the framework's use of the correspondence to obtain Born-rule statistics and unitary evolution from the §2.3 process; they bound what may be claimed to follow *from* it.

Independent corroboration of the trace-out mechanism. The technical move from a deterministic substrate to non-Markovian visible dynamics via the trace-out of inaccessible degrees of freedom — central to both routes — is now established at theorem level in the open systems literature. Brandner [17, 18] proves that for autonomous linear evolution equations, integrating out inaccessible degrees of freedom (whether external environments or internal subsystems) yields well-defined non-Markovian visible dynamics in a controlled weak-memory regime, with explicit error bounds and a convergent perturbation scheme. This is the mathematical apparatus the present construction relies upon, derived independently for general open systems. Direct experimental demonstrations of the mechanism include Mehl et al. [19], who showed that hidden slow degrees of freedom in a colloidal system produce non-Markovian visible dynamics violating naive Markovian fluctuation theorems, and Gröblacher et al. [20], who observed non-Markovian Brownian motion in a macroscopic micromechanical oscillator. On the quantum side, Kim [21] shows that monitored quantum systems are formally quantum hidden Markov models, with a rigorous correspondence to classical hidden Markov models for setups involving Haar-random unitaries and measurements — the same formal structure as the stochastic-quantum bridge. The framework's central technical move is therefore standard rather than speculative.

From phase space to configuration space. The transition probabilities T_{ij} are projections of the full phase-space flow onto the configuration sector of Γ_V, with momenta and hidden-sector degrees of freedom absorbed into the Liouville marginalization. The resulting process lives on the discrete configuration space of Lemma 1.

Continuous spectra and the finite-substratum commitment. Lemma 1 fixes the visible configuration space as finite, and the Barandes correspondence as applied here produces an emergent Hilbert space of finite dimension. Standard quantum mechanics makes heavy use of continuous-spectrum observables — position on $\mathbb{R}^3$, momentum, continuous-spectrum scattering — which strictly require infinite-dimensional Hilbert spaces and unbounded self-adjoint operators. The framework's position on this is that continuous spectra are the infrared limit of the discrete spectrum produced by Lemma 1, not a deeper reality that the framework fails to capture. The quantitative separation is extreme: for a visible sector bounded by the cosmological horizon with discreteness scale $\epsilon = 2l_p$, the number of distinguishable positions is $|\mathcal{C}_V| \sim (L_H/\epsilon)^3 \sim 10^{183}$, and the holographic bound on emergent Hilbert-space dimension is $e^{S_{\rm dS}} = e^{10^{122}}$. The discretization error separating this finite-dimensional structure from a continuum approximation is $\lesssim 1/\sqrt{\dim \mathcal{H}} \sim e^{-10^{121}}$ per eigenvalue — operationally zero at any accessible resolution. Deviations from continuum QM at scale E are suppressed by $(E/M_{\rm Pl})^2$, the same scaling that controls Lorentz-invariance violation and trans-Planckian corrections; no independent continuous-spectrum prediction is needed or claimed. Calvo [13] extends the Barandes correspondence to countable-infinite dimensions for completeness, but the finite-dim correspondence is the load-bearing one for the framework — every result in §2–§3 runs on Lemma 1 directly.

Three features emerge. The Schrödinger equation arises from the differentiability of $U(t)$. The Born rule is the equilibrium distribution of the indivisible process [11, 12]. The action scale $\hbar$ enters when defining $\hat{H}(t) \equiv i\hbar\,\partial_t U \cdot U^\dagger$, converting dimensionless rates to energy units; its value requires additional physical input from the partition.

Phase uniqueness from continuous-time data. The relation $T_{ij}(t) = |U_{ij}(t)|^2$ discards phase information at any single time. Doukas [22] identifies this gap for discrete-time data. The resolution: continuous-time data $\{T_{ij}(t)\}_{t\in\mathbb{R}}$ provides strictly more information.

Lemma (phase-locking). *Let H be Hermitian on $\mathbb{C}^n$ with eigenbasis $\{|k\rangle\}$, eigenvalues $\{E_k\}$, and $V_{ik} = \langle i|k\rangle$. Assume: (G1) non-degenerate spectrum; (G2) non-degenerate energy gaps; (G3) $V_{ik} \neq 0$ for all i, k. Then $T_{ij}(t) = |\langle i|e^{-iHt}|j\rangle|^2$ for all i, j, t uniquely determines H up to an overall energy shift, basis phase conventions, and the antiunitary conjugation $H \to -H^*$ — the complex-conjugate representation, which satisfies $|\langle i|e^{-i(-H^*)t}|j\rangle|^2 = |\langle i|e^{-iHt}|j\rangle|^2$ identically and, for generic H, is not shift- or rephasing-related (its spectrum is negated).*

Proof. Write:

$$T_{ij}(t) = \sum_{k,l} V_{ik}\, V_{jk}^*\, V_{jl}\, V_{il}^*\; e^{-i(E_k - E_l)t}$$

By (G2), the frequencies $\omega_{kl} = E_k - E_l$ are distinct for distinct pairs with $k \neq l$; since $T_{ij}(t)$ is real, the components at $\pm\omega_{kl}$ are conjugate-paired, and the simultaneous conjugation $a \to a^*$, $\omega \to -\omega$ (i.e. $V \to V^*$, $E \to -E$: the $H \to -H^*$ ambiguity of the statement) is invisible to the data. Fourier-transforming yields, up to that pairing:

$$a_{ij}^{kl} = V_{ik}\, V_{jk}^*\, V_{jl}\, V_{il}^*$$

The moduli $|V_{ik}|$ follow from $a_{ii}^{kl} = |V_{ik}|^2 |V_{il}|^2$ (non-zero by (G3)): the rank-1 outer-product structure determines $|V_{ik}|^2$ up to a global scale, and unitarity $\sum_k |V_{ik}|^2 = 1$ fixes the scale. The phases carry a double-difference structure:

$$\arg(a_{ij}^{kl}) = (\varphi_{ik} - \varphi_{il}) - (\varphi_{jk} - \varphi_{jl})$$

The gauge freedom preserving all double differences is $\varphi_{ik} \to \varphi_{ik} + \alpha_i + \beta_k$ (physically irrelevant basis rephasing). Fixing $\varphi_{1k} = 0$, $\varphi_{i1} = 0$, each remaining phase is $\varphi_{ik} = \arg(a_{i1}^{k1})$, directly extracted from Fourier data. $H = V\,\mathrm{diag}(E_k)\,V^\dagger$ is unique up to energy shift and phase conventions. □

The ancilla-marginal form of the reconstruction — where the visible transition data constrains a Hamiltonian on $\mathcal{H}_V \otimes \mathcal{H}_{\mathrm{anc}}$ through the block projectors P_i

— is governed by the linearised idempotency system, and admits a reformulation that makes its solution space transparent.

Lemma (ancilla-marginal reformulation). *Let $M_i = V^\dagger P_i V$ be the block projectors in the eigenbasis, θ real antisymmetric, and set $K_i := i\,(\theta \circ M_i)$ with $\circ$ the entrywise product. Then each K_i is Hermitian, and the linearised idempotency constraint $\sum_b (\theta_{ab} + \theta_{bc}) M_i[a,b] M_i[b,c] = \theta_{ac} M_i[a,c]$ holds for all a, c if and only if*

$$M_i K_i M_i = 0 \qquad \text{and} \qquad (I - M_i) K_i (I - M_i) = 0,$$

that is, if and only if K_i is purely off-diagonal in the splitting induced by M_i. In particular the coboundaries $\theta_{ab} = \alpha_a - \alpha_b$ satisfy it identically: $\theta \circ M = [\Delta, M]$ with $\Delta = \mathrm{diag}(\alpha)$, so $K = i[\Delta, M]$ and $MKM = i(M\Delta M - M\Delta M) = 0$ for every projector M and every V.

Theorem (ancilla-marginal phase-locking; certified block structures). *Fix (n, m_a) with $D = n\,m_a$. The kernel of the linearised system contains the $(D-1)$-dimensional coboundary space for every V, and for $(n, m_a) \in \{(2,2), (3,2), (2,3), (4,2), (2,4), (3,3), (3,4)\}$ it equals that space for all V outside a proper closed algebraic subset of $\mathrm{U}(D)$ — hence for Haar-almost-every $\hat{H}$. The reconstruction is therefore locally identifiable up to eigenbasis rephasing at those structures.*

Proof. Containment is the Lemma. For equality: $\dim\ker$ is upper semi-continuous in V and $\mathrm{U}(D)$ is irreducible, so it suffices to exhibit a single V at which the kernel has dimension exactly $D-1$; the maximal-rank locus is then Zariski-open and dense. Such a V is given explicitly over the Gaussian rationals — a product of Pythagorean Givens rotations across all coordinate planes times a diagonal of unit-modulus Gaussian rationals — and the kernel dimension is computed there in exact arithmetic: $3 = D-1$ at $D = 4$, and $5 = D-1$ at $D = 6$ in both block shapes (`phaselock_probes.py`). $\square$

Remark (genericity is load-bearing). The hypothesis cannot be dropped: at $V = I$ the constraint is vacuous and the kernel is the whole parameter space (six of six at $D = 4$, fifteen of fifteen at $D = 6$), and a real Hadamard V degenerates likewise. *Remark (scope).* The semi-continuity argument runs per block structure; $(3,3)$ and $(4,2)$ are confirmed numerically by the same reformulated system but are not exactly certified here, and no uniform-in-(n, m_a) statement is claimed.

Scope of the lemma under the ancilla-marginal form, and its extension. The lemma as stated assumes the strict form $T_{ij}(t) = |\langle i|e^{-iHt}|j\rangle|^2$ on an n-dimensional space — the trivial-ancilla case of the definition in §3.4. The emergent object is the ancilla-marginal (§3.1), and the lemma extends to it in a stronger form. Writing $U(t) = e^{-i\hat{H}t}$ on the dilated space of dimension $D = n\,m_a$ with eigenbasis $\{|a\rangle\}$ and configuration projectors P_i, the marginal takes the exact form

$$T_{ij}(t) = \frac{1}{m_a} \sum_{a,b} e^{-i(E_a - E_b)t} \langle a|P_i|b\rangle \langle b|P_j|a\rangle,$$

so under non-degenerate gaps of the *dilated* spectrum, Fourier analysis of the continuous-time visible data recovers, for each eigenvector pair (a, b), the rank-one coefficient matrix $A^{ab}_{ij} = \frac{1}{m_a}\langle a|P_i|b\rangle\langle b|P_j|a\rangle$, hence the coherence vector $\langle a|P_i|b\rangle$ up to one phase per pair. The residual gauge is larger than the diagonal rephasing of the original lemma — one phase per pair, $D(D-1)/2$ parameters, under which $T(t)$ is exactly invariant — but the requirement that the reconstructed P_i be orthogonal projectors summing to the identity is a nonlinear constraint tying the pairs together, and it eliminates precisely the excess. The reduction can be shown directly. Writing a per-pair rephasing as the linearized perturbation $M_i[a,b] \mapsto M_i[a,b](1 + i\theta_{ab})$ with θ antisymmetric (where $M_i[a,b] = \langle a|P_i|b\rangle$), the linearized idempotency constraint $\delta(M_i^2) = \delta(M_i)$ reads $\sum_b(\theta_{ab} + \theta_{bc})\, M_i[a,b]\, M_i[b,c] = \theta_{ac}\, M_i[a,c]$ for all i, a, c. The diagonal coboundary $\theta_{ab} = \alpha_a - \alpha_b$ satisfies this identically: its left side telescopes to $(\alpha_a - \alpha_c) \sum_b M_i[a,b] M_i[b,c] = (\alpha_a - \alpha_c)\, M_i^2[a,c] = (\alpha_a - \alpha_c)\, M_i[a,c]$ using $P_i^2 = P_i$, which is the right side. This exhibits the $D-1$ diagonal-gauge directions (the D shifts α_a modulo a global constant that acts trivially) as an analytic null space of the constraint, for every projector family. That these are the *only* null directions — that the constraint admits no pair-phase deformation beyond the coboundary — holds for generic $\widehat{H}$ at each block structure, by the genericity argument set out below; the constructive Fourier extraction has been demonstrated end-to-end. The marginal data therefore determine — generically, and at least locally — the *dilated* Hamiltonian and the configuration projectors up to an energy shift, diagonal phase conventions, and the antiunitary conjugation (statement above); the reconstruction reaches the dilation itself, not only an n-dimensional effective generator. The status of each component: the Fourier structure, its rank-one property, and the coboundary null directions are exact algebra; and the coboundary *exhausts* the null space for generic projector families, $\ker L_{\{P_i\}} = \{\theta_{ab} = \alpha_a - \alpha_b\}$, at each fixed (n, m_a). The families of n rank-m_a projectors summing to the identity are a single $U(D)$ orbit, hence an irreducible variety; $\dim\ker L$ is upper semi-continuous on it, its entries being polynomial in the family; and the coboundary lies in the kernel identically, so $\dim\ker L \geq D-1$ throughout. A single family with $\dim\ker L = D-1$ therefore confines the excess locus to a proper closed algebraic subset, of empty interior and Haar measure zero. Witnesses over $\mathbb{Q}(i)$ are certified in exact arithmetic at $(n, m_a) = (2,2), (3,2), (2,3), (4,2), (2,4), (3,3), (3,4)$ (`phaselock_probes.py`); the argument runs shape by shape and supplies no statement uniform in D. The genericity conditions (G1)–(G3) transfer to the dilated spectrum, gaps, and coherence vectors. One caveat applies in sharpened form: the idealized permutation dynamics has dilated eigenphases at roots of unity — equally spaced, hence maximally gap-degenerate — so the lemma's genericity holds for the physical continuous-time dynamics away from that idealized limit, not at it; this is the same idealization boundary carried by the finite-ϵ displacement $\mathcal{D}$.

Conditions (G1)–(G3) fail only on a measure-zero set of Hamiltonians. A stronger argument applies whenever the partition itself breaks symmetries of H_{tot}. The genericity conditions concern the *effective* visible-sector Hamiltonian $\hat{H}_{\mathrm{eff}}$ — the operator emerging after the trace-out — not the total Hamiltonian H_{tot}. The relevant distinction is whether a symmetry of H_{tot} *preserves the visible/hidden factorization* or not. A symmetry that fails to preserve the factorization — which includes the spatial symmetries an observer-centered partition breaks, since a partition centered on a specific observer maps boundary visible sites to hidden sites under translation, rotation about distant points, and boost — has no well-defined action on the visible factor alone, and the degeneracies it protects in H_{tot} are lifted in $\hat{H}_{\mathrm{eff}}$. (By contrast, a symmetry that *does* factorize as $S_V \otimes S_H$ descends to the visible symmetry S_V of $\hat{H}_{\mathrm{eff}}$, because the uniform-bath partial trace is invariant under any unitary on the hidden factor; such factorization-preserving symmetries are not lifted, and the visible-internal ones among them — those acting entirely within the visible sector and commuting with the trace-out — are precisely the gauge symmetries of the emergent description. Gauge degeneracies correspond to physically equivalent states and do not affect the phase-locking argument, which operates on gauge-inequivalent transition data.) Because the spatial symmetries of an observer-centered partition are factorization-breaking, $\hat{H}_{\mathrm{eff}}$ satisfies G1 (non-degenerate spectrum) not by typicality in an abstract ensemble but because the partition acts as a symmetry-breaking mechanism. Condition G2 (non-degenerate energy gaps) is not symmetry-protected — gap coincidences are a codimension-one condition rather than a group-theoretic one — so it is not delivered by the partition mechanism and rests on genericity directly. G2 holds for generic $\hat{H}_{\mathrm{eff}}$; it can fail only on a measure-zero set, and we are not aware of a structural feature of the substratum that forces such failure systematically. (In particular, single-cycle hidden dynamics does not: although the unitary cycle operator has equally-spaced phases on the unit circle, the Hermitian effective generator $\hat{H}_{\mathrm{eff}}$ has real eigenvalues that are not equally spaced, and retains non-degenerate gaps after the trace-out.)

3.2 Independent Derivation via Stinespring Dilation

An independent derivation of the emergent quantum description uses only Stinespring's dilation theorem [15] and standard properties of the partial trace [16], without invoking the stochastic-quantum correspondence of §3.1. The two routes are independent in their inputs and in the results they invoke (the correspondence's own proof also proceeds via Stinespring-type dilation, so the shared bedrock is Stinespring's 1955 theorem); either alone suffices.

Setup. The finite configuration spaces $\mathcal{C}_V = \{x_1, \dots, x_n\}$ and $\mathcal{C}_H = \{h_1, \dots, h_m\}$ (Lemma 1) embed into Hilbert spaces $\mathcal{H}_V = \mathbb{C}^n$ and $\mathcal{H}_H = \mathbb{C}^m$ via $|i\rangle \leftrightarrow x_i$ and $|k\rangle \leftrightarrow h_k$. This introduces no quantum postulates: it is the canonical identification of probability distributions on a finite set with diagonal density matrices.

Lemma (permutation unitarity). *Any bijection* $\varphi : \mathcal{C}_V \times \mathcal{C}_H \to \mathcal{C}_V \times \mathcal{C}_H$ *defines a unitary* U_φ *on* $\mathcal{H} = \mathcal{H}_V \otimes \mathcal{H}_H$.

Proof. Define $U_\varphi|i, k\rangle = |\varphi(x_i, h_k)\rangle$. Since φ is a bijection, U_φ permutes the orthonormal basis, hence is unitary. □

A finite permutation does not by itself generate a continuous flow: on a finite discrete configuration space the dynamics is the discrete group $\{U_\varphi^k\}$, and any continuous one-parameter interpolation is *additional structure*, chosen rather than derived. When such an interpolation is introduced — a strongly continuous one-parameter unitary group U_t with $U_1 = U_\varphi$, which always exists but is not unique (any branch of $\log U_\varphi$, differing by 2π shifts of eigenphases, defines one) — the exponential $U_t = e^{-i\hat{H}t}$ then supplies the continuous unitary interpolation for Hermitian $\hat{H}$; the framework's continuous-time statements are conditional on that choice, with the branch ambiguity part of the phase-retrieval gauge catalogued below.

Lemma (Reverse direction, scoped). *Any single one-step doubly stochastic matrix on a finite configuration space* $\mathcal{C}_V$ *can be realised as the one-step marginal of a deterministic bijection on* $\mathcal{C}_V \times \mathcal{C}_H$ *with uniform prior on* $\mathcal{C}_H$*, for some finite* $\mathcal{C}_H$ *— exactly when the transition probabilities are rational, and to arbitrary precision otherwise.* The restriction to doubly stochastic matrices is forced, not elective: a uniform-prior bijection marginal preserves the uniform distribution and is therefore doubly stochastic, so a merely row-stochastic matrix — one sending both states of a two-state space to the first, say — is not realisable in this form. The emergent statistics this converse serves lie inside the realisable class: ancilla-uniform marginals of unistochastic matrices are doubly stochastic. The scope is the matrix, not the process: realising prescribed *multi-time* statistics — a family of joint distributions over trajectories — with a single bijection and a single uniform prior is a stronger demand and is not claimed by this lemma; where multi-time structure is used below, it enters through the dilation actually constructed, not through this converse.

Proof. Any marginal of a bijection with uniform prior is a doubly stochastic matrix, and by Birkhoff–von Neumann is a convex combination of permutation matrices. A bijection on $\mathcal{C}_V \times \mathcal{C}_H$ realising the required mixture is obtained by letting $\mathcal{C}_H$ enumerate the permutations; for multi-step processes, the construction extends by history dilation — made precise, with a structured prior, by the process-dilation theorem of §3.4. □

Remark. This is the correspondence at the stochastic-process level: bijection $\leftrightarrow$ stochastic matrix via marginalization, both directions. It is *not* a full CPTP-channel correspondence — channels that create coherences from diagonal inputs have no permutation-unitary realisation with uniform ancilla. For the characterization theorem (§3.4), which is stated in terms of transition probabilities, this is the appropriate scope.

The quantum channel. The observer's ignorance of the hidden sector (Lemma 3) corresponds to $\rho_H = I_m/m$. The visible-sector quantum channel is

$$\Phi(\rho_V) = \mathrm{Tr}_H\left[U_\varphi\,(\rho_V \otimes \rho_H)\,U_\varphi^\dagger\right]$$

This is CPTP by a standard result [16, Theorem 8.1], with Kraus representation $\Phi(\rho_V) = \sum_{k,l} K_{kl}\,\rho_V\,K_{kl}^\dagger$ where $K_{kl} = m^{-1/2}\langle l|U_\varphi|k\rangle_H$. The triple $(\mathcal{H}_H, U_\varphi, \rho_H)$ is a *unitary realization* of Φ ([Structure] Definition 9.2). The reference state here is the uniform mixed state, so the triple is a unitary realization rather than a Stinespring dilation: with a pure reference the Stinespring dilation is the associated isometry $\psi \mapsto U_\varphi(\psi \otimes |0\rangle)$, not the unitary, and obtaining one from a mixed reference requires purifying ρ_H. Nothing below depends on the distinction; it is kept because the uniqueness statements of [Structure] Propositions 9.7a–9.7b do.

Theorem (Born-form representation of the transition probabilities). *The classical transition probabilities T_{ij} (§1.4) equal the Born-rule probabilities of Φ — a representation identity: it exhibits the Born form and does not select the quadratic exponent (§3.4).*

Proof. $P(j|i) = \langle j|\Phi(|i\rangle\langle i|)|j\rangle = m^{-1}\sum_{k,l}|\langle j,l|U_\varphi|i,k\rangle|^2$. Since U_φ is a permutation, $\langle j,l|U_\varphi|i,k\rangle = \delta_{(j,l),\varphi(i,k)}$. Thus $P(j|i) = m^{-1}\sum_k \delta_{j,\pi_V(\varphi(x_i,h_k))} = T_{ij}$. $\square$

Remark (failure of (C1)). Condition (C1) does not suffice. For $|V| = |H| = 2$ and $\varphi(x,h) = (h,x)$ one has $T_{ij} = 1/2$ for all i,j, yet $\Phi(\rho) = I/2$ identically: a constant channel, hence measure-and-prepare. Exhaustive enumeration gives the same verdict for every (C1)-satisfying bijection at $|V| = |H| = 2$, and for 612 of 648 at $|V| = 2$, $|H| = 3$ (verification code: `papers/oi_lattice_code/coherence/`). The obstruction is structural: (C1) constrains the diagonal of Φ, whereas entanglement-breaking is a property of $\mathrm{id} \otimes \Phi$ and is not fixed by the action on single-system inputs.

What replaces (C1) is not a further condition on T but a condition on the partition, and it is established for a specific realization rather than for arbitrary bijections. Take the substratum of §1.2 on a lattice Λ with $q = 2$, state $(u,v) \in \mathbb{F}_2^\Lambda \times \mathbb{F}_2^\Lambda$ and the reversible nearest-neighbour update

$$u'_x = \sum_{z\sim x} u_z + v_x, \qquad v'_x = u_x,$$

the visible sector being the sites of a region $R \subseteq \Lambda$. Write M_{HV} and M_{VH} for the blocks of this update carrying visible input to hidden output and hidden input to visible output, $\kappa = \dim\ker M_{HV}$, $w = \operatorname{rank} M_{VH}$, and $s = \dim V = 2|R|$. The inner and outer boundaries of R are $\partial^- R = \{x \in R : \exists\, z \notin R,\ z \sim x\}$ and $\partial^+ R = \{y \notin R : \exists\, z \in R,\ z \sim y\}$.

Lemma 1 (boundary bound). $\operatorname{rank} M_{HV} \le |\partial^+ R|$ *and* $\operatorname{rank} M_{VH} \le |\partial^- R|$.

Proof. For $y \notin R$ the component $v'_y = u_y$ is a hidden input, and $u'_y = \sum_{z\sim y} u_z + v_y$ involves a visible input only if some $z \sim y$ lies in R, that is only if $y \in \partial^+ R$; so M_{HV} has non-zero rows only among $\{u'_y : y \in \partial^+ R\}$. Dually, for $x \in R$ the component $v'_x = u_x$ is a visible input, and u'_x involves a hidden input only if $x \in \partial^- R$. □

One update step transports information one lattice site, so the sectors resolve one another only across the cut. What the argument establishes is a factorization: each cross-block has non-zero rows only at the boundary sites named, so it factors through the corresponding boundary coordinates. That statement uses no property of the coefficient ring and holds verbatim over $\mathbb{Z}_q$; over a field — in particular over the $\mathbb{F}_2$ used here — it gives the displayed rank bounds, ordinary vector-space rank being the relevant notion there. For a general nearest-neighbour bijection the same support argument yields the weaker conclusion that the hidden output depends on the visible input only through $\partial^- R$. Since adjacency is symmetric, the same reading applies to the columns: M_{HV} reads visible input only through $\{u_x : x \in \partial^- R\}$ and M_{VH} only through $\{u_y : y \in \partial^+ R\}$, so each rank is in fact bounded by $\min(|\partial^- R|, |\partial^+ R|)$ — a sharpening that uses the symmetry where the row statement does not, and that Corollary 2 consumes.

Lemma 2 (channel normal form). *Write $A = M_{VV}$, $B = M_{VH}$, $C = M_{HV}$ for the blocks of the update, and L_{RR} for the adjacency of Λ restricted to R. Then: (i) A is invertible, with $A^{-1}(a_u, a_v) = (a_v,\ a_u - L_{RR}\, a_v)$, so the visible block alone induces the permutation (Clifford) unitary $V_A|i\rangle = |Ai\rangle$; (ii) $\Phi(|i\rangle\langle i'|)$ vanishes unless $C(i - i') = 0$; (iii) on the Weyl operators the channel acts, with no phase, as*

$$\Phi(X^a Z^b) = [\, Ca = 0\,]\,[\, B^T A^{-T} b = 0\,]\; X^{Aa}\, Z^{A^{-T} b},$$

so that $\Phi = \mathrm{Ad}_{V_A} \circ \Phi_G$, where $\Phi_G = |G|^{-1} \sum_{g\in G} g \,\cdot\, g^\dagger$ is the uniform mixture over the group G of Weyl operators $X^\alpha Z^\beta$ with $\alpha \in \operatorname{im}(A^{-1}B)$ (translations) and $\beta \in \operatorname{im} C^T = (\ker M_{HV})^\perp$ (phases); (iv) G is abelian, of order $2^{\,s-\kappa+w}$.

Proof. (i) is the displayed inverse, read off from $v'_x = u_x$ and $u'_x = (L_{RR}u)_x + v_x$ on R; invertibility of A is what makes $i \mapsto Ai$ a relabelling. (ii) The hidden outputs of i and i' agree for some hidden input exactly when $C(i-i') = 0$, and then for every hidden input. (iii) With the hidden input uniform, $\Phi(|i\rangle\langle i'|) = [\, C(i - i') = 0\,]\ \mathbb{E}_{t\in \operatorname{im} B}\, |Ai + t\rangle\langle Ai' + t|$. Summing this against $X^a Z^b = \sum_i (-1)^{b\cdot i}\, |i + a\rangle\langle i|$ and substituting $i \mapsto i - A^{-1}t$ gives the displayed rule: the character sum over $\operatorname{im} B$ produces the second indicator and cancels every phase. The right-hand side is Ad_{V_A} applied to the Hilbert–Schmidt-orthogonal projection onto $\operatorname{span}\{X^a Z^b : Ca = 0,\ B^T A^{-T} b = 0\}$, and the uniform mixture over the symplectic complement of that index set — which is exactly G — is that projection. (iv) An element $X^\alpha Z^\beta$ with $\alpha \in \operatorname{im}(A^{-1}B)$, $\beta \in \operatorname{im} C^T$ commutes with another iff the cross-pairings $\alpha \cdot \beta'$ vanish; over all of G this is the single condition $CA^{-1}B = 0$, which holds identically for this update: B feeds hidden u into the visible u'-components, A^{-1} carries the u'-subspace onto the v-subspace, and C reads only the visible u-components. The translation and phase families intersect only in the identity, so $|G| = 2^{\operatorname{rank} B} \cdot 2^{\operatorname{rank} C} = 2^{\,w+(s-\kappa)}$. □

Theorem (separability threshold). *Let G be an abelian subgroup of the Weyl group on s qubits, of order 2^t modulo phases. Then $\Phi_G = |G|^{-1} \sum_{g \in G} g \cdot g^\dagger$ is entanglement-breaking if and only if $t = s$.*

Proof. Modulo phases the Weyl operators form the symplectic space $\mathbb{F}_2^{2s}$, on which abelian subgroups are exactly the isotropic subspaces; isotropy bounds $t \le s$. Any isotropic subspace of dimension t is carried to $\langle Z_1, \dots, Z_t \rangle$ by a symplectic transformation (Witt's extension theorem for the symplectic space $\mathbb{F}_2^{2s}$), and every symplectic transformation is implemented by a Clifford unitary U. Hence $U \Phi_G U^\dagger$ is complete dephasing on t qubits tensored with the identity on the remaining $s - t$. Complete dephasing is measure-and-prepare and therefore entanglement-breaking, and a tensor product of channels is entanglement-breaking exactly when every factor is — separability of the joint Choi state survives the partial trace onto any single factor pair, so one entangled factor Choi rules out separability of the whole; the identity on a non-trivial factor is not entanglement-breaking. Since conjugation by a unitary preserves the property, Φ_G is entanglement-breaking precisely when $s - t = 0$. □

Corollary 1. *For one step of the displayed update on $\mathbb{F}_2$ — $u'_x = \sum_{z \sim x} u_z + v_x$, $v'_x = u_x$, with the hidden sector initially uncorrelated and maximally mixed — the visible channel Φ admits the Clifford-conjugated abelian Weyl-mixture normal form of Lemma 2 and is not entanglement-breaking if and only if $w < \kappa$.* Immediate from Lemma 2 and the Theorem with $t = s - \kappa + w$: entanglement breaking is invariant under composition with the unitary Ad_{V_A}, so Φ is entanglement-breaking exactly when Φ_G is, and $t = s$ reads $w = \kappa$ (Lemma 2 gives $w \le \kappa$, matching $t \le s$).

Corollary 2. *If $\min(|\partial^- R|, |\partial^+ R|) < |R|$ then Φ is not entanglement-breaking.*

Proof. $s = 2|R|$. Lemma 1's column sharpening bounds each of $\operatorname{rank} M_{HV} = s - \kappa$ and $\operatorname{rank} M_{VH} = w$ by $\min(|\partial^- R|, |\partial^+ R|)$, so $w + (s - \kappa) \le 2 \min(|\partial^- R|, |\partial^+ R|) < 2|R| = s$, that is $w < \kappa$. □

Remark. For an exact cubic block of side $\ell \ge 2$ in $d = 3$: $|\partial^- R| = \ell^3 - (\ell - 2)^3$ and $|\partial^+ R| = 6\ell^2$, the inner boundary being the smaller of the two, so the hypothesis $\min(|\partial^- R|, |\partial^+ R|) < |R|$ reads $\ell^3 - (\ell - 2)^3 < \ell^3$, that is $(\ell - 2)^3 > 0$: the criterion holds exactly when the block has a non-empty bulk, first at $\ell = 3$ (at $\ell = 2$ every site lies on the cut and $8 = 8$); the threshold is structural, and a big-O statement alone could not name it. Within this single-step result the criterion carries no dependence on $|H|$, since the coupling reads only the boundary; the counterexample above is the degenerate case, every visible site lying on the cut so that the region has no bulk. Whether the same holds over many steps is not settled here: repeated interaction can propagate boundary information into the hidden bulk, and a multi-time statement would require a process-tensor argument rather than a single-channel one. The suggestion that coherence is a bulk quantity and decoherence an area quantity — the accounting that reappears in the horizon entropy of §7 — is at this stage an interpretation of the one-step result, not a theorem about the asymptotic dynamics.

Remark (scope). Lemma 1's factorization holds over any $\mathbb{Z}_q$, the displayed rank bounds being its reading over a field, and in weakened form the support argument covers arbitrary nearest-neighbour bijections. Lemma 2, the Theorem and both Corollaries are stated for $q = 2$; the symplectic and Clifford arguments carry over to prime q with $2^{\bullet}$ replaced by $q^{\bullet}$, while composite q requires module-theoretic treatment of rank and kernel and is not covered. All of this concerns the realization displayed above and is not a property of arbitrary finite bijections. For a general linear bijection the abelian property of Lemma 2(iv) can fail ($CA^{-1}B \neq 0$); the mixture is then over a non-abelian Weyl family, and entanglement breaking is governed by the isotropy of the surviving index set rather than by its order alone — not treated here. Two boundary markers on interpretation. Coherence preservation is a property of this realization, not an additional structural condition: (C1)–(C4) constrain coupling, timescales, and capacity, and neither contain nor imply a coherence requirement — the swap example satisfies (C1) while its channel is entanglement-breaking, and which realizations preserve one-step coherence is the geometric question answered above for the displayed update alone. And non-entanglement-breaking is a statement about a single channel, not about quantum mechanics: it supplies neither the observable algebra, nor preparations and interventions, nor tensor composition, nor multi-time statistics — the operational reconstruction claims rest on the dilation and correspondence arguments, not on Corollary 1 — and the entanglement-breaking status of one step does not determine whether the visible multi-time process is P-divisible: the hidden sector can carry memory even when a step's channel is entanglement-breaking, as a two-state system with a two-state hidden register already shows. The enumeration and the boundary bounds are reproduced by the deterministic tests in `papers/oi_lattice_code/coherence/` (`coherence_enumeration.py`); the transfer rule and normal form of Lemma 2, the threshold Theorem exhaustively at $s = 2, 3$, and both directions of Corollary 1 — a negative partial transpose wherever $w < \kappa$, an explicit measure-and-prepare form wherever $w = \kappa$ — are verified for $q = 2$ and $q = 3$ by `coherence_theorem_tests.py` in the same directory; the strictness of the dephasing/entanglement-breaking containment and the compatibility of an entanglement-breaking step with a P-indivisible visible process are certified by `eb_pindivisibility_tests.py` there as well.

Lemma (diagonal preservation). *The channels $\Phi_t(\rho) = \mathrm{Tr}_H[U^t(\rho \otimes \mu_H)U^{-t}]$ with U a permutation unitary map computational-diagonal states to computational-diagonal states.*

Proof. A permutation unitary conjugates a diagonal state to a diagonal state;

the partial trace of a diagonal state is diagonal. □ (Certified exactly for $t = 1, \dots, 6$ in `c1_cp_scope_probes.py`.)

Theorem (CP-indivisibility, framework scope). *For the permutation-dilation family $\{\Phi_t\}$ above, the P-indivisibility of §2.3 implies CP-indivisibility: there exist $t_2 > t_1 > 0$ with no CPTP map Λ satisfying $\Phi_{t_2} = \Lambda \circ \Phi_{t_1}$.*

Proof. Suppose such a Λ exists. $M_{kj} := \langle j|\Lambda(|k\rangle\langle k|)|j\rangle$ is a stochastic matrix (positivity and trace preservation of Λ). By the Lemma, $\Phi_{t_1}(|i\rangle\langle i|) = \sum_k T_{ik}(t_1)\,|k\rangle\langle k|$, so $T_{ij}(t_2) = \langle j|\Lambda(\Phi_{t_1}(|i\rangle\langle i|))|j\rangle = \sum_k T_{ik}(t_1)\,M_{kj}$ — P-divisibility of the population process. The contrapositive gives the result. □

Remark (scope: P-indivisibility is not quantum non-Markovianity). For a general CPTP family the reduction fails: $\Phi_{t_1} = \mathrm{Ad}_H$ (Hadamard), $\Phi_{t_2} = \mathrm{id}$ is CP-divisible with a unitary intermediate map, while the computational-basis populations $\frac{1}{2}J \to I$ are P-indivisible — the intermediate map consumes coherences invisible to $T(t_1)$. Stochastic P-indivisibility must therefore not be equated with open-system non-Markovianity outside the diagonal-preserving class; the framework's channels are in that class, which is what licenses the theorem. (Certified exactly over $\mathbb{Q}(\sqrt{2})$ in `c1_cp_scope_probes.py`.)

By the Breuer-Laine-Piilo criterion [3, 4], CP-indivisibility is witnessed by non-monotonic distinguishability, but the correct witness is the ANCILLA-ASSISTED one: monotonicity of system-plus-ancilla distinguishability characterizes CP-divisibility, while ordinary system trace distance is a strictly weaker witness and can stay monotone on a CP-indivisible family. The classical analogue is exact and is certified here: a family can be P-indivisible while total-variation distance is non-increasing for every pair of visible distributions (`translation_probes.py`, §2.3). Backflow in the system alone is therefore sufficient evidence of indivisibility, never necessary.

Approximate unitarity. On observable timescales $t \ll \tau_B$, the hidden-sector state is approximately frozen (conditions (C2)–(C3)). At $t = 0$ the channel generator is exactly Hamiltonian, $d\Phi_t/dt|_{t=0}(\rho_V) = -i[\hat{H}_{\mathrm{eff}}, \rho_V]$ with $\hat{H}_{\mathrm{eff}} = H_V + \mathrm{Tr}_H[V_{\mathrm{int}}(\mathbb{1} \otimes \rho_H)]$ the bath-averaged coupling; the dissipative part of the generator vanishes at $t = 0$. At finite time the visible channel acquires two distinct dissipative contributions, which it is important to separate. (i) Even in the strictly frozen-bath limit ($\tau_B \to \infty$) the channel is not unitary: it is a mixture over the bath ensemble of bath-conditioned visible unitaries $e^{-i(H_V + \langle h|V_{\mathrm{int}}|h\rangle)t}$, which dephases by the *spread* of those conditioned Hamiltonians — a contribution of order $\|V_{\mathrm{int}}\|^2$ (the coupling strength), independent of τ_B. (ii) The correction from the bath's actual motion away from the frozen limit enters at order $(\tau_S/\tau_B)^2$. Consequently the dynamics is approximately the Schrödinger equation generated by $\hat{H}_{\mathrm{eff}}$ to the extent that *both* the slow-bath condition $\tau_S \ll \tau_B$ *and* a weak-coupling condition (small conditioned-Hamiltonian spread, $\|V_{\mathrm{int}}\|\,\tau_S \ll 1$) hold; of the two contributions, the bath-motion term is fixed by the timescale hierarchy at $\mathcal{O}((\tau_S/\tau_B)^2) \sim 10^{-64}$, while the conditioned-Hamiltonian spread is not — its magnitude for the cosmological partition awaits the substratum interaction structure. Where these hold, the phase-locking lemma (§3.1) determines $\hat{H}_{\text{eff}}$ from continuous-time transition data up to the lemma's full gauge — an overall energy shift, the residual diagonal-unitary rephasing freedom that is physically trivial (basis convention), and the antiunitary conjugation $\hat{H} \to -\hat{H}^*$, which is not shift- or rephasing-related and is a genuine twofold ambiguity, not a convention; the residual dissipation $\mathcal{D}$, if measurable at sufficient precision, is a prediction of the framework rather than an approximation artifact.

Comparison of routes.

	Barandes route (§3.1)	Stinespring route (§3.2)
Input	Finite-configuration member of the source's class, under (T)	Bijection on $\mathcal{C}_V \times \mathcal{C}_H$ + marginalization
Bridge	Stochastic-quantum correspondence [11, 12]	Stinespring dilation [15] + partial trace [16]
Output	Dilated unitary whose ancilla-marginal Born probabilities reproduce T_{ij}	CPTP channel with $T_{ij} = \langle j\|\Phi(\|i\rangle\langle i\|)\|j\rangle$
Born rule	Equilibrium of indivisible process	Partial trace structure (above)
Scope	The source's class, Markov special cases included	Processes from marginalized bijections

The Barandes route is more general (any finite-configuration member of the source's class); the Stinespring route requires only textbook results [15, 16] and delivers additional structure at three separated levels. *Level one (generic, proved):* the tensor product $\mathcal{H}_V \otimes \mathcal{H}_H$, a unitary representation of the enlarged basis dynamics, and a CPTP reduced channel — the representation of the basis statistics. *Level two (specific, proved):* non-entanglement-breaking one-step coherence for the linear lattice realization under the $w < \kappa$ criterion (§3.2) — not generic: the swap-class dilations are entanglement-breaking. *Level three (open):* coherent preparation, noncommuting interventions, and the instrument algebra — the operational bridge (§3.2 checklist). Within level one, the frozen-limit error analysis stands: the leading departure from the frozen-limit description enters at $\mathcal{O}(\tau_S/\tau_B) \sim 10^{-32}$ as an absorbable Hermitian drift of $\hat{H}_{\text{eff}}$, with genuine non-unitarity $\mathcal{D}$ entering through the conditioned-Hamiltonian spread and the second-order bath-motion term $\mathcal{O}((\tau_S/\tau_B)^2) \sim 10^{-64}$ (§3.2). The first-order drift is moreover common-mode: the constraint coupling (C1) acts on total energy–momentum, so the slow change of the traced-out horizon state rescales $\hat{H}_{\text{eff}}$ uniformly and cancels in all dimensionless ratios — which is why the 10^{-32} figure is absorbable rather than observable — with the composition-differential remainder suppressed by the tidal factor $(L_{\text{lab}}/R_H)^2 \sim 10^{-52}$ and independently capped by equivalence-principle tests. The two routes agree on the configuration-transition probabilities $T_{ij}(t)$; equivalence of the complete operational observable structure is part of level three. The internal permutation-unitary route now carries (i) for realized processes (the equivalence theorem, §3.4), with the Barandes route supplying the compressed form under (T); the Stinespring route supplies the level-one representation of (iii) independently.

Dependency scope of the correspondence. The equivalence theorem of §3.4 fixes the framework's exposure to the correspondence precisely, and it is narrower than earlier versions of this paper claimed. *In-house:* the finite-horizon equivalence $S \iff D \iff Q_{\text{fb}}$ — existence of a fixed-basis unitary Born representation for every finite-horizon law, the framework's own processes included, by the permutation unitary of any realization — together with the fixed-$\widehat{H}$ lemma, the phase-locking gauge, and the universal hidden-memory theorem. *Imported, under (T):* the compressed representation — the source's $\leq n^3$ dilation bound for its finite-configuration class — and the interpretive apparatus that travels with it (basis independence, the general-observable reading). *Additional, with neither route supplying it:* the operational instrument algebra and coherent preparation — supplied exactly by the five operational completion conditions of the characterization in [GR §3.3] (§3.4 Remark), none of which bare finite OI provides — and representation consistency across growing horizons (the obstruction theorem of §3.4). A failure of the correspondence would therefore cost the framework the compressed form and its interpretation, and nothing else: existence of the representation, the equivalence, and the necessity results are internal. P-indivisibility appears nowhere in this division of labor — it gates neither membership in the source's class nor the representation's existence (§3.4, Remark) — and its role is confined to nontriviality at the recurrence scale and to the Bell-class constructions of §3.3.

An independent necessity statement. Both routes above are constructive: they exhibit quantum mechanics as a description of the marginalized process. **Operational-reconstruction route: target conditions and the outstanding bridge.** The reconstruction theorems [Hardy, arXiv:quant-ph/0101012 (2001); Masanes and Müller, New J. Phys. **13**, 063001 (2011); Chiribella, D'Ariano and Perinotti, Phys. Rev. A **84**, 012311 (2011)] derive quantum theory from operational axiom sets. The finite physical layer supplies two useful ingredients: every fixed finite representative has finite predictive dimension, and the substratum dynamics admits reversible dilations. Neither statement by itself discharges the corresponding hypothesis for an **exact operational completion**. The classical-dimension obstruction below forces any literal continuum-QM endpoint beyond one fixed finite carrier into a compatible closure/limit, and stagewise finiteness does not prove that this completion retains finite predictive rank. The purification, continuous-transitivity, and tomographic-locality axioms of the cited reconstructions are hypotheses of their continuum endpoint, which the classical-dimension obstruction places beyond any fixed finite carrier; they are not what the finite characterization uses. On a fixed finite carrier and its composites the intervention side is settled as a characterization rather than a derivation ([GR §3.3], kernel-verified with no external premise): for every nonempty finite visible system, an operational theory's available outcome families on the system and on every ancilla level are exactly the normalized finite Kraus instruments — the full instrument algebra, complete positivity of every branch, noncommuting composition, and consistency across composites — if and only if five conditions hold: valid probabilities, trivial-ancilla consistency, inert spectators, full reversible control, and iterated composition. Each condition is independent of the other four and of the observation process: for each there is a qubit theory realizing the same finite C1–C4 process with its visible readout that satisfies the other four and fails it. Bare finite OI therefore does not select quantum mechanics; the route classifies the OI-compatible completions by those conditions. The intervention dilation's classical rung — permutation instruments reproducing the comb statistics $P(X_{k+1} \mid X_{\leq k}, A_{\leq k})$ — is constructed (Theorem, §3.4) and reaches the intervention statistics of any finite instrument set (Leggett-Garg violations included, §3.4); the general form of (T) is instantiation-checked for representative finite processes, not established for the framework's realizations at arbitrary horizon. (T) alone licenses the imported stochastic–quantum representation [11, 12] — P-indivisibility gates neither membership nor representation (§3.4, Remark) — and P-indivisibility does not select the nonclassical branch of these reconstructions either: a classical stochastic system can be P-indivisible. The route is therefore a classification conditional on the five completion conditions, not an independent derivation.

3.3 Bell Inequality Violations

The framework is a deterministic hidden-state theory, so its Bell content must be stated at the correct level. Bare finite OI representability is too broad to select the quantum Bell set: if the two wings are allowed to share globally coupled setting-dependent hidden dynamics, the same reversible response-table construction that realizes arbitrary finite stochastic laws also realizes PR-box behavior. That fact is not in tension with the exact unitary/Born representation of §3.4; a global stochastic law can be represented unitarily while failing the *local subsystem* conditions of standard quantum mechanics. The selector in a Bell experiment is therefore causal locality, not representability.

Coupling-graph causal cone. Write the spatial substratum as $S = \prod_{i \in V} S_i$ and define the coupling graph G_φ by one-step dependency: $i \sim j$ whenever the ith output component of φ depends on the jth input component for some state (symmetrized for the spatial graph). Then, by induction, $(\varphi^k(s))_i$ depends only on initial components in the graph ball $B_{G_\varphi}(i, k)$. Thus a setting intervention supported in region A cannot alter a readout in region B before the corresponding causal cones meet. This result uses the full spatial OI constitution — site factorization plus the graph read from the actual dynamics — but does not require the stronger simple-cubic nearest-neighbor restriction used elsewhere to obtain a particular dispersion relation.

The causal-cone statement is pointwise in the pre-setting microstate. It must be distinguished from **operational setting screening**. In a closed deterministic universe the setting generators may be statistically correlated with preparation-relevant hidden variables; conditioning on a setting can then reweight the hidden ensemble and create apparent remote statistical dependence even though no influence crossed the graph cone. **Theorem (Bell ceiling for a screened local completion).** *Let Ω be a pre-setting microstate that fixes each wing's response to every setting. Suppose (i) the dynamics is deterministic; (ii) the setting operations are interventions supported in the respective regions; (iii) readout occurs before the post-setting cones meet; and (iv) the interventions act on one common pre-setting ensemble, so $p(\Omega \mid a, b) = p(\Omega)$. Then there are deterministic local response functions $q_a(\Omega)$, $r_b(\Omega)$ with*

$$P(q, r \mid a, b) = \sum_{\Omega} p(\Omega)\, \mathbf{1}[q = q_a(\Omega)]\, \mathbf{1}[r = r_b(\Omega)],$$

so the model is Bell-local and $|S_{\mathrm{CHSH}}| \leq 2$.

Proof. The cone statement makes the A-readout independent of the B-intervention at fixed Ω, and conversely; determinism makes the conditionals point masses; the common ensemble supplies measurement independence. Pointwise $q_0(r_0 + r_1) + q_1(r_0 - r_1) \in \{-2, +2\}$; average. ☐ Certified in `equivalence_recovery_probes.py` (check 11).

This is Bell's theorem applied to the framework's own hypotheses. **A deterministic completion that keeps the pointwise graph cone and a setting-independent pre-setting ensemble cannot reproduce quantum Bell correlations at all.** The Barandes–Hasan–Kagan bound is not contradicted: their causal-independence condition constrains the stochastic law and is strictly weaker than factorization, which is what lets a causally local indivisible process occupy the quantum interval. A deterministic response-complete completion supplies more than they assume, and the surplus costs precisely the interval between 2 and $2\sqrt{2}$.

Determinism is constitutional here, so exactly two hypotheses remain available and the framework must choose between them.

(a) Ontic parameter dependence — the option taken. The Bell-relevant response structure is not finite-range at the separation of the wings: at the microstate level, one wing's response depends on the remote setting. This is not an operational signal, and the no-signaling statement below is untouched. In the general finite response construction the dependence is explicit: the composite realizing PR correlations carries the joint history in each wing. For the specific SM/GR representative the compatible implementation is more constrained. [SM §3.1] already uses a state-dependent coupling graph $G(x)$; the natural branch-(a) target is therefore **preparation-indexed adjacency**, in which the reference vacuum/uniform graph remains geometric and bounded-range while a Bell preparation supplies bounded-degree long edges or an equivalent nonlocal response map. The open cost is not the graph-boundary area law by itself, but the continuum-curvature limit of those prepared graphs.

Corollary (Bell–lattice obstruction and state-dependent escape). *If the Bell-relevant degrees of freedom are governed entirely by the nearest-neighbor cubic wave-equation update used in [SM], with maximum lattice propagation speed identified with c, and the setting interventions and readouts are local to spacelike-separated wing regions, then the theorem above applies to every measurement-independent response-complete deterministic completion and* $|S_{\mathrm{CHSH}}| \leq 2$. Hence the **reference nearest-neighbor graph alone** cannot supply option (a). This is not a no-go for the state-dependent-graph framework: [SM §3.1] permits $G(x)$ to depend on the state, and its mod-q area-law proof is explicitly for uniform initial conditions. A measurement-independent Bell completion may therefore use preparation-indexed nonlocal edges while leaving the reference uniform/vacuum graph local. What remains to be proved is H-Bell: the prepared graph family must preserve operational no-signaling and satisfy a curvature/metric convergence condition strong enough for the Ollivier–Ricci continuum step used by the Einstein reconstruction. That condition is now downstream of a harder one. The cited convergence theorem is arbitrary-dimensional only for the manifold-distance-weighted graph metric, and for the degree-6 cubic reference graph the intrinsic hop metric is exactly ℓ_1 — a $\sqrt{2}$ diagonal stretch that does not decay at any scale — so the hop-metric curvature route FAILS for the reference family, before any Bell edge is added, and no $D = 3$ extension of that theorem would rescue it. The emergent geometry itself is not in question: the substratum's dispersion relation is isotropic at leading order. What must be re-founded is the curvature functional, on the propagation/Laplacian geometry rather than on shortest-path counts [SM §3.2]. The sparse-neighborhood diagnostic in [SM §3.1] gives a tractable locality target but is not itself a global convergence theorem.

(b) Ontic measurement dependence — recorded, not adopted. If instead the pre-setting ensemble is allowed to depend on the settings, then with $Z = (A, B)$ uniform and $I_{\mathrm{ont}} = I(\Omega; Z)$ in bits, data processing, Pinsker's inequality and Jensen's inequality give

$$\boxed{S_{\mathrm{CHSH}} \leq \min\Big\{4,\ 2 + 4\sqrt{2\ln 2\, I_{\mathrm{ont}}}\Big\}},$$

so reaching the quantum value requires $I_{\mathrm{ont}} \geq (3 - 2\sqrt{2})/(8 \ln 2) = 0.0309409\ldots$ bits. This is a **model-independent necessary lower bound from the stated inequality, not an achievable cost and not claimed tight**. For comparison, Hall and Branciard [57] construct the optimized causal measurement-dependent simulation in their CHSH model class at approximately 0.080 bits for the maximal quantum violation; the approximately 0.046-bit optimum corresponds instead to their retrocausal causal structure. The framework's one-way constitution does not adopt that retrocausal branch.

Measurement dependence is therefore retained as a quantified alternative but not adopted.

Response-completeness requirement for the information bound. The quantitative measurement-dependence estimate must use a response-complete ontic variable and therefore has Bell-local reference baseline 2. If Λ is coarser than the response-relevant microstate, $I(\Lambda; Z)$ alone does not control CHSH: a constant coarse Λ has $I(\Lambda; Z) = 0$ while omitted response-relevant data remains setting-correlated and a local deterministic model reaches the algebraic maximum 4 (check 11). Thus the stated information bound is a bound in terms of the response-complete Ω, not an arbitrary preparation-level coarse graining.

This formulation also fixes the scope of the no-signaling claim. Outcome correlations are nonfactorizable operationally through the shared preparation and the indivisible joint process, and under (a) at the ontic level as well; but a later local setting cannot produce an operational signal at the remote wing. Dropping that spatial/intervention structure returns the unrestricted OI behavior class, including PR, which is the mandatory negative control.

3.4 The Characterization Theorem: Necessity of the Conditions

The logical structure is easiest to state by keeping representation and selection separate. The finite-horizon theorem below proves internally that S, D, and Q_{fb} are the same observable-law class; the existence of a unitary/Born representation therefore requires neither C4 nor indivisibility. C4 supplies the nontrivial content: universal hidden predictive memory and, by the recurrence theorem of §2.3, global indivisibility of a fixed finite physical representative. The converse process-dilation theorem constructs a finite OI realization for every finite stochastic target. The remaining question is not whether a quantum representation exists, but whether an entire controlled family of preparations, coherent instruments, and composites is represented simultaneously by the standard local operational quantum theory.

Dilation existence. Any single one-step doubly stochastic matrix on a finite configuration space can be realized as marginalization of a deterministic process on a larger state space — furnishing Lemmas 1–3. The scope is the one scoped at the lemma above: the matrix, not prescribed multi-time trajectory statistics.

Lemma (non-permutation witness). *If the one-step matrix T is not a permutation, the marginalized process is P-indivisible — and in particular hidden-to-visible influence (C1) is present at the first step.*

Proof. The indivisibility is §2.3's recurrence theorem; and a non-permutation T means some visible transition has a nondegenerate conditional law, which requires hidden dependence. □

Remark (witness, not equivalence). The converse fails: the delayed-revival family $X_1 = X_0$, X_2 uniform and independent, $X_3 = X_0$ has $T^{(1)} = I$ — a permutation — while the family is P-indivisible ($T^{(3)} = T^{(2)}\Lambda$ is impossible: $\frac{1}{2}J\,\Lambda$

has identical rows for every stochastic Λ), and it is realized exactly by the §3.4 dilation construction (`review4_probes.py`). C1 is therefore formulated globally — influence at some accessible step (§1.3) — with the non-permutation one-step matrix its cheapest witness. The witness is not an equivalence: the reverse direction would assume the semigroup property $\Lambda = T^{(k_2-k_1)}$, which hidden memory is precisely there to break.

Remark (influence, not bidirectionality). The equivalence is with hidden-to-visible influence, which is C1's defining content (§1.3); it does not force bidirectional coupling. $\varphi(v,h) = (v \oplus h,\ h)$ on two visible and two hidden states has $T^{(1)} = \frac{1}{2}J$ and $T^{(2)} = I$ — P-indivisible — while the hidden update $h' = h$ is independent of v: the coupling is strictly one-way. Bidirectionality is a property of the physical realization (the bidirectional horizon coupling of §1.3), recorded at C1's realization clause there. (Certified in `c1_cp_scope_probes.py`.)

Physical motivation (ETH dephasing). Assume the hidden-sector Hamiltonian H_H satisfies the eigenstate thermalization hypothesis (ETH) [17, 18]. Then in the fast-bath regime ($\tau_B \ll \tau_S$), the hidden sector dephases to its diagonal ensemble between coupling events, so the marginal dynamics on $\mathcal{C}_V$ is Markovian on accessible timescales ($T^{(k)} = T^k$) and hence P-divisible; contrapositively, observable P-indivisibility in an ETH hidden sector requires $\tau_S \ll \tau_B$. This is the physical mechanism behind the mixing hypothesis; the load-bearing necessity statement is the quantitative theorem below, whose contrapositive makes the requirement operational. The dephasing picture imports from ETH a conditional-distribution equilibration that ETH's few-body form does not by itself deliver (Remark, *Status of ETH*), which is why it is carried as motivation rather than theorem.

The mechanism: between coupling events (separated by τ_S), the hidden sector evolves under its own Hamiltonian H_H. The substratum dynamics is reversible, so there is no relaxation in the strict sense — finite-dimensional Hamiltonian flow has purely imaginary spectrum and Poincaré recurrence at t_R exponentially large in the system size. What does occur, under the ETH assumption, is *dephasing*: starting from any distribution $\mu_H(\cdot|x_i)$ conditioned on the previous visible state, the off-diagonal matrix elements in the energy basis decay on the timescale $\tau_B \sim \hbar/\Delta E$, where ΔE is the hidden sector's typical energy spread. After dephasing, the diagonal ensemble agrees with the microcanonical distribution to corrections exponentially small in the hidden-sector size, so for any few-body visible-sector observable O_V and any time $\tau_S \gg \tau_B$:

$$\left|\langle O_V\rangle_{\mu_H(\tau_S|x_i)} - \langle O_V\rangle_{\mathrm{eq}}\right| \lesssim \tau_B/\tau_S$$

In the fast-bath regime ($\tau_B \ll \tau_S$) this is parametrically small: from the visible sector's perspective the hidden sector "looks" thermalized at every coupling event regardless of its post-coupling state, even though the underlying trajectory remains reversible and quasi-periodic with recurrence only at $t \sim t_R$. Each single-step transition matrix T is therefore computed against the same effective equilibrium distribution, so $T^{(k)} = T^k$ — a homogeneous Markov chain on accessible timescales $t \ll t_R$, hence P-divisible in the observable sense.

Contrapositively, within the mixing class this analysis presupposes — hidden dynamics that relax toward the reference prior between couplings — observable P-indivisibility requires $\tau_S \lesssim \tau_B$: the visible sector must couple back into the hidden sector before dephasing erases the memory. (Fast *non-mixing* hidden dynamics evade the premise, not the theorem: conserved circulation retains records with no timescale separation, `fastbath_probes.py`, §1.3.) For the strong, persistent P-indivisibility of the characterization theorem, the separation must be $\tau_S \ll \tau_B$. The continuous-time extension to general Hamiltonian flow replaces the collision-model picture with the full ETH/dephasing condition; the relevant machinery for chaotic open quantum systems in the weak-memory regime is established in [3, 8, 17, 18].

Theorem (C2 necessity, quantitative mixing form). *Model one coupling step as the uniform-prior bijection marginal of §3.2's construction, and between coupling events let the hidden sector evolve by any map on the hidden conditional. Suppose the mixing hypothesis holds: with π_H the* reference prior defining T^* — the uniform hidden prior of the model above — every realized post-relaxation hidden conditional, given the visible record to date, lies within ε of π_H in total variation. Then the k-step visible marginal satisfies*

$$\|T^{(k)} - T^k\|_{\mathrm{TV}} \ \le\ (k-1)\,\varepsilon$$

in the max-row total-variation norm, and the constant is optimal: no bound $C\varepsilon$ with $C < 1$ holds at $k = 2$. Contrapositively, departure of magnitude δ from the homogeneous Markov benchmark at step order k forces $\varepsilon \ge \delta/(k-1)$ — the benchmark form; the operative contrapositive in condition (ii)'s currency is the physical-memory theorem's $\varepsilon \ge \varepsilon_{\min}(I^*, n_V)$ below.

Proof. For $k = 2$: with $\mu'_{x_0x_1}$ the post-relaxation conditional given the visible record (x_0, x_1) and $\Delta = \mu'_{x_0x_1} - \pi_H$,

$$T^{(2)}_{x_0x_2} - (T^2)_{x_0x_2} = \sum_{x_1} T_{x_0x_1} \sum_h \Delta(h)\, P_\varphi(x_2 \mid x_1, h),$$

where $P_\varphi(x_2 \mid x_1, h) \in \{0,1\}$ is the bijection's visible step. Taking the row's total variation and the triangle inequality, $\frac{1}{2}\sum_{x_2} |\cdot| \le \sum_{x_1} T_{x_0x_1} \cdot \frac{1}{2}\sum_h |\Delta(h)| \sum_{x_2} P_\varphi(x_2 \mid x_1, h) = \sum_{x_1} T_{x_0x_1}\, \mathrm{TV}(\mu'_{x_0x_1}, \pi_H) \le \varepsilon$ — pushing two hidden distributions through the same channel contracts total variation. For general k, telescope through the hybrids $A_j = T^{(j)}\, T^{k-j}$: $A_1 = T^k$, $A_k = T^{(k)}$, and $\|A_{j+1} - A_j\| = \|(T^{(j+1)} - T^{(j)}T)\, T^{k-j-1}\| \le \|T^{(j+1)} - T^{(j)}T\| \le \varepsilon$, since right-multiplication by a stochastic matrix contracts the norm and the middle difference is the $k = 2$ argument applied at slot $j + 1$. Summing the $k - 1$ differences gives the bound. Optimality: the two-hidden-state revival bijection of §3.4's corollary, taken with zero relaxation, has $\varepsilon = \frac{1}{2}$ (its post-step conditionals are point masses, maximally far from uniform at $m = 2$) and $\|T^{(2)} - T^2\| = \frac{1}{2}$, saturating the bound. □

Remark (operational form of $\tau_S \ll \tau_B$). Define the *conditional mixing time* $\tau_B(\varepsilon_0)$ as the least inter-coupling relaxation time after which every realized hidden conditional lies within ε_0 of π_H in total variation. (The theorem's bound $\|T^{(k)} - T^k\|$ measures departure from the homogeneous Markov benchmark; it is not itself a measure of P-indivisibility — two-notions Remark, §3.4.)

Theorem (physical memory: C2, process form). *Under the conditional-mixing hypothesis — every realized hidden conditional lies within ε of π_H in total variation at the coupling time — every history-conditional next-step law satisfies $\|P(X_{k+1} \mid X_{\le k}) - \bar{K}_{x_k}\|_{\mathrm{TV}} \le \varepsilon$ with $\bar{K}_{x_k} := \sum_h \pi_H(h)\, K(x_k, h; \cdot)$; hence the departure $\|P(X_{k+1} \mid X_{\le k}) - P(X_{k+1} \mid X_k)\|_{\mathrm{TV}} \le 2\varepsilon$ uniformly in the history, and for $\varepsilon \le 1 - 1/n_V$,*

$$I(X_{<k}; X_{k+1} \mid X_k) \;\le\; 2\left[\varepsilon \log(n_V - 1) + h(\varepsilon)\right],$$

h the binary entropy (for $n_V = 2$ the bound is $2h(\varepsilon)$). Contrapositive: observed conditional memory $I^* > 0$ at an accessible order forces $\varepsilon \ge \varepsilon_{\min}(I^*, n_V) > 0$ — the hidden sector has not mixed by the coupling time. This is (C2)'s necessity in the currency of condition (ii).

Proof. The hidden conditional given any visible past is within ε of π_H by hypothesis; pushing through the one-step visible kernel contracts total variation, giving the first display. $P(X_{k+1} \mid X_k)$ is a mixture of history-conditionals, hence also within ε of $\bar{K}_{x_k}$, giving the second. For the information bound, write $I = \mathbb{E}[H(P(\cdot \mid X_k)) - H(P(\cdot \mid X_{\le k}))]$ and apply Fannes–Audenaert continuity to each term against the common reference $\bar{K}_{x_k}$. □ (Certified against exact conditional mutual information on a parametric relaxation family, `c2_mixing_probes.py`.)

Remark. The matrix-power bound above measures departure from the homogeneous Markov benchmark and is blind to conditional memory — the XOR family has $T^{(k)} = T^k$ exactly with one full bit of it (`review4_probes.py`). The process form is the statement that feeds the characterization's C2 clause.

Remark (coherent extension). The equivalence holds between the operational, dilated, and fixed-basis descriptions and is exact at that level; it does not by itself determine a coherent extension of the fixed-basis theory, and at least one conservative coherent extension preserves the fixed-basis theory exactly as a quotient while adding coherent structure not fixed by the equivalence. *Remark ((C2)'s role: two theorems).* The characterization's realization clause needs (C2) only in its limit form — the canonical dilation's bath is static, $\tau_B = \infty$ — so the structural equivalence is carried by (C1), (C3), (C4), with (C2) automatic in that limit. The physical content of (C2) is the memory theorem above: under the conditional-mixing hypothesis, observed conditional memory I^* lower-bounds the bath's mixing time. The structural equivalence and the physical-memory bound are distinct results with distinct hypotheses; (C1)–(C4) are not four necessities of equal standing — (C2)'s necessity is the conditional bound above.

Remark (passive scope). The theorem concerns the closed multi-time law $P(X_0, \dots, X_K)$. Conditioning on interventions — $P(X_{k+1} \mid X_{\le k}, A_{\le k})$ — requires the instrument-augmented realization $(S, \varphi, \mu_H, \{\mathcal{I}_a\})$, which is not constructed here: the *intervention dilation* is a named target of the operational bridge (§3.2) — and its classical rung is constructed below.

Theorem (intervention dilation: the finite classical action-labelled comb). *Let a finite action alphabet $\mathcal{A}$ and a finite-horizon action-labeled family $\{P(x_{k+1} \mid x_{\le k}, a_{\le k})\}_{k<K}$ be given. There exist a finite set S, a bijection φ, a fixed prior μ_H, and bijections $\{\mathcal{I}_a\}_{a\in\mathcal{A}}$ on S such that, for every action sequence, alternating $\mathcal{I}_{a_k}$ then φ from the initial ensemble reproduces the target conditional law at every step — the classical process-tensor (comb) statistics — with one fixed realization serving all action sequences and cross-history consistency automatic.*

Proof (construction). Extend the response-table realization: contexts $c = (x_{\le k}, a_{\le k})$; hidden variable $u = (u_c)_c$ with independent components $u_c \sim P(\cdot \mid c)$ under the product prior μ_H; state $= (x, \text{history ledger}, \text{action ledger}, u, \text{clock})$. $\mathcal{I}_a$ writes a into the action ledger's next blank slot — injective on blank-bearing states, completed to a bijection on the finite remainder as in the base construction. φ reads the ledgers and u, outputs $x' = u_{c(\text{state})}$, appends x to the history ledger, and advances the clock — bijective by the base theorem's ledger argument. For any action sequence, the conditional law of X_{k+1} given $(x_{\le k}, a_{\le k})$ is exactly $P(\cdot \mid x_{\le k}, a_{\le k})$: precisely the coordinate u_c is read, and its marginal is the target. One table, one prior: consistency across histories and across action sequences is automatic. □ (Certified exactly — all four action sequences, forty-eight conditionals in exact arithmetic, both instruments and φ verified bijections on $|S| = 124{,}416$ at $n_V = |\mathcal{A}| = K = 2$ — in `intervention_probes.py`.)

Remark (scope of the closure). The theorem closes the classical rung of the intervention bridge: classical action labels, permutation instruments, comb statistics of the visible configurations. Its reach is wider than it may appear. Because the instruments are *invasive* — they write to the action ledger — the construction is not bound by the non-invasiveness assumption behind temporal inequalities, and it reproduces quantum instrument statistics that violate them. A worked case is certified exactly (`lg_comb_probes.py`): sequential σ_z measurements on a qubit with $\pi/3$ rotations between them give $C_{12} = C_{23} = 1/2$, $C_{13} = -1/2$ and a Leggett-Garg value $K = 3/2 > 1$, and the realization — twelve contexts, one fixed prior, permutation instruments — reproduces every conditional and returns $K = 3/2$. The open remainder is therefore not the numbers.

Proposition (no uniform finite realization). A fixed finite S with a fixed prior and a fixed bijection supports only finitely many distinct action-labelled conditional families, whereas the instrument space of any non-trivial system is a continuum. Hence no fixed finite realization carries the whole instrument algebra: the theorem above is a statement per finite instrument set, and $|S|$ grows with the set chosen. □

Remark (what the remainder is). What stays open on the operational bridge is accordingly the instrument *algebra* rather than the statistics: multilinearity of the process tensor over the full instrument space, complete positivity of the induced slot maps, composition of noncommuting interventions, and the requirement that the intervened description use the same Hilbert space and the same $\hat{H}$ as the passive one (§3.2).

That last clause can be settled, and it is settled negatively at fixed dimension.

Proposition (no horizon-independent intervened representation). *Fix a representation dimension d, a Hamiltonian $\hat{H}$, and the slot structure. An instrument with n_{out} outcomes contributes $n_{out}d^4 - d^2$ real parameters per slot, so the intervened statistics reachable by CP instruments on that fixed representation are the image of a polynomial map from a space of dimension LINEAR in the horizon K. The action-labelled comb family with a given passive marginal has dimension growing like $(n_V|\mathcal{A}|)^K$. Hence for K beyond a small threshold the reachable set has measure zero in the family, and since the theorem above realizes the entire family, generic action-labelled combs are not hostable on any fixed passive representation.* □

The threshold is not asymptotic. At $n_V = d = |\mathcal{A}| = 2$ the counts are 70 against 168 at $K = 3$ and 310 against 224 at $K = 4$; at $|\mathcal{A}| = 3$, 244 against 252 then 1524 against 336; at $n_V = d = 3$, 438 against 1404 then 2868 against 1872. Keeping pace forces $d \sim (n_V|\mathcal{A}|)^{K/4}$ — concretely $d \geq 3$ at $K = 4$, 4 at $K = 6$, 8 at $K = 8$, 14 at $K = 10$ (`repconsistency_probes.py`).

Remark (the per-horizon qualifier is load-bearing). The characterization is stated per finite horizon (§3.4), and (T) gates the quantum representation per horizon likewise. The Proposition shows that this is not a technical convenience awaiting a better proof: a horizon-independent representation of the *intervened* process does not exist for generic combs, because the intervened statistics carry exponentially more information than any fixed representation can encode. The remaining question at Layer 4 is therefore not which fixed Hilbert space to choose — no such choice works — but how the representation must grow with the horizon, and whether that growth can be made compatible with the passive $\hat{H}$. Three successor problems replace it, and the framework must eventually choose among them: (a) identify the *quantum-compatible subclass* of action-labelled combs — those a fixed representation can host — and argue that physical interventions fall in it; (b) accept $d = d(K)$, a representation dimension growing with the horizon, and characterize the growth; or (c) allow $(\hat{H}, \mathcal{H})$ to depend on the intervention context, at the cost of the single-Hamiltonian reading of the passive dynamics. Each is a definite mathematical programme; none is supplied here. The quantitative-mixing theorem's contrapositive (§2.3) reads as a timescale statement in the homogeneous-Markov benchmark: memory of magnitude δ at step order k requires the coupling to return before the bath's conditional mixing to $\varepsilon < \delta/(k-1)$ completes — the benchmark form. For condition (ii)'s currency the operative statement is the physical-memory theorem's $\varepsilon \geq \varepsilon_{\min}(I^*, n_V)$ (§2.3); the benchmark inequality is not the general process-memory bound.

Remark (role of ETH, restated). The necessity direction rests on the mixing hypothesis above — an operational, in-principle-measurable condition — not on ETH itself. ETH is the physical case for the hypothesis: dephasing to the diagonal ensemble is the mechanism by which a generic chaotic hidden sector drives ε small between coupling events (previous theorem). A hidden sector failing ETH but satisfying the mixing bound still delivers the necessity direction; a hidden sector satisfying ETH for few-body observables while retaining conditional-distribution memory (conserved quantities, slow modes) is exactly the case the quantitative form guards against. The bound, its optimality, and the fact that the revival counterexample violates the hypothesis maximally are certified deterministically by `papers/oi_lattice_code/foundations/c2_mixing_probes.py` *(preregistered ensemble: 21{,}000 exact rational evaluations, zero violations; all three controls green).*

Remark (Status of ETH). *The eigenstate thermalization hypothesis is a well-supported conjecture, not a theorem, for generic chaotic many-body Hamiltonians. It is proved for specific models (random matrix ensembles, certain integrable-chaotic transitions) and supported by extensive numerical evidence across many systems [17, 18]. For the OI framework's hidden sector — the cosmological-horizon complement, a generic chaotic many-body system — ETH is the standard assumption. The C2 necessity direction is conditional on the quantitative mixing hypothesis (the theorem above); ETH is the physical case for that hypothesis rather than the load-bearing assumption, so relaxations or failures of ETH bear on how the hypothesis is argued for the framework's hidden sector, not on the theorem that uses it. For the framework's specific substratum, moreover, the chaoticity underlying ETH is not merely the standard assumption: the realized bijection's discretization nonlinearity is numerically demonstrated to be chaotic — two-trajectory Lyapunov measurement, $\lambda_{\max} > 0$; spectral-continuity, thermalization, cycle-consolidation, and orbit diagnostics concur, the quantitative rate being ensemble-dependent — so the hidden sector is a measured-chaotic system rather than an idealized one, with the analytic characterization of its cycle structure the open residual recorded there. The C1 and C3 necessity theorems do not depend on ETH.*

Remark (Consequence for the characterization theorem). *Within the characterization below, only the (ii) $\Longrightarrow$ (iii) direction's C2 clause is conditional on the mixing hypothesis (the process-form theorem above; ETH its physical motivation). The equivalence's other content — the readback direction, the dilation existence, and the P-indivisible-subclass representation clause — does not depend on ETH.*

Theorem (C3 necessity). *Let $m = |\mathcal{C}_H|$. The non-Markovian mutual information satisfies:*

$$I(X_{<t}; X_{>t} \mid X_t) \le \log_2 m$$

Proof. The total system is deterministic: $X_{>t}$ is a function of (X_t, H_t). Conditioning on X_t: $X_{<t} \to H_t \to X_{>t}$ is a Markov chain. By data processing:

$$I(X_{<t}; X_{>t} \mid X_t) \le I(X_{<t}; H_t \mid X_t) \le H(H_t \mid X_t) \le \log_2 m \quad \square$$

Corollary (per-process capacity bound). Any deterministic-bijection realization of a process whose conditional past–future information reaches $I^* = \sup_t I(X_{<t}; X_{>t} \mid X_t)$ requires $m \ge 2^{I^*}$: the hidden sector must be at least as large as the backflow it carries. The bound is process-quantitative and does not reduce to the visible count — P-indivisibility alone does not force $m \ge n$: a two-hidden-state, three-visible-state bijection already exhibits full distinguishability revival ($m = 2 < n = 3$), so the necessity content of (C3) is the capacity to hold the *observed* backflow, not a bound set by n. Sustained backflow at rate $I_0 > 0$ per coupling event over K events requires $m \gtrsim 2^{KI_0}$ *under the additional hypothesis* that the per-event contributions accumulate without hidden-state reuse or compression; that independence hypothesis is not a consequence of P-indivisibility, and without it only the per-process bound above is forced. The framework's qualitative (C3) ("hidden sector large enough") is the conservative restatement of the per-process bound. $\square$

The revival counterexample and the doubly-stochastic obstruction of §3.2 are certified by the deterministic probes in `papers/oi_lattice_code/foundations/` (`c3_dilation_probes.py`); the process-dilation construction below, the one-way-coupling counterexample, and the diagonal-preservation lemma are certified in `process_dilation_probes.py` and `c1_cp_scope_probes.py` in the same directory.

Theorem (finite-horizon process dilation). *Let $\mathcal{S}$ be a stochastic process on $\mathcal{C}_V$ ($n = |\mathcal{C}_V| \ge 2$) with rational multi-time law on a finite accessible horizon K ($t \ll t_R$). Then there exist a finite $\mathcal{C}_H$, a bijection φ on $\mathcal{C}_V \times \mathcal{C}_H$, and an initial hidden prior μ_H independent of the visible initial state, such that the marginal process of (φ, μ_H) reproduces $\mathcal{S}$'s full multi-time joint law for all $k \le K$. If the target process is non-Markovian, the construction necessarily contains hidden-to-visible influence at some accessible step — hence satisfies (C1); a non-permutation first-step matrix is a sufficient first-step witness only. The realization realizes (C2) in the static-bath form ($\tau_B = \infty$ between couplings, hence $\tau_S \ll \tau_B$), and satisfies (C3) with $|\mathcal{C}_H| \ge 2^{I^*}$. Rationality tracks the prior, not the mechanism: with the uniform-sector prior used here the marginal law is rational, and conversely every rational finite-horizon law is realized; an irrational prior can produce an irrational law, so the equivalence is between rational laws and finite realizations with rational priors.*

Extension (arbitrary priors — the response-table form). The rationality restriction is a property of the uniform-tape presentation, not of realizability. Replace the tape by a response coordinate $u = (u_c)_c$ indexed by the horizon's finitely many history contexts, with product prior $\mu(u) = \prod_c \mathrm{kern}_c(u_c)$ and deterministic step $x_{k+1} := u_{c(x_{\le k})}$; the ledger and clock complete the bijection exactly as above. Conditioning on the visible past exposes u only at visited contexts, so the current context's conditional law is kern_c verbatim, and the full multi-time joint is reproduced for arbitrary real kernels (`review4_probes.py`, P-G). The characterization below therefore quantifies over general finite-horizon laws.

Proof (constructive). Let D be a common denominator of the conditional kernels $P(x_{k+1} \mid x_0 \dots x_k)$. Take $\mathcal{C}_H = (\mathcal{C}_V \cup \{\square\})^K \times \mathbb{Z}_D^K \times \mathbb{Z}_{K+1}$ — a history ledger, a randomness tape, and a clock — with μ_H uniform on $\{\text{ledger} = \square^K,\ \text{clock} = 0\} \times \mathbb{Z}_D^K$. On well-formed states the step is $(x,\ (x_0 \dots x_{c-1}),\ \mathbf{s},\ c) \mapsto (Q_{c+1}(x_0 \dots x_{c-1}, x;\ s_{c+1}),\ (x_0 \dots x_{c-1}, x),\ \mathbf{s},\ c+1)$, where Q's tape fibers have sizes $D \cdot P(\cdot \mid \text{past})$ (the conditional-quantile representation). The step is injective structurally — the ledger's last entry recovers the pre-image regardless of the kernels — and extends to a bijection of the finite product by completing the partial permutation on the complement. Marginalizing over the uniform tape reproduces each conditional exactly, hence the full joint by the chain rule. When $\mathcal{S}$ is non-Markov on the horizon, the realization inherits (C4): distinct visible pasts with differing target kernels condition the ledger differently and are read back through Q exactly. (C1): if the target is non-Markovian, hidden-to-visible influence is present at some accessible step (the global argument, as in the characterization proof); a deterministic first kernel need not give a permutation marginal — many-to-one visible steps are realizable, the ledger retaining global reversibility. (C2): the ledger and tape have no autonomous dynamics between couplings — a static bath, the slow-bath limit. (C3): $\log_2 |\mathcal{C}_H| \ge I^*$, consistent with the per-process bound above. Certified exactly on P-indivisible targets — including the §3.2 revival family and a non-Markovian kernel set with $T^{(2)} \ne (T^{(1)})^2$ — by `process_dilation_probes.py` (full-joint equality in exact rational arithmetic; permutation control silent). $\square$

Remark (prior structure). The realization uses a structured prior — uniform on the initialized ledger/clock sector times the tape. The uniform-prior one-step lemma of §3.2, with its doubly stochastic scope, is the special case and is unchanged: double stochasticity is a property of uniform-prior marginals, not of deterministic realizability as such.

Definition (unitarily evolving QM). A stochastic process $\mathcal{S}$ on a finite configuration space $\mathcal{C}_V = \{x_1, \dots, x_n\}$ is *mathematically equivalent to unitarily evolving quantum mechanics* if there exists a Hilbert space $\mathcal{H} = \mathcal{H}_V \otimes \mathcal{H}_{\mathrm{anc}}$ of total dimension $\le n^3$, a Hermitian operator $\hat{H}$ on $\mathcal{H}$, a unitary family $U(t) = e^{-i\hat{H}t}$, and orthogonal projectors $P_i = |i\rangle\langle i| \otimes \mathbb{1}_{\mathrm{anc}}$ onto the configuration sectors, such that the transition probabilities are the ancilla-uniform marginal of the dilated Born-rule probabilities: $T_{ij}(t) = \mathrm{Tr}\big[P_j\, U(t)\, (|i\rangle\langle i| \otimes \rho_{\mathrm{anc}})\, U(t)^\dagger\big]$ for a fixed ancilla state ρ_{anc} on $\mathcal{H}_{\mathrm{anc}}$ (a diagonal ρ_{anc} suffices throughout; the uniform ancilla $\rho_{\mathrm{anc}} = I/m_a$ recovers $\frac{1}{m_a} \sum_{k,l} |\langle j, l|\, U(t)\, |i, k\rangle|^2$ — a special case, and the one that forces doubly stochastic T; general priors, as the response-table construction of §3.4 requires, need the general form).

Two objects. The definition covers two distinct targets, delivered by separate results. **(Q1) Barandes-representable:** the process corresponds to a unitarily evolving quantum system in the source theorem's own language [11, 12] — the $\leq n^3$ dilation bound is the source's result for the source's finite-configuration class, under its hypotheses. What connects the framework to the source is an *embedding*, and stating it as such is more accurate — and less demanding — than the identification of criteria it replaces. The source's object is a tuple: a configuration space, target and conditioning times, transition maps $\Gamma(t \leftarrow t_0)$ carrying first-order conditional probabilities, an initial distribution, and a random variable [11, 12]. Divisibility failure is *generic* within that class but is not what defines membership in it — a Markovian-homogeneous dynamical system is already a unistochastic process and sits inside the class — and the theorem is correspondingly broad: every finite-configuration process of that form can be regarded as a subsystem of a composite unistochastic process, on a Hilbert space of dimension $\tilde{N} \leq N^3$.

The framework's side supplies $\mathcal{C}_V$, the accessible times, the per-horizon transition family $T(t \leftarrow t_0)$, the visible prior, and the visible random variable. **Translation hypothesis (T) is the claim that this data instantiates the source's tuple, so that its theorem applies.** The instantiation is checked directly for representative processes: the tuple is constructed from marginalized bijections at $(n_V, |\mathcal{C}_H|) = (2,2), (2,3), (3,2)$ and the source's structural requirements — stochasticity of every $\Gamma(t \leftarrow t_0)$, trivialization at $t = t_0$, marginal consistency with the deterministic dynamics, and well-definedness of the random variable — hold in every case (`tuple_probes.py`). Two things follow. First, (T) is a structural check rather than an unproved equivalence between two notions of indivisibility, and it is correspondingly weaker than the corpus previously asked it to be. Second, the source theorem does not apply only to a "P-indivisible subclass": it applies to the class, with P-indivisibility governing whether the resulting representation is *nontrivially* non-Markovian rather than whether the representation exists. What (T) still does not license is transporting the multi-time content of (C4) into the source's framework, whose dynamical data is first-order by construction. **(Q2) autonomous OI representation:** a fixed Hermitian $\hat{H}$ with the ancilla-marginal form above, exact at integer horizon times, delivered in-house by the fixed-$\hat{H}$ lemma on an uncompressed space of dimension $n \cdot |\mathcal{C}_H|$. Condition (i) of the characterization means (Q1); the in-house chain proves (Q2); time-homogeneity and continuous-t interpolation belong to (Q2)'s autonomous form and are not imported claims (§3.1). (This is the form the correspondence delivers — the visible matrix is the ancilla-marginal of the dilated unistochastic matrix, per the scope statement in §3.1. The strict form $T_{ij}(t) = |U_{ij}(t)|^2$ on visible indices is the trivial-ancilla special case $m_a = 1$ and only that case: a doubly stochastic marginal need not be unistochastic at every t, let alone realized by a single one-parameter unitary group on $\mathcal{H}_V$.) This definition captures the Hilbert space, unitary dynamics, and the Born rule.

Additional quantum-mechanical structures — the tensor product decomposition for spatially separated visible-sector subsystems, state update, and the measurement formalism — are addressed in the remarks following the theorem.

Lemma (fixed-$\hat{H}$ form for realized processes). *Let a process arise as in (iii) below: the marginal of an iterated fixed bijection φ with a fixed hidden prior. Then the fixed-$\hat{H}$ ancilla-marginal form of (i) holds constructively at every horizon time: take U_φ the permutation unitary of φ on $\mathcal{C}_V \times \mathcal{C}_H$ and $\hat{H} := i \log U_\varphi$; for the uniform hidden prior, $T^{(k)}_{ij} = \frac{1}{|\mathcal{C}_H|} \sum_{h,l} |\langle j, l | U^k_\varphi | i, h \rangle|^2$ exactly, and for the §3.4 construction's structured prior the same identity holds with the ancilla prepared uniform on the initial register slice (final sum over the full space); and for a general fixed prior μ_H — the response-table construction's product prior included — with $\rho_{anc} = \sum_h \mu_H(h)\, |h\rangle\langle h|$, so that $T^{(k)}_{ij} = \mathrm{Tr}[P_j\, U^k_\varphi (|i\rangle\langle i| \otimes \rho_{anc}) U^{k\dagger}_\varphi]$ exactly; $U(t) = e^{-i\hat{H}t}$ matches U^k_φ at integer times.*

Proof. $|\langle j, l | U^k_\varphi | i, h \rangle|^2 = \mathbb{1}[\varphi^k(i, h) = (j, l)]$; summing l over the final space and h uniformly over the prepared set counts exactly the defining fraction of $T^{(k)}$. □ (Certified exactly — uniform, slice-prepared, and weighted-prior preparations — in `review4_probes.py`, P-F(a)–(c).)

Consequence. The chain (ii) $\implies$ (iii) $\implies$ fixed-$\hat{H}$ form is in-house per horizon *in the uncompressed sense*: the internal representation is exact at integer horizon times on a space of dimension $n \cdot |\mathcal{C}_H|$, which generally exceeds the n^3 bound in (i)'s definition. The $\leq n^3$ compression is the imported correspondence's result for the source's finite-configuration class, under its hypotheses; continuous-t interpolation beyond integer times likewise carries no internal claim. The import therefore retains the compressed representation and the interpretive apparatus — basis independence and the general-observable reading; the Born-rule *form* itself is internal, being the marginal formula of the permutation unitary.

Characterization theorem. *Let $\mathcal{S}$ be a time-homogeneous stochastic process on a finite configuration space $\mathcal{C}_V$ with $|\mathcal{C}_V| \geq 2$ and consider dynamics on accessible timescales $t \ll t_R$ where t_R is the Poincaré recurrence time of any realizing substrate. Then (ii) $\iff$ (iii); and, under (T), (i) holds for $\mathcal{S}$ — as it does for every finite-configuration member of the correspondence's class, P-indivisibility determining not whether the representation exists but whether its dynamics is nontrivially indivisible (Remark below). Condition (C2) is deliberately absent from (iii): the existential realization satisfies it trivially with a static bath ($\tau_B = \infty$), so it is not needed for the equivalence, and its physical content is carried separately by the conditional physical-memory theorem of §2.3:

(i) $\mathcal{S}$ is mathematically equivalent to unitarily evolving QM (in the sense of the preceding definition).

(ii) $\mathcal{S}$ is non-Markovian on accessible timescales: at some order k with $k\tau_S \ll t_R$, the conditional mutual information $I(X_{<k}; X_{k+1} \mid X_k)$ is strictly positive — the

visible past carries information about the visible future beyond the present.

(iii) For every finite accessible horizon long enough to contain a memory witness — every $K \geq k+1$, with k an order at which (ii)'s conditional mutual information is positive — $\mathcal{S}$ arises from marginalizing a deterministic bijection on $\mathcal{C}_V \times \mathcal{C}_H$ — the hidden space may grow with the horizon — with (C1) non-trivial coupling, (C3) sufficient capacity, and (C4) history readback at such a k — with a fixed hidden prior μ_H carried as part of the realization datum (§1.4 Remark).

The two theorems the equivalence turns on follow, with their own proofs; the characterization's proof comes after them.

Theorem (finite-horizon stochastic–reversible–unitary equivalence). *Fix a finite visible alphabet and a finite accessible horizon K. The following three classes of finite-horizon observable law $P(x_0, \dots, x_K)$ coincide:*

(S) the finite stochastic laws; (D) the visible marginals of finite reversible deterministic systems under incomplete observation; (Q_{fb}) the laws realizable by finite-dimensional unitary systems with Born-rule readout in a fixed basis — the fixed-basis unitary Born representations.

Proof. (S $\Rightarrow$ D) is the response-table construction of §3.4, which reproduces the whole multi-time law, not merely the one-step matrices. (D $\Rightarrow$ Q_{fb}) is internal and needs no imported correspondence: for a bijection φ on $\mathcal{C}_V \times \mathcal{C}_H$ set $U_\varphi|x,h\rangle = |\varphi(x,h)\rangle$ and take the hidden prior diagonal. A permutation unitary preserves diagonality, so sequential projective measurements in the configuration basis do not disturb the state, and the measured record reproduces $P(x_0, \dots, x_K)$ exactly; $U_\varphi = e^{-i\hat{H}}$ for a Hermitian $\hat{H}$, giving a fixed autonomous generator at integer times. (Q_{fb} $\Rightarrow$ S) is the Born rule: the record of any finite quantum experiment is a finite stochastic law. $\square$ (D $\Rightarrow$ Q_{fb} certified over 1{,}488 realizations at horizons 3 and 4 in `sdq_probes.py`.)

Remark (what this equivalence does and does not say). The theorem says the three classes **coincide** at the level of finite-horizon fixed-basis observable statistics. A Born-form unitary representation is therefore exact and internal, but universal: Markov and memory-bearing laws alike lie in Q_{fb}, so representability alone is not the framework's discriminating content. That content is the readback sector isolated below. C4 forces nonzero predictive memory and, for a fixed finite recurrent representative, global indivisibility. The subscript on Q_{fb} marks the remaining boundary precisely. The theorem does not by itself prove that all coherent preparations, noncommuting interventions, and composites are represented simultaneously by one standard local quantum instrument theory. The finite operational gluing theorem crosses much of that boundary for bounded action-labelled experiments, and the uniform continuum theorem gives arbitrary finite experimental accuracy; exact coherent/composite compatibility is the residual operational-lifting problem. This residual problem is not the exponent question of §3.1, and the two should not be conflated. Inside Q_{fb} the Born dictionary $p = |U|^2$ is not an open choice at all: it is part of the definition of the class, so the equivalence's quantum side already carries it and nothing here is left unresolved by it. What §3.1's scope clause says is the different and compatible point that the construction does not SELECT the quadratic exponent — the unitaries it supplies are permutations, whose entries are 0 or 1, so the recovered table is the same for $|U|^p$ at every $p > 0$ and no step distinguishes $p = 2$ (check 6, `equivalence_recovery_probes.py`). Why the quadratic functional is the physical one is therefore a frontier item about the dictionary's origin, not a gap in the equivalence. The residual operational-lifting problem is instead a question about extending the exact description consistently to the full operational algebra.*

Remark (interventions). The same construction indexed by intervention histories realizes any FIXED finite adaptive family (§3.4's comb theorem), which narrows the operational gap for fixed instrument menus. It does not close it: the obstruction of §3.4 — no fixed representation hosts generic combs as the horizon grows, with the dimension forced as $(n_V|\mathcal{A}|)^{K/4}$ — is untouched, and the all-horizons fixed-representation question remains open.

Proposition (finite-substratum operational obstruction — classical-dimension form). Fix one realization: a substratum of $N < \infty$ ontic configurations with its observation map. Its full prepare-and-measure behavior factors as $B = \Xi M$ — preparations as columns of a column-stochastic M (distributions over the N ontic states), effects as rows of $\Xi \in [0,1]^{\cdot \times N}$ — so every behavior matrix the realization generates has nonnegative factorization through N states. For the qubit this is impossible uniformly: qubit prepare-measure behaviors have PSD rank 2 but nonnegative rank growing without bound over finite prepare–measure sets, so for every N there is a finite qubit experiment no N-state model reproduces exactly (Heinosaari–Kerppo–Leppäjärvi–Plávala [55]; an alternative route under procedural-independence assumptions: Garner–Müller [56]). The factorization interface assumed is the ordinary preparation $\to$ finite ontic system $\to$ measurement one; the same work's coordinated version extends the obstruction against finite shared coordination between the sides [55]. Hence no fixed finite realization is exactly operationally equivalent to even a single qubit's full theory — the elementary factorization half proved here, the unboundedness imported. The obstruction is independent of C1–C4 and of horizon growth. It does not touch the equivalence theorem — there the realization is chosen per finite-horizon law, and a fixed finite table of statistics always has one — and it is precisely why the equivalence carries its subscript. □

Remark (correction of record; why the geometry alone cannot do it). An earlier version argued through state-space geometry: finitely many ontic states, it said, force a polytope state space, and a polytope is not a ball. The inference is invalid — a convex subset of a simplex need not be a polytope — and the SIC embedding is the certified counterexample: tetrahedral a_i, $p_i(r) = (1 + a_i \cdot r)/4$ maps the whole Bloch ball affinely and injectively into Δ_3, so a four-state model with restricted preparations carries the qubit's state GEOMETRY intact. What it cannot carry is the sharp effects: the affine extension of $|0\rangle\langle 0|$

takes the value $\frac{1}{2}+\frac{\sqrt{3}}{2}>1$ at an ontic vertex, no valid response function exists, and the dimension obstruction above is the correct general form (embedding, curvature, effect violation, and the factorization half all certified exactly over $\mathbb{Q}(\sqrt{3})$: `nogo_probes.py`).

Remark (the operational-extension boundary). The exact finite representation theorem and the finite-substratum obstruction answer different questions and should not be turned into a new 'selection' paper. At fixed observable horizon, $S \iff D \iff Q_{\mathrm{fb}}$ is exact. For a bounded family of interventions, the gluing theorem below gives one reversible realization for the entire classical action-labelled experiment; the uniform-continuum theorem then covers bounded quantum instrument envelopes to arbitrary requested finite accuracy. What no fixed finite carrier can do is reproduce the entire exact continuum operational theory of even a qubit at all preparations/effects and all precisions, by the classical-dimension obstruction above. The compositional target is settled as a characterization ([GR §3.3]): on any nonempty finite carrier, the operational theories whose available families on the system and on every ancilla level are exactly the finite Kraus instruments are precisely those satisfying five conditions — valid probabilities, trivial-ancilla consistency, inert spectators, full reversible control, iterated composition — each independent of the others and of the C1–C4 process, so that what the OI architecture must supply is named condition by condition rather than as one lifting hypothesis. Any exact-completion reconstruction beyond a fixed finite carrier must additionally show that its chosen completion retains finite predictive dimension; local tomography/composite completeness and all-level operational purification are sufficient reconstruction routes only after their hypotheses are actually established for that completion. The Bell side is now separate rather than analogous: the graph supplies a causal cone for finite-range substrata, but a measurement-independent deterministic completion with that pointwise cone is Bell-local. The framework's adopted Bell branch therefore requires ontic parameter dependence while retaining operational no-signaling (§3.3).

Theorem (finite operational realization and gluing — ε-form). *Fix a quantum experiment: a system of dimension d with an initial state — alternative preparations entering, when wanted, as outcome-free channels in the family — a finite instrument family $\mathcal{F}$ — each instrument a finite family of CP maps $\{\mathcal{I}_o\}$ summing to a channel, each $\mathcal{I}_o$ with finitely many Kraus operators, so conditional states may be mixed — and a finite horizon K; fix a grid size $G \in \mathbb{N}$. There is a finite reversible deterministic embedded realization — visible sector: step counter, intervention record, outcome registers; hidden sector: the unnormalized digitized branch matrix (a density register), a predecessor stack, and K pre-loaded seeds uniform on $\{0,\dots,G-1\}$; one FIXED injective step map φ_a per instrument, extended to a bijection by finite padding — with:*

(1) ***Completeness, adaptive.*** *For every adaptive protocol over $\mathcal{F}$ through K, the realized outcome-sequence law is within $K(|\mathcal{O}_{\max}|-1)/G$ of the quantum law in total variation, $|\mathcal{O}_{\max}|$ the largest per-instrument outcome count — K/G for*

binary-outcome instruments; in the fixed-point b-bit regime an additional state-update term joins the same telescoping sum, giving the unified bound $\|P_R - P_Q\|_{\mathrm{TV}} \le K(|\mathcal{O}_{\max}|-1)/G + 12\,K^2 C_{d,R}\,2^{-b}$*, absorbed by* $G \ge 2K(|\mathcal{O}_{\max}|-1)/\varepsilon$ *and* $b \ge \log_2(24\,K^2 C_{d,R}/\varepsilon)$*, each term at most* $\varepsilon/2$ *(*$R := \max_{a\in\mathcal{F}} \sum_o r_{a,o} \le |\mathcal{O}_{\max}|\,d^2$*, the family-wide maximum total Kraus count, so one* $C_{d,R}$ *covers every adaptive step; the* K^2*, carried by the Corollary's negativity ledger, costs one extra* $\log_2 K$ *bit); the b-bit term is supplied by the instrument-level Rounding Lemma of clause (5) — proved there in the fixed-point dyadic model and telescoped over subnormalized cq branches — so both regimes are unconditional, the exact-ring one additionally probe-certified end-to-end.*

(2) ***Gluing by restriction.*** *For every* $\mathcal{F}' \subseteq \mathcal{F}$ *and* $K' \le K$*, the same realization with the agent confined to* $\mathcal{F}'$ *and stopped at* K' *realizes the subexperiment — overlapping families agree on shared subexperiments identically, not approximately, because they run the same maps.*

(3) ***Context-independent implementation.*** *Each labelled instrument is one fixed map, applied whenever chosen, in every context; context enters only through the state. The outcome VALUES remain contextual — the only combination Kochen–Specker permits of any account reproducing these statistics. This is implementation-fixedness of labelled instruments, not Spekkens noncontextuality, which concerns operationally equivalent procedures and is not claimed.*

(4) ***Composition.*** *Sequential composition by construction; for spatially separated experiments the product construction implements the joint family by applying the local instruments' CP maps as* $\mathcal{I}_a \otimes \mathcal{I}_b$ *on the joint branch register, visible sectors in product — necessarily joint for entangled preparations, in the factorizability-failing, locality-retaining pattern of §3.3.*

(5) ***Capacity.*** *With exact-arithmetic registers over the instruments' entry ring,* $\log_2|\mathcal{C}_H| \le (K{+}1)\log_2|V_{\mathrm{reach}}(\mathcal{F},K)| + K\log_2 G$*, where* V_{reach} *is the finite set of unnormalized branch vectors reachable in* $\le K$ *steps; with fixed-point dyadic registers (b fractional bits and* $O(\log d)$ *integer bits per entry, all counted),* $G = \lceil 2K(|\mathcal{O}_{\max}|-1)/\varepsilon\rceil$*,* $b \ge \log_2(24\,K^2 C_{d,R}/\varepsilon)$*, and a canonical renormalization — the branch register replaced by* $\rho^\sharp = 2^{s(\tilde\rho)}\tilde\rho$ *only when its norm falls below the window's lower edge, the rescaling one-sided and upward, hence an exact left shift in fixed point (a downward shift, the only one that could round, never occurs: trace-nonincreasing steps plus the small accumulated error keep the norm under the generous upper edge; the exact-ring variant extends the ring by* $\frac{1}{2}$*); no shift is stored anywhere: the thresholds are scale-invariant, so the realized law is unchanged, and reversibility needs no exponent, the predecessor stack holding the predecessor itself — the fractional precision needs only* $b = O(\log(Kd\,|\mathcal{O}|/\varepsilon))$ *bits — plus the counted* $O(\log d)$ *integer bits per entry, no exponent anywhere, giving* $\log_2|\mathcal{C}_H| = O(K\,d^2\log(Kd\,|\mathcal{O}|/\varepsilon) + K\log_2 G)$ *in general — linear in K up to logarithms, the former exponent-range* K^2 *term removed (rescaling invariance certified: check 10,* `opglue_probes.py`*). The lower bound is of a different scope:* $\log_2|\mathcal{C}_H| \ge I^*$ *holds universally, for every faithful realization (memory*

theorem above), and is $\Omega(K)$ on explicit memory-chain families — a constructive upper bound for this realization and a universal floor, deliberately not a matched pair. For the constructed realization on those families, $c_1 K \le \log_2|\mathcal{C}_H| \le c_2 K\,(d^2\log(Kd\,|\mathcal{O}|/\varepsilon)+\log_2 G)$ — the former $+E$ on the right removed with the exponent range by the canonical renormalization above.

Rounding Lemma (instrument-level, fixed-point dyadic model). Floating-point entries carry uncounted per-entry exponents, so the model is fixed-point: every register entry is a dyadic rational with b fractional bits and $O(\log d)$ integer bits, all counted; Hermiticity is exact by storing the upper triangle and defining the lower by conjugation; each operation rounds to b fractional bits with absolute error $\le 2^{-b}$. The lemma is stated for the whole instrument as the classical–quantum map $\mathfrak{I}(\tilde\rho)=\bigoplus_o \mathcal{I}_o(\tilde\rho)$, with $R=\sum_o r_o \le |\mathcal{O}_{\max}|\,d^2$ the instrument's total Kraus count (the theorem takes R as the family-wide maximum $\max_a\sum_o r_{a,o}$):

$$\sum_o \left\|\hat{\mathcal{I}}_o(\tilde\rho)-\mathcal{I}_o(\tilde\rho)\right\|_1 \;\le\; C_{d,R}\,2^{-b}\,\|\tilde\rho\|_1, \qquad C_{d,R}=48\,R\,d^4.$$

Proof. Kraus quantization, by operator norms — the earlier componentwise step is abandoned as unjustified: $\|\Delta_\alpha\|_{\mathrm{op}} \le \|\Delta_\alpha\|_F \le d\,2^{-b}$ and $\|M_\alpha\|_{\mathrm{op}} \le 1$ give, per Kraus term, $\|(M{+}\Delta)\tilde\rho(M{+}\Delta)^\dagger - M\tilde\rho M^\dagger\|_1 \le (2d\,2^{-b}+d^2 4^{-b})\|\tilde\rho\|_1 \le 3d\,2^{-b}\|\tilde\rho\|_1$, summing to $3Rd\,2^{-b}\|\tilde\rho\|_1$ over the instrument. Fixed-point arithmetic with propagation: the first product injects at most $2d\,2^{-b}$ per entry, the second matmul propagates that with gain $\le 2d$ and injects again, within-block sums accumulate — componentwise $\le 7r_o d^2\,2^{-b}$ per block — and $\|A\|_1 \le d^2\max_{ij}|a_{ij}|$ with the window $\|\tilde\rho\|_1 \in [\frac{1}{2},4]$ converts and sums the blocks within the stated constant, which carries slack for the clipping below. Window management is one-sided: the register is rescaled upward, by an exact left shift, only when its norm falls below $\frac{1}{2}$; trace-nonincreasing steps grow the norm per step by at most $(1{+}\eta)$, and each upward shift lands it in $[\frac{1}{2},1)$, so with $K\eta \le \frac{1}{2}$ it stays below $e^{1/2}<2<4$ — a downward shift, the only one that could round, never occurs. Telescoping: the exact cq instrument is a trace-norm contraction on Hermitian inputs ($\sum_o\|\mathcal{I}_o(A)\|_1 \le \|A\|_1$, via $A=A_+-A_-$ and trace-nonincrease), so along the process σ_t of subnormalized cq branches, $\|\hat\sigma_{t+1}-\sigma_{t+1}\|_1 \le \|\hat\sigma_t-\sigma_t\|_1 + C_{d,R}2^{-b}\|\hat\sigma_t\|_1$, whence $\|\hat\sigma_K-\sigma_K\|_1 \le 2K\,C_{d,R}\,2^{-b}$ — rare outcomes stay weighted by their own small mass, and no per-branch conditioning ever occurs. The simulator does not sample the raw traces: the digitized Kraus family is not exactly trace-preserving, so it clips and normalizes. Corollary (clipping and normalization).* *For a positive branch with $s=\sum_o \operatorname{Tr} B_o > 0$ and raw drift $E=\sum_o\|\widehat{B}_o - B_o\|_1 \le \eta s$: clipping ($C_o=\widehat{B}_o$ if $\operatorname{Tr}\widehat{B}_o>0$, else 0) does not increase the aggregate error, since a clipped outcome's exact block has $\operatorname{Tr} B_o \le \|\widehat{B}_o{-}B_o\|_1$; with $Q=\sum_o \operatorname{Tr} C_o \ge (1{-}\eta)s$, the normalized digitized kernel $\tilde{\mathfrak{I}}(\rho)=\bigoplus_o (s/Q)\,C_o$ — exactly the simulator's pre-grid probabilities, the blocks retained up to the irrelevant power-of-two scale — satisfies $\|\tilde{\mathfrak{I}}(\rho)-\mathfrak{I}(\rho)\|_1 \le \frac{2\eta}{1-\eta}\,s \le 4\eta s$ for $\eta \le \frac{1}{2}$; a parent whose clipped total vanishes takes a fixed deterministic* *fallback — outcome o^* is written and the register set to the reference state P_0 at window scale, the predecessor pushed — so the step stays a bijection and no probability mass is lost; the entire mass of such parents is already bounded inside the same error term, so the arbitrary assignment costs nothing further.* $\square$ The Corollary is exact for positive inputs; digitized branches are Hermitian but not positive, so the telescope carries a negativity ledger — now proved rather than stated. Define $\nu_t := \sum_h \|(\hat{\rho}_h)_-\|_1$ on the UNSCALED analytic representatives: the subnormalized cq objects, related to the physical registers by undoing the accumulated power-of-two rescalings — an analysis device only, no shift being stored, the stack alone carrying reversibility. Recurrence: exact CP maps contract negative mass ($\|(X-Y)_-\|_1 \leq \operatorname{Tr} Y$ for $X, Y \geq 0$, summed over outcomes), rounding injects at most η per unit trace norm, and clipping or the fallback only removes or relabels mass, so

$$\nu_{t+1} \;\leq\; \nu_t + \eta\, \|\hat{\sigma}_t\|_1 \;\leq\; \nu_t + 2\eta, \qquad \nu_0 = 0, \qquad \nu_t \leq 2t\eta.$$

The Hermitian-input application of the Corollary costs at most $8\nu_t$ extra across the parents — write $\hat{\rho} = \hat{\rho}_+ - \hat{\rho}_-$, apply the positive-branch estimate to $\hat{\rho}_+$, and charge the exact-instrument contraction and the clipping/renormalization perturbation of $\hat{\rho}_-$ by triangle inequality — so $\delta_{t+1} \leq \delta_t + 4\eta(1+\delta_t) + 8\nu_t \leq \delta_t + 8\eta + 16t\eta$, and the one-line summation $\sum_{t<K}(8\eta + 16t\eta) = 8K^2\eta$ delivers the classical-marginal rounding term of clause (1) as $12\, K^2 C_{d,R}\, 2^{-b}$, with slack (WLOG $0 < \varepsilon \leq 1$: the choice of b gives $8t^2\eta \leq \frac{1}{3}$ inductively, closing the bootstrap $\delta_t < 1$ behind $4\eta(1+\delta_t) \leq 8\eta$) — the K^2 carried by the ledger, entering the bit requirement only through one extra $\log_2 K$; tightening it away is a named polish, not a gap. Positivity of the register is never needed — Hermiticity is exact by construction, and contraction holds on Hermitian arguments.* $\square$ Status: proved at instrument level; per-outcome and cq-aggregate inequalities instance-certified in exact rational simulation of the b-fractional-bit rounding, Hermiticity enforced by construction, at $b \in \{8, 16, 24\}$, with the telescoping conclusion instanced over a $K{=}3$ branch ensemble including rare outcomes, and the ACTUAL clipped-and-normalized simulator kernel — deterministic fallback included, mass conserved exactly — evaluated against the exact law with the negativity ledger tracked (checks 14–16, `opglue_probes.py`).

Proof (construction and bound). The step map φ_a reads the counter t; an outcome-free instrument applies its CP map to the branch matrix and pushes the predecessor onto the stack; an instrument with outcomes computes $q_o = \operatorname{Tr} \mathcal{J}_o(\tilde{\rho})$ exactly in the ring, compares the t-th integer seed by exact threshold ($r \cdot \sum_o q_o < G\, q_{<o}$ cumulatively), writes the outcome to the visible register, sets $\tilde{\rho} \leftarrow \mathcal{J}_o(\tilde{\rho})$ — unnormalized, so the true conditionals are exact ring ratios even when conditional states are mixed — and pushes the predecessor; the stack, not map injectivity, carries invertibility, so multi-Kraus maps cost nothing. Injectivity: the stack makes every step invertible on the reachable set, and a finite injective map extends to a bijection on a finite superset. The bound: each step realizes the grid rounding of the true conditional through cumulative integer thresholds, off by at most $(|\mathcal{O}|-1)/G$ in total variation — $1/G$ for two

outcomes, and — because the branch register is carried unnormalized — the true conditionals are computed exactly in the ring, so in the exact-arithmetic regime the grid is the only error source; the chain rule over K steps gives (1). In the finite-precision regime the comparison runs at the level of subnormalized cq branches: one digitized instrument step differs from the exact step by at most $C_{d,R}\,2^{-b}$ in trace norm on the full outcome direct sum, the exact instrument is trace-norm contractive on Hermitian inputs, so errors telescope additively over K steps with no per-branch conditioning — rare outcomes stay weighted by their own small mass — and the classical marginal delivers the TV term (the instrument-level Rounding Lemma below). Adaptivity is free: an adaptive protocol's trajectory probability is the product of the same per-step conditionals along its realized history, and (2) is the observation that the construction never asks which protocol is running. *Certified* at $(d, K, |\mathcal{F}|) = (2, \{2, 3\}, 4)$ — projective Z and X, the Hadamard intervention, and the 3-4-5 rotation, whose non-dyadic branch probabilities make the grid error strictly positive — with the K/G bound and $\sim 1/G$ scaling observed at $G \in \{8, 128, 1024\}$; operational noncommutativity reproduced at total variation $\frac{1}{2}$ between the XZ and ZX outcome-string laws, with Z-repeatability exact; restriction identity; per-instrument injectivity over all 672 reachable states; capacity floor — exact arithmetic over $\mathbb{Q}(\sqrt{2})$; the general multi-Kraus case certified via the density register against a two-Kraus-per-outcome instrument with exactly mixed conditionals (checks 11–12) — `opglue_probes.py`. □

Corollary (single controlled dynamics). The per-instrument family assembles into one map: on the enlarged hidden sector carrying a read-only program register — the protocol table — alongside the visible record, the seeds, and the branch register, the policy-controlled step $\Phi(x) = \varphi_{\pi(x_{\mathrm{vis}})}(x)$, the instrument selected by the protocol from the retained record, is a single reversible deterministic map realizing the entire adaptive experiment; and one $\Phi_{\mathcal{F},K}$ serves every protocol over $\mathcal{F}$ at once, the protocol entering as the preparation of the program register rather than as a change of dynamics. Invertibility: strip the appended outcome from the record, recompute the control from the restored record, undo the controlled step. One law, initial conditions vary — the substratum ontology stated as a theorem at finite resolution (check 13, `opglue_probes.py`: one map, two protocols including an adaptive one, all steps injective, laws exact). Resource accounting: the capacity clause counts branch register, stack, and seeds — the controlled model, intervention labels arriving as visible inputs. Here the read-only program register is additional state: an arbitrary adaptive policy over $\mathcal{F}$ through K costs up to $\Theta(|\mathcal{O}_{\max}|^K \log_2 |\mathcal{F}|)$ bits of table — exponential in K, charged separately and deliberately outside the linear-in-K estimate. One dynamics; the program is paid for as an initial condition, not obtained free. □

Remark (what the gluing theorem does and does not deliver). At finite resolution it crosses most of the gap between Q_{fb} and operational quantum experiments: incompatible measurements, adaptive interventions, preparation and instrument statistics, fixed per-instrument mechanisms, and gluing by construction. It is an **operational realization theorem**, not a uniqueness theorem:

the same reversible machinery can realize non-quantum finite instrument families, so the theorem by itself does not identify the standard local quantum instrument category as the only coherent completion. The exact-continuum boundary is separately sharp: no fixed finite realization can reproduce the full prepare-and-measure theory of even one qubit exactly, while the constructed quantum-shadow family is dense on every finite test. Thus the finite realization direction is closed at the stated accuracy. The uniqueness side of the coherent/composite lift is settled by the coherent-completion classification ([GR §3.3]): completions carrying the same complete operational data are equivalent up to unitary gauge and one global antiunitary orientation reversal. Its existence side is settled by the operational-completion characterization ([GR §3.3]): exact finite operational quantum mechanics on a carrier and its composites is equivalent to five named conditions, none supplied by bare finite OI, so what remains is orientation and the physical question of which conditions the substratum supplies, not the size of the family and not a lifting theorem. Bell is not an extra rung of that lift: §3.3 separately fixes the deterministic Bell branch as measurement-independent but ontically parameter-dependent, with operational no-signaling. The Kochen–Specker inheritance (§3.2) remains conditional exactly at that operational-algebra layer.*

The operational-completion theorem also admits a physically layered formulation. Keeping the original OI core unchanged, define quantum-complete OI (OI^+) by adding well-formedness together with observational independence, reversible richness, and observer recursion. On every nonempty finite carrier these three principles are respectively equivalent, on the relevant class, to inert spectators, sufficient reversible control, and iterated composition; hence

$$\mathrm{OI}^+ \iff \text{exact finite endomorphic operational QM.}$$

Each added principle is independently necessary relative to the OI core, well-formedness, and the other two. This is a characterization of a quantum-complete extension of OI, not a claim that bare OI entails the added principles; the OI-core clause becomes logically redundant once the reversible-control condition is present. See [GR §3.3] for the full statement and scope (kernel: `carrier_general_oiPlus` in `OIBridge/CarrierGeneralOIPlus.lean`; `oiPlus_independence` in `OIBridge/CompletedOI.lean`).

Lemma (adaptive diamond telescoping). Let π be an adaptive protocol of horizon K and π' the protocol with each instrument I_t replaced by I'_t. Then the outcome-sequence laws obey $\mathrm{TV}(P_\pi, P_{\pi'}) \leq \sum_{t\leq K} \max_a \|I_t^a - I_t'^a\|_\diamond$. *Proof.* Hybrid: swap one step at a time. An instrument is a channel into a classical–quantum register, the adaptive continuation is itself a channel, and the diamond norm is monotone under pre- and post-composition and under tensoring — so each single swap moves the final classical law by at most that step's diamond distance; summing the hybrids gives the bound. □

Theorem (uniform continuum realization — mutual ε-density). *Fix* *d, an outcome bound m, a horizon K, and* $\varepsilon > 0$*. Let N be a δ-net,* $\delta = \varepsilon/(2K)$*, in diamond norm on the compact set of* $\leq$ *m-outcome instruments on* $\mathbb{C}^d$ *(preparations included as channels; a standard volumetric bound gives* $|N| \leq \exp(C\, m\, d^4 \log(1/\delta))$*, the set having real dimension* $\leq 2md^4$*), and apply the gluing theorem to* $\mathcal{F} = N$ *with* $G = \lceil 2K(m-1)/\varepsilon \rceil$*. The single resulting finite reversible realization R satisfies: (i) for every adaptive quantum protocol* π *of horizon* $\leq K$ *with arbitrary continuum instruments there is an R-protocol* $\hat{\pi}$ *— the net rounding — with* $\mathrm{TV}(P^{\mathrm{QM}}_{\pi}, P^{R}_{\hat{\pi}}) \leq K\delta + K(m-1)/G \leq \varepsilon$*, by the telescoping Lemma plus clause (1); (ii) conversely every R-protocol runs genuine quantum instruments (net members), so its law is within* $K(m-1)/G \leq \varepsilon/2$ *of a quantum law. The two operational theories are therefore mutually ε-dense at horizon K, in the finite-test metric the density Corollary below makes explicit. And in the controlled model the hidden cost of uniformity is nil: the mechanisms are per-instrument, so the hidden registers — branch register, stack, seeds — do not scale with* $|N|$ *(the autonomous program register is charged separately: single-dynamics Corollary above) — hidden-STATE capacity is independent of* $|N|$*, while the DESCRIPTION complexity of the controlled transition law itself scales with the instrument net and is deliberately outside this capacity measure; the net inflates only the visible intervention record, by* $\lceil \log_2 |N| \rceil = O(m\, d^4 \log(2K/\varepsilon))$ *bits per step, and clause (5)'s hidden bound applies verbatim at the net's entry precision.* □ *(Telescoping and off-net density certified on the qubit instance — the 3-4-5 against the 5-12-13 rotation, and a mean-angle off-net target realized through its nearer net member within the predicted bound — checks 8–9,* `opglue_probes.py`*.)*

Corollary (finite-test density of deterministic realizations in operational quantum theory). Along any $K_n \to \infty$*,* $\varepsilon_n \to 0$*, the quantum-shadow realizations* R^Q_n *of the theorem satisfy: every finite-horizon adaptive quantum protocol's law is the limit of its net-shadows' realized laws, and every* R^Q_n*-law is within* $\varepsilon_n/2$ *of a quantum law — so, in the finite-test (Hausdorff-on-behavior-sets) topology,* $Q^{(d,m)}_{\mathrm{operational}} \subseteq \overline{D^{(d,m)}_{\mathrm{finite}}}$ *at each fixed system dimension d and outcome bound m — the scope the theorem fixes; the union over* (d, m) *is a further limit, not taken here — with two-sided ε-density holding for the constructed subclass* $\{R^Q_n\}$*. As classes this is density, not identity:* D_{finite} *is strictly larger — the response-table construction realizes arbitrary finite stochastic behavior, quantum or not — so its closure contains non-quantum operational theories too, and the remaining question (route (ii)) is the operational-lifting problem: which additional structure makes one controlled family of preparations, coherent instruments, and composites a single standard local quantum operational theory. The exact continuum theory stays unreachable at any fixed stage (the classical-dimension Proposition).* □

Summary (the equivalence at maximum proved strength). Four claims, each with its exact status. (1) $S \iff D \iff Q_{\mathrm{fb}}$ — proved, all finite laws. (2) $M_t > 0 \iff$ unavoidable readout-relevant hidden predictive memory — proved, every faithful completion. (3) No fixed finite classical carrier reproduces full qubit operational behavior — proved under the stated operational interface, the unbounded-dimension half imported [55]. (4) Every d-dimensional quantum experiment of horizon K has a finite reversible deterministic ε-realization — proved unconditionally at exact arithmetic AND at finite b (the instrument-level Rounding Lemma, fixed-point dyadic model), covering general instruments, mixed conditionals, incompatible measurements, and adaptive protocols, with explicit controlled-model resources and one controlled dynamics serving all protocols. Per (3), density is the maximum: the equivalence cannot be strengthened to exact class identity at any fixed carrier, so what remains is precisely selection — route (ii). This is the maximal equivalence available to a fixed finite substratum under the stated operational interface.

Theorem (unavoidable hidden predictive memory). *Let $\mathcal{S}$ be a visible process and let $\mathcal{R}$ be ANY faithful deterministic realization of it — a bijection φ on $\mathcal{C}_V \times \mathcal{C}_H$ with a hidden prior reproducing $\mathcal{S}$. Write $f_x(h) = \pi_V(\varphi(x, h))$ for the next visible state, and for a visible history p ending in x write μ_p for the induced hidden posterior $P(H_t = \cdot \mid X_{<t} = p, X_t = x)$. Then:*

(a) ***Pushforward identity.*** *$P(X_{t+1} \mid p, x) = (f_x)_\# \mu_p$ exactly.*

(b) ***Distinguishability floor.*** *For histories p, p' sharing the endpoint x, the observed gap obeys $\delta = \|P(X_{t+1} \mid p, x) - P(X_{t+1} \mid p', x)\|_{\mathrm{TV}} \leq \|\mu_p - \mu_{p'}\|_{\mathrm{TV}}$.*

(c) ***Capacity floor.*** *$M_t := I(X_{<t}; X_{t+1} \mid X_t) \leq I(X_{<t}; H_t \mid X_t) \leq \log_2 |\mathcal{C}_H|$.*

Proof. (a) Determinism makes X_{t+1} a function of (X_t, H_t), so conditioning on p and pushing μ_p through f_x IS the conditional law. (b) A deterministic map is a channel and total variation is monotone under channels. (c) $X_{<t} \to H_t \to X_{t+1}$ is a Markov chain given X_t, by the same functional dependence; data processing gives the first inequality and $I \leq H(H_t) \leq \log_2 |\mathcal{C}_H|$ the second. □ (All three certified over every realization at $(n_V, |\mathcal{C}_H|) = (2, 2), (2, 3), (3, 2)$ — 1{,}464 cases — in `memory_probes.py`.)

Remark (what this strengthens). The equivalence of §3.4 is existential: some finite reversible realization exists. That statement is satisfied by Markov processes too, so it cannot by itself distinguish the framework's regime. The theorem above is UNIVERSAL — it constrains every deterministic completion — and it is the sense in which observation incompleteness forces hidden structure rather than merely permitting it: **if the visible process remembers its past, then every deterministic completion reproducing it must carry that memory in the hidden sector, with distinguishability surviving to readback and capacity at least M_t.** And the statement is a biconditional: if in some faithful realization two positive-probability histories sharing an endpoint induce differing pushforwards $(f_x)_\# \mu_p \neq (f_x)_\# \mu_{p'}$, then by (a) those pushforwards *are* differing next-step conditional laws, so $M_t > 0$ — hence $M_t > 0$ holds exactly when every faithful deterministic realization — of any horizon $K \geq t+1$, the least on which M_t is defined — contains readout-relevant hidden predictive memory at t. The conditions follow rather than being assumed: (C1) because a hidden distinction invisible to f_x produces identical pushforwards; (C2-structural) as the retention required by (b) — the slow-bath form (C2-slow) one mechanism among several (§1.3); (C3) as (c) itself.

Remark (a variational identity, open — and false in its naive form). It is natural to conjecture $M_{\det} := \inf_{\mathcal{R}} I(X_{<t}; H_t \mid X_t) = M_t$: that conditional mutual information is exactly the minimum hidden predictive memory a reversible deterministic completion requires. The lower bound is (c). The reading in which the infimum ranges over realizations of FIXED hidden dimension is FALSE: at $(n_V, |\mathcal{C}_H|) = (2, 3)$ the minimum equals M_t for only one of fifteen distinct processes with $M_t > 0$, with residuals up to 0.667 bits (`memory_probes.py`). Any proof must therefore let the hidden dimension vary — the response-table construction, which preloads exactly the next-response variable, is the candidate attainer — and must quotient inert ledger bits that cannot influence the future; that quotient is now constructed and shown canonical (the predictive-completion theorem below), leaving open only the variational VALUE question: whether the post-quotient minimum of $I(X_{<t}; H_t \mid X_t)$ attains M_t. Stated here so the naive form is not re-derived.

Remark (the conditions are diagnostics, not hypotheses). The reduction extends one step further. (C4) is not an independent condition on the realization either: it asserts a readback gap *mediated through the hidden state*, and in a faithful realization every correlation is so mediated, so the mediation clause is automatic and (C4) coincides with clause (ii) itself. Exhaustive enumeration confirms it — the (C4) gap and $I(X_{<k}; X_{k+1} \mid X_k) > 0$ agree on all 24, 720 and 720 realizations at $(n_V, |\mathcal{C}_H|) = (2, 2), (2, 3), (3, 2)$, with no disagreement (`primitive_probes.py`). Taken together with the dependencies above, this does NOT license dropping readback from the right-hand side. The bare form

$$\text{accessible non-Markovianity} \iff \text{a per-horizon finite reversible deterministic realization exists}$$

is **false**: the process-dilation construction of §3.4 realizes Markov laws as well, so the right-hand side is satisfied by processes with no memory at all. What the reduction does license is the *universal* statement — the theorem below — in which readback appears not as a hypothesis but as an unavoidable feature of every faithful realization. What C1–C4 supply is not hypotheses but a *diagnostic checklist* describing what such a realization looks like — which is precisely what the physical applications need, since one cannot inspect a cosmological horizon or a Boltzmann brain for "a realization exists" but can check coupling, capacity, readback and persistence directly. The substance of the theorem lives in the construction (§3.4) — the response-table realization, the hidden space growing with the horizon, the prior carried as realization datum — and is untouched by this reading.

Theorem (canonical predictive quotient — a purification ingredient, substratum form). *Let P be a finite-horizon law on visible trajectories. Define the tail object $T(P)$: hidden states the stage-indexed pairs (t, τ) of a time and a positive-probability length-$(K{-}t)$ visible future-tail, prior P itself at stage 0, dynamics the shift $(t, \tau) \mapsto (t{+}1, \tau_{2:})$. Then: (existence) $T(P)$ completes P, and* *adjoining a history ledger — the emitted prefix, with each tail's chain closed into a cycle — makes it a reversible (bijective) realization; (universality) every faithful deterministic realization R of P carries, on its prior-supported reachable part, the canonical morphism $q_R : (x, h) \mapsto (x,$ its visible future-tail$)$ — a morphism of realizations here being stage-preserving, visible-preserving ($\pi_V \circ q = \pi_V$), and dynamics-intertwining ($q \circ \varphi_R = \sigma \circ q$, σ the shift) — which pushes the prior forward to P and is the unique morphism $R \to T(P)$; (canonical quotient) $T(P)$ is the canonical terminal predictive quotient of the category of faithful realizations — itself a staged predictor, bijective only once the ledger is adjoined — and any two faithful realizations of the same P induce the identical canonical quotient, so the predictive content of every completion is unique — while the hidden fibers above the quotient may differ: full completions are NOT claimed unique up to reversible relabeling, and same-law realizations with non-relabel-equivalent fibers exist (`fiber_freedom`, certified).* □ *Proof.* Determinism makes the visible future from (x, h) a single string, so q_R is well-defined; one step of φ shifts that string, giving the intertwining; faithfulness makes the pushforward P. Uniqueness of the morphism: visible-preservation propagated through the intertwining forces the image's visible trajectory to equal the source's, and a $T(P)$-state IS its remaining visible trajectory, so any morphism sends each state to its tail — without visible-preservation the claim would fail, since law symmetries can permute equal-probability tails; the definition excludes exactly that. Terminality: the same argument against any tail-injective target. Reversibility of the ledger extension: the prefix-and-cycle construction is a permutation — injective along each chain, chains disjoint, terminals rotated to roots. □ (All clauses certified exhaustively — intertwining, pushforward, canonicity across same-law realizations, quotient idempotence, ledger reversibility, fiber freedom, and the predictive floor $\log_2 |\text{tails}_t| \geq I(X_{<t}; X_{>t} \mid X_t)$ — over all 1{,}464 φ-realizations at $(n_V, |\mathcal{C}_H|) = (2, 2), (2, 3), (3, 2)$ under two priors, 2{,}928 instances, exact: `purification_probes.py`.)

Remark (relation to the purification axiom). In the Chiribella–D'Ariano–Perinotti reconstruction [53], purification demands existence and essential uniqueness — up to reversible transformations of the purifying system — within the operational theory. The theorem above supplies the substratum-level existence half and, toward uniqueness, the canonical predictive quotient every completion shares — an ingredient, not the axiom: CDP's uniqueness (any two purifications related by a reversible transformation on the purifier alone) is strictly stronger, and the certified fiber freedom shows completions here are not unique as wholes. For the operational-lifting route, two questions remain: translate this substratum construction into the operational composite language, and determine whether quotient-level uniqueness is strong enough to play purification's reconstruction role. The result is an in-house ingredient toward that lift, not yet the all-level operational purification principle itself.

Remark (which condition is primitive). Of the three structural conditions, only (C4) is logically independent. (C1) follows from it — a readback gap requires hidden mediation, and zero-coupling realizations exhibit no gap at any order —

and (C3) follows by data processing, with the bound $\log_2 |\mathcal{C}_H| \geq I^*$ saturated by realizations that attain it; both are certified by exhaustive enumeration in `primitive_probes.py`. The equivalence could therefore be stated with (C4) alone. It is not, deliberately: (C1) and (C3) are independently *checkable* — Bell-class dynamics force (C1) directly, capacity arguments use (C3) directly — and a characterization that lists checkable consequences alongside its primitive is more useful than a minimal one, in the way that axiom lists for groups are conventionally redundant. The dependency is recorded at each condition's definition (§1.3).*

Remark (two notions of memory). Non-Markovianity of the process — condition (ii) — and P-indivisibility of the transition-matrix family are distinct, and the distinction mirrors the quantum-side gap between process-tensor non-Markovianity and CP-divisibility (§3.2). Markov implies P-divisible (chain factorization), so P-indivisibility is the strictly stronger property; the strictness is witnessed by an explicit family — X_1 uniform independent of X_0, $X_2 = X_0 \oplus X_1$ — with maximal conditional memory ($I(X_0; X_2 \mid X_1) = 1$ bit) whose matrix family factors through $\Lambda = I$ (`review4_probes.py`). The framework's accessible-timescale claim is (ii). P-indivisibility enters as the stronger property in two ways: unconditionally at the recurrence scale from a non-permutation one-step witness (§2.3), and empirically — P-indivisible dynamics are documented in open-system experiments, the imported correspondence reads Bell-class dynamics as indivisible, and (ii) itself is certified by directly observed memory effects.

Proof. (iii) $\Longrightarrow$ (ii): the readback lemma of §2.3 — (iii) includes (C4), whose gap forces $I(X_{<k}; X_{k+1} \mid X_k) \geq p_0\delta^2/\ln 2 > 0$ at an accessible order. (ii) $\Longrightarrow$ (iii): existence, per accessible horizon containing the witness order k, by the finite-horizon process-dilation theorem above, which reproduces the full multi-time law; the realization inherits (C4) at k directly — the ledger-conditioned kernels reproduce $\mathcal{S}$'s differing past-conditionals exactly (`c4_backflow_probes.py`) — and satisfies the remaining conditions: (C1) because a Markov-breaking kernel dependence requires hidden influence at that step (with none anywhere, every visible transition is a fixed function of the visible present and the process is Markov, contradicting (ii)); (C2) trivially in the construction — the bath is static, $\tau_B = \infty$, satisfying $\tau_S \ll \tau_B$; for *physical* hidden sectors the physical-memory theorem's contrapositive (§2.3, the C2 process form) supplies the necessity direction — observed conditional memory bounds the mixing residual below — conditional on the conditional-mixing-curve hypothesis with finite $\tau_B(\varepsilon_0)$ (Remark, §3.4; ETH [17, 18] the motivation; no smallness of ε at the coupling time is assumed); (C3) from the data-processing bound in per-process form ($m \geq 2^{I^*}$). Both directions of the equivalence are theorems of this paper. Finally, a unitarily evolving representation is available for $\mathcal{S}$ regardless — internally by the permutation unitary of any realization (the equivalence above), and under (T) by the imported correspondence [11, 12], which represents it by a unitarily evolving quantum system; under the time-homogeneity restriction the representation takes the fixed-$\hat{H}$ ancilla-marginal form of (i) (scope Remark) — the paper's single external import, sitting outside the equivalence. $\square$

Remark (the horizon restriction is forced). The witness-anchored quantifier in (iii) is not a courtesy. A horizon ending before the first memory witness restricts $\mathcal{S}$ to a Markovian law, and by the universal theorem's biconditional no faithful realization of a memory-free law carries a readback gap — so the unrestricted every-horizon form of (iii) is false, while every horizon containing a witness supports the construction. The strong per-witness form is exact: $M_t > 0 \iff$ for every $K \geq t+1$ there is a faithful realization of the law through K with readback biting at t — the forward direction by dilation of the K-law, whose restriction contains the t-witness, the converse because a readback gap in any faithful realization *is* a differing next-step conditional (universal theorem, (a)).

Remark (retired direction). (i) alone does not imply (ii): a diagonal Hamiltonian satisfies the definition with $T(t) = I$ for all t — perfectly P-divisible — and a nearly degenerate $\hat{H}$ is permutation-close throughout an accessible window (`review3_probes.py`). The equivalence therefore runs (ii) $\iff$ (iii) with representability following for the source's finite-configuration class; (ii) is certified empirically by directly observed memory effects in open-system dynamics, and the further reading of Bell-class dynamics as temporally indivisible is the imported framework's — the necessity argument uses neither that reading nor any converse.

Remark (existential and universal readings). The equivalence is existential: (ii) holds iff *some* per-horizon (C1)–(C4) realization exists. The necessity analysis is universal and per-condition: any deterministic realization of a non-Markovian process has hidden influence at some step (C1), memory capacity $m \geq 2^{I^*}$ (C3), and read-back differing past-conditionals (C4); C2's necessity holds within the conditional-mixing class. The two readings are logically independent.

Remark (representation, not explanation). The equivalence is a representation theorem: every accessible non-Markovian law *admits* a per-horizon C1–C4 realization. It does not, by itself, explain why the physical universe exhibits such a law; that explanatory burden — the specific substratum satisfying C4 quantitatively, with the horizon's capacity and timescales — is carried by the cosmological construction and tested there.

Remark (consolidation). The equivalence is per-horizon, and the universal all-time form is settled *negative*: a bijection on a finite set has finite order L, so any fixed realization with a fixed prior has an exactly L-periodic multi-time law ($\varphi^{k+L} = \varphi^k$) — a nonperiodic target admits no exact all-time realization by one fixed finite substratum. The honest open questions are (a) exact gluing for recurrent target families — which compatibility conditions on the per-horizon realizations suffice — and (b) the physically operative form: quantitative approximation of the target on windows $t \ll t_R$ by one fixed substratum before its recurrence. Form (b) coincides with the accessible-timescale scoping of §4.5.

Remark (scope: time-homogeneity). The restriction to time-homogeneous processes is load-bearing for the fixed-$\hat{H}$ form of (i). A non-autonomous quantum evolution such as $U(t) = e^{-i\sigma_x t^2}$ (generated by $H(t) = 2t\,\sigma_x$) has P-indivisible computational-basis populations, but $\cos^2(t^2)$ is not almost-periodic and therefore not any fixed-$\widehat{H}$ ancilla-marginal — (ii) without homogeneity does not deliver (i) as defined. Autonomy is not homogeneity: a single fixed bijection applied every step can still produce time-dependent visible kernels through a hidden clock — the §3.4 construction is the explicit example. Homogeneity is an additional property of the visible process; the imported correspondence is invoked in its fixed-$\widehat{H}$ form only where it holds, and there the restriction costs the framework nothing. The fixed-$\widehat{H}$ form itself is constructive for realized processes — the realization's permutation unitary with $\widehat{H} = i \log U$ (Lemma above) — at the horizon's integer times; the almost-periodicity obstruction of this Remark concerns the all-real-time reading.

At the level of finite-horizon fixed-basis observable laws, embedded observation and unitary Born-rule representation are not merely *compatible* — they are mathematically *equivalent* ($S \iff D \iff Q_{\mathrm{fb}}$, the theorem above). The scope restriction is the content: the equivalence identifies the representation class exactly and stops short of operational quantum mechanics: the instrument algebra is reached exactly when the five operational completion conditions hold ([GR §3.3]; Remark below), none of them supplied by the fixed-basis layer, and cross-horizon representation consistency is bounded by the classical-dimension obstruction.

Remark (scope of the equivalence). The characterization theorem establishes the Hilbert space, unitary dynamics, and Born-rule transition probabilities. Four further structures of operational quantum mechanics merit comment.

(i) Tensor product structure (visible–hidden). The Stinespring route (§3.2) constructs $\mathcal{H} = \mathcal{H}_V \otimes \mathcal{H}_H$ and derives the CPTP channel Φ by partial trace. The visible–hidden tensor product is therefore given by the construction, not postulated.

(ii) Tensor product structure (visible-sector subsystems). Spatially separated visible regions $V_A, V_B \subset V$ inherit an exact **dynamical causal separation** from the dependency graph: after k updates, a local state can depend only on the graph ball of radius k (§3.3). This establishes the causal-cone part of locality without the stronger simple-cubic nearest-neighbor stipulation. It does not alone establish statistical product structure after conditioning, local tomography, or one common tensor-product instrument category. Bell adds a sharper restriction: if that pointwise cone is kept at the wings' separation and the presetting ensemble is measurement-independent, determinism gives Bell factorization and $|S_{\mathrm{CHSH}}| \leq 2$. The framework therefore does not use graph locality as the hidden-variable explanation of quantum Bell violation; it adopts ontic parameter dependence while requiring operational no-signaling. The common local quantum composite/instrument structure is the content of the inert-spectator and iterated-composition conditions of the operational-completion characterization (Remark on the operational-extension boundary; [GR §3.3]).

(iii) State update, measurement, and multi-time predictions. Projective mea-

surement in the emergent description corresponds to conditioning on a visible-sector outcome and re-marginalizing over the hidden sector. The Lüders rule $\rho \to P_k \rho P_k / \mathrm{Tr}(P_k \rho)$ transcribes Bayesian updating on the classical substratum (Lemma 3) for the represented diagonal sector — which is the transition-statistics layer, not the instrument algebra: the general operational update for noncommuting interventions belongs to that algebra, reached exactly under the operational completion conditions (§3.4; [GR §3.3]); its extension to arbitrary coherent density matrices belongs to the operational bridge (§3.2). No independent measurement postulate is required for the represented statistics. Multi-time correlation functions of the passive law are standard calculations within the delivered formalism ($\mathcal{H}_V$, $U(t)$, Born rule); weak values and Leggett-Garg protocols are intervention-rooted and belong to the intervention dilation (§3.4 Remark; §3.2 target) — the delivered formalism underdetermines them. The multi-time statement carries one qualification. The correspondence is engineered to reproduce rooted two-time transition data, which under-determines multi-time joint statistics [22]; the substratum, by contrast, fixes a unique multi-time law. Agreement of the substratum's conditioned multi-time statistics with the quantum formalism's intervention-rooted predictions is supported by the same chaotic-mixing suppression argument used to motivate decorrelation of controllable setting generators from preparation-relevant hidden data (§3.3); that decorrelation is not the Bell-violation mechanism, because the adopted Bell branch instead relinquishes ontic parameter independence. Residual pre-division memory, if present, would appear as deviations from quantum multi-time predictions — a falsifiable texture of the same standing as the residual dissipation $\mathcal{D}$.

(iv) Classical P-indivisibility and quantum mechanics. The representation clause is not a claim that all P-indivisible processes are "secretly quantum" in some metaphysical sense; it is a theorem that the mathematical structures are identical. Classical systems exhibiting P-indivisibility — renewal processes, semi-Markov processes on finite state spaces — admit a unitary description, generically constrained up to the phase-locking gauge (shift, rephasing, the antiunitary conjugate), whose consequences separate into two layers: continuous transition data constrain *generator structure* beyond the raw data up to that gauge — energy-gap structure, under the phase-locking hypotheses — while *operational* interference and entanglement predictions require in addition the instrument and observable structure of the operational bridge, stated as open (§3.2, §3.4). The processes can be realized as marginalization of deterministic dynamics under C1–C4 (necessity theorems). The framework locates the observer's process inside the correspondence's domain, where the imported theorem supplies a unitarily evolving representation for every finite-configuration member — including divisible and Markovian ones. P-indivisibility does not gate that supply; it distinguishes representations whose dynamics is genuinely indivisible from those that are trivially unistochastic (§3.4). The physical content of the unitary is not that of a mere mathematical embedding: it is a Hamiltonian determined by the transition data up to the stated gauge freedom — energy shift, rephasing, and the antiunitary conjugate $H \to -H^*$ (phase-locking lemma) — arising from the

actual dilation structure of the system (§3.2).

(v) The single external dependency, scoped. The paper imports one external result: the Barandes stochastic-quantum correspondence [11, 12], which establishes that every finite-configuration process in its class — a broad class in which Markov chains are special cases — can be regarded as a subsystem of a composite unistochastic process on dimension $\leq n^3$, so that the stochastic and unitary descriptions are inter-translatable in compressed form. What the import is load-bearing for has narrowed as the internal results grew: existence of a fixed-basis unitary Born representation is internal (the equivalence theorem — the permutation unitary of any realization), as are the dilation construction, the C1/C3/C4 necessity theorems, and the ETH-conditional C2 bound; the import retains the compressed ($\leq n^3$) representation and the interpretive apparatus that travels with it, under (T). This gives the dependency a clean falsification profile: if the correspondence were shown to be incomplete or to require additional hypotheses, the framework would lose the compressed form and its interpretation — and nothing else, since existence, equivalence, and necessity are internal. A referee skeptical of the import should direct that skepticism at the compression and interpretation claims specifically; the framework's own results no longer route through it.

3.5 Generative and limitative faces of the observation map

The characterization theorem establishes what embedded observation *produces*. The same trace-out map has a complementary structural property — what it *fails to fix* — and stating it makes precise a limitation that recurs in the gravitational and Standard-Model sectors (the residual-input status of spatial isotropy in the framework's gravitational and Standard-Model treatments).

Lemma (rest-frame selection). *Let the embedded-observation partition $S = V \sqcup B$ be realized across a causal horizon $\mathcal{H}$, with induced coarse-graining map T. The horizon fixes a simultaneity foliation of the substratum, hence a distinguished timelike direction n (the frame in which the update steps are simultaneous). Then the invariance group of T within the local frame transformations $O(3,1)$ is the stabilizer of n — the spatial rotation group $O(3)$, the little group of a timelike vector. Consequently local frame transformations split into those preserved by T (spatial rotations, which fix n) and those not preserved by T (boosts, which move n).*

Proof. T is defined by the horizon partition, which fixes the simultaneity foliation and hence n up to sign. A frame transformation $\Lambda \in O(3,1)$ is an invariance of T only if it preserves the data defining T, i.e. only if $\Lambda n = n$. The stabilizer of a timelike vector in $O(3,1)$ is isomorphic to $O(3)$ (the standard little-group classification). Boosts map n to a distinct timelike vector and so are not invariances; spatial rotations fix n and so are. □

Corollary (boost/rotation inequivalence). Because T selects the timelike direction n but no spatial direction, the emergent theory inherits from the map a physical rest frame but no physical spatial axis. Lorentz-violating operators therefore have structurally inequivalent status by sector: boost-sector violation is tied to the map-selected object n (and is correspondingly exposed to map-level structure — the local-boost-invariance prerequisite of the framework's horizon-thermodynamic construction, the custodial mechanism of its lattice Standard-Model construction), whereas rotational-sector violation is referred to no map-selected object and is inherited from the substratum constitution alone, where the cubic point group protects it only at tree level. This is the structural origin of the boost/rotation asymmetry recorded in the framework's Lorentz-violation accounting; it explains the asymmetry without resolving either sub-sector.

Converse, and the resulting biconditional. The lemma admits a converse, which upgrades it to an equivalence and sharpens the corollary. *If the invariance group of T within $O(3,1)$ is a maximal compact subgroup (isomorphic to $O(3)$), then it is the stabilizer of a unique timelike direction n, so T selects a rest frame.* This is the standard maximal-compact-subgroup structure of the Lorentz group: the maximal compact subgroups of $O(3,1)$ are the conjugates of the rotation group $O(3)$, and they stand in bijection with the timelike directions (equivalently, with the points of the rest-frame space $O^+(3,1)/O(3) = H^3$), each subgroup fixing exactly the one-dimensional timelike line it leaves invariant. Combining the two directions:

> *The coarse-graining map T selects a unique preferred timelike direction n **if and only if** its invariance group within the local frame group $O(3,1)$ is a maximal compact subgroup $O(3)$.*

The content of the converse is not decorative. It says that **rotational invariance of the observation map and rest-frame selection are the same fact**: one cannot have T be rotation-invariant without its selecting a preferred frame that boosts violate, and conversely. The boost/rotation asymmetry is therefore not a contingent feature of how T happens to act — it is necessary. The residual rotational symmetry *is* the boost-breaking frame selection; spatial isotropy of the emergent description and the existence of a cosmic rest frame are two descriptions of one structure, not independent facts. (As with the forward direction, the converse rests on the maximal-compact-subgroup classification — standard Lie theory, not a framework-specific input — and requires the invariance group to be the full $O(3)$; it completes the equivalence at the same elementary depth as the lemma, not at the depth of the characterization theorem.)

The two faces, and a caution against over-reading them. The characterization theorem and this lemma are two properties of the *same* map T: what it produces (its image: the quantum kinematics) and what it leaves invariant (its symmetry: spatial rotations only). It is tempting to cast these as opposite operations — generation versus limitation — but they are the *same* operation, a projection of the observed statistics under information missing to the embedded observer; what differs is the *type* of missing information. In the quantum case the missing information is the hidden-sector *state*, which is dynamically resampled (the bath thermalizes) and therefore fluctuates; averaging over a fluctuating quantity is productive and yields the memory-bearing non-Markovian statistics that quantum mechanics represents. In the isotropy case the "missing" information is the *orientation* of the lattice constitution, a fixed global parameter that is not resampled; there is no ensemble to average over, and the observer's ignorance of the lattice axes is not a statistical mixture. The discriminant is whether the missing information fluctuates: fluctuating missingness is averaged into emergent structure, fixed missingness is inherited. Both faces trace to the temporal asymmetry of embedded observation — the persistence condition (C2) that sustains the memory-bearing sector — the slow bath one mechanism for it (§1.3) — is the same feature that selects the simultaneity surface and hence the timelike n — but the quantum/isotropy contrast is properly attributed to fluctuating-versus-fixed missingness, not to a dichotomy of operations.

Two honest qualifications bound this subsection. First, the lemma is one-directional and near-elementary (it follows from the little-group classification once the horizon is granted to select n); it is far less deep than the characterization theorem — proved in-house, the imported correspondence retaining only the compressed form under (T) (§3.4) — and it is labelled a lemma for that reason. Second, it is a structural statement about the map's reach, not a derivation of a physical law: it explains *why* the framework can constrain the boost sector and must take spatial isotropy as input, but it neither supplies that isotropy nor forbids its violation. (Whether the fixed orientation could, in some régime, be effectively resampled — restoring isotropy generatively with a calculable residual anisotropy of the $\ell = 4$ cubic-harmonic form — is a separate open question pursued in the book's open-problems treatment; the present lemma is agnostic on it.)

4.1 Interpretive Consequences

The degrees of freedom involved in quantum experiments — photons, electrons, slits, detectors — are all visible-sector objects. Their quantum behavior is a downstream consequence of the trace-out: the quantum representation, once established by the theorem, is available for all visible-sector dynamics — the operational theory's unique selection remaining open (§3.4).

In the double-slit experiment, the particle traverses a single slit in the deterministic substratum. The interference pattern arises because opening or closing the second slit changes the boundary conditions of the transition matrix, altering the distribution of detection events. A which-path detector at one slit couples the trajectory to additional visible-sector degrees of freedom, changing the transition matrix and eliminating the interference terms. The Everettian measure problem dissolves: the system evolves as a single reality; "branches" are features of the compressed description. The Born rule, often treated as an independent postulate, is here part of the exact fixed-basis representation dictionary ($S \iff D \iff Q_{\text{fb}}$, §3.4), rather than an additional stochastic axiom. What remains open is the common coherent operational extension of that dictionary to all interventions and composites. The equilibrium-of-the-cycle reading is the cosmological realization's mechanism proposal rather than a theorem of this paper.

Observer covariance and nested observers. The framework treats the observer partition $V \subsetneq S$ as a choice, not an absolute. Different partitions correspond to different observers; each partition satisfying the characterization conditions produces its own C1–C4-quantum description of the remaining degrees of freedom. This is not a multiplicity of incompatible physics — it is a covariance property of the framework under observer-partition choice, which resolves the standard nested-observer paradoxes.

Consider the Wigner's-friend setup: two disjoint partitions $V_W, V_F \subset S$ (Wigner and Friend) with $V_W \cup V_F \subsetneq S$, and shared external hidden sector $H = S \setminus (V_W \cup V_F)$. The framework assigns three descriptions, all derived from the same φ by the same trace-out procedure:

- **Friend's description.** Trace over $S \setminus V_F = V_W \cup H$. Under C1–C4 applied to $(V_F, S \setminus V_F)$, Friend's reduced description is quantum-mechanical. Friend observes definite outcomes in her visible sector, including any system she measures.
- **Wigner's description.** Trace over $S \setminus V_W = V_F \cup H$. Friend is part of Wigner's hidden sector. Under C1–C4 applied to $(V_W, S \setminus V_W)$, Wigner's reduced description is also quantum-mechanical. Wigner does not see a definite Friend-outcome within his own description — Friend's state is traced out.
- **Joint description.** The marginal transition matrix $T_{WF}(\alpha, \beta \to \alpha', \beta') = |\{h \in H : \pi_{V_W \cup V_F}(\varphi(\alpha, \beta, h)) = (\alpha', \beta')\}|/|H|$ exists and is unique given φ. Its marginals reproduce Wigner's T_W and Friend's T_F respectively.

The three descriptions are simultaneous rather than competing. Friend's claim that she has a definite outcome, and Wigner's description of Friend as a traced-out subsystem, are statements in different reduced descriptions, not conflicting claims about the substratum state. At the substratum level, Friend, Wigner, and the system all have definite configurations; neither Friend nor Wigner has epistemic access to the complete state, and this epistemic limitation produces their respective quantum descriptions through the same mechanism in §2–§3. The apparent "paradox" dissolves into a covariance property: observer identity is partition-gauge, not absolute, and framework predictions are covariant under choice of observer partition. No-go theorems assuming a single universal quantum description (such as Frauchiger–Renner) do not apply, because the framework provides multiple simultaneous reduced descriptions, each internally consistent with respect to its own partition, with all derived from the same deterministic substratum.

4.2 Scope

The characterization theorem's equivalence is (ii) $\iff$ (iii) under (C1)–(C4), both directions in-house; the quantum representation follows for the source's finite-configuration class, and observed Bell-class dynamics — indivisible, hence non-Markovian a fortiori — certify (ii) directly; representability alone is strictly weaker (diagonal-$\hat{H}$, §3.4 Remark). The theorem does not identify which physical systems satisfy the conditions; this is an empirical question. Universality in our universe follows from the shared causal partition defined by null geodesics.

4.3 The Status of the Discreteness Scale

ϵ does not smuggle in a quantum assumption, but the reason is foundational rather than field-theoretic. Finiteness of the *visible* configuration space is a lemma of the observation axiom itself: a single tokened differentiation registers determinate — and therefore finitely structured — content, so what is distinguishable at the locus per moment is finite; invoking holographic bounds at the foundational layer would be circular, those being downstream physical results. Finiteness of the hidden sector holds in the forward construction with a two-part status (A1, as scoped in the companion paper: the boundary layer $\mathcal{C}_B$ is bounded by the same finite-entropy argument that bounds $\mathcal{C}_V$, while the deep-sector cardinality beyond it is a gauge choice — the minimal finite representative under deep-sector enlargement), and is *constructed* rather than assumed in the reconstruction direction (§3.2's dilation supplies a finite hidden space). Holographic entropy bounds [2] then enter not as motivation but as a self-consistency check: the framework derives $\hbar$ and the Bekenstein–Hawking coefficient downstream, and the derived bound is compatible with the finiteness assumed here — a closed loop, not a circular one.

4.4 Relation to Prior Work

't Hooft [1]: Differs in mechanism (epistemic trace-out vs. information loss), generality (any embedded observer vs. particular rules), and Bell placement (outcome independence violation vs. superdeterminism). **QBism/relational QM:** Share observer-relative epistemic states; this framework provides a structural *why* and quantitative predictions.

Relation to prior stochastic-quantum programs. Three earlier programs sought to derive quantum mechanics from classical-looking substrates. **Nelson's stochastic mechanics [32]** postulates a continuous Brownian-motion-like process on configuration space with drift determined by the imaginary part of the log wave function; the Schrödinger equation follows from the Fokker-Planck structure. Wallstrom [33] showed that Nelson's program cannot uniquely select the quantization condition $\oint p \cdot dq = nh$ — the stochastic process admits a broader class of solutions than standard QM, and the quantization must be reimposed by hand. The present framework is not subject to this critique because it does not derive QM from a continuous stochastic process on ψ or on configuration space; its exact identification is of the quantum REPRESENTATION — universal at the fixed-basis layer, with the common coherent operational lift stated separately — and proceeds through the characterization theorem for embedded observers on a deterministic bijection, with stochasticity emerging from the trace-out over the hidden sector rather than being fundamental. **Adler's trace dynamics [34]** derives QM from statistical mechanics of matrix-valued dynamical variables with a trace-class Hamiltonian, with the canonical commutation relations emerging as expectation values over a specific equilibrium ensemble. The framework here requires neither matrix-valued variables nor an equilibrium-ensemble derivation; the substratum is a finite configuration space with a deterministic bijection, and the emergence mechanism is the Stinespring dilation applied to observer-marginalized transition probabilities. **Barandes's stochastic-quantum correspondence [11, 12]**, used as a technical tool in §3.1, is not a competitor but a building block: the correspondence maps its finite-configuration tuple class — a broad class in which Markov chains are members and P-indivisibility marks the nontrivially indivisible dynamical sector, not a membership criterion — to unitary quantum evolutions in compressed ($\leq n^3$) form under (T); the present framework supplies the physical mechanism (embedded observation under C1–C4) that produces the memory-bearing processes occupying that nontrivial sector — P-indivisibility in the stronger regimes established separately — and supplies the fixed-basis representation itself internally ($S \iff D \iff Q_{\text{fb}}$, §3.4), retaining the import for compression and interpretation. The three earlier programs identified emergent QM as a legitimate research target; what the present framework adds is an equivalence with necessary and sufficient conditions — (ii) $\iff$ (iii) with representability as a one-way consequence (§3.4) — and a constructive derivation chain that does not require independent quantization conditions or equilibrium ensembles.

Relation to the post-2022 observer-essentiality wave. A substantial body of work since 2022 has independently established that incorporating observers as physical degrees of freedom produces structural changes in the formalism of quantum field theory in gravitational settings, with results that converge on the foundational claim of the present framework — that observation cannot be treated as an external operation on physics — while differing in mechanism. **Chandrasekaran, Longo, Penington, and Witten [35]** showed that adding an observer to QFT in a gravitational subregion promotes the von Neumann algebra of observables from type III to type II_∞, with a well-defined trace and entropy; Witten [36] extended the construction to de Sitter space. **Maldacena's recent dS observer-cancellation [37]** demonstrates that the unphysical phase of the de Sitter sphere partition function cancels exactly when the observer-with-clock degrees of freedom are properly incorporated, removing a long-standing obstruction to a clean dS thermodynamic interpretation. **Harlow, Usatyuk, and Zhao [38]** argue that the closed-universe Hilbert-space dimension is determined by the observer's degrees of freedom rather than being a free parameter; the **AAIL construction** [**39**] provides a complementary calculation in which the relevant dimension scales as $d_{Ob} \cdot e^{2S_0}$, refining HUZ's heuristic bound. **DEHK partition-relativity** [**40**] establishes that the algebra of observables is partition-dependent and that physical predictions are encoded in equivalence classes under partition refinement, paralleling the present framework's commitment to partition-defined observation. **Slagle and Preskill** [**41**] constructed an explicit emergent-QM model in which boundary quantum dynamics arise from a classical bulk lattice with stochastic dynamics, demonstrating the logical possibility of QM-from-classical at the cost of an exponentially long extra spatial dimension and explicit stochasticity.

The present framework converges with these programs on the foundational substance — observation is not external to physics; the algebra of observables depends on the observer; the partition structure carries physical content — while differing in mechanism and scope. CLPW achieves type II algebra by adding observer degrees of freedom to QFT; the present framework derives the characterization of embedded observation directly from the deterministic substratum, with the type II structure as a consequence rather than a postulate. Maldacena requires an explicit observer-with-clock; here the partition trace-out is the *defining* operation, so the cancellation is structurally forced. HUZ argues that dimension is observer-determined; the present framework provides a microscopic realization in which the Hilbert-space dimension is computed directly from the partition geometry, with AAIL's $d_{Ob} \cdot e^{2S_0}$ formula matching the form of the substratum count. Slagle-Preskill demonstrates the logical possibility of QM-from-classical with an extra spatial dimension and stochastic dynamics; the present framework achieves the same emergence with neither — pure determinism on a finite substratum, no extra dimensions. The convergence of multiple independent programs on the foundational observer-essentiality move strengthens the empirical case for all of them; the present framework's distinguishing feature is that it derives this content from a finite deterministic substratum and, in doing so, produces a quantitative empirical record that the comparison programs do not currently match. The convergence is not a competition for priority — the deposit record is clear — but a triangulation: when several independent investigations of the foundations of physics arrive at structurally similar conclusions through different routes, the conclusions deserve more weight than any one of them alone would warrant.

Witten's Type II algebra program achieves finite entropy without finite-dimensional Hilbert spaces; this is compatible, since the framework's predictions depend on finite entropy (the physical content of Lemma 1), not on dimensional finiteness per se.

4.5 Assumptions, Limitations, and Falsifiability

The theorem requires Lemma 1, the stochastic-quantum bridge (§3.1 and §3.2; two routes reach its transition-statistics layer, neither reaching the operational instrument algebra), and genericity conditions (non-degenerate spectrum, non-vanishing overlaps), which hold for all but a measure-zero set. The P-indivisibility proof uses finiteness via the recurrence argument ($\varphi^N = \mathrm{id}$). The accessible-timescale backflow lemma (§2.3) provides the quantitative route to accessible non-Markovianity, with P-indivisibility retained at the recurrence scale, establishing observable information backflow on the timescales accessible to laboratory experiments — physically indistinguishable from a timeless identity for all experiments well below the recurrence time.

Falsifiability. The framework makes a specific, experimentally testable structural claim: *the hidden sector is a classical substratum* — a finite set with a deterministic bijection. Every fundamental non-Markovian open-system process in nature — dynamics arising from embedded observation, not from engineered quantum environments — must therefore be classical-memory simulable in the operational sense of Bäcker, Beyer, and Strunz [28]: the environment's memory can be captured by a classical variable, and the non-Markovian dynamics reconstructed by conditioning on that variable. A physically realized fundamental open-system process whose memory provably exceeds the classical bound — demonstrated via the entanglement-structure protocols of [28, 29] — would falsify the framework's foundational ontology. This is a current-technology test whose experimental tools are under active development; a recent unified review [30] surveys the rapidly evolving classical-vs-quantum memory distinction in open-system dynamics. The claim is sharpest in the *C2-marginal regime* ($\tau_S \sim \tau_B$ rather than $\tau_S \ll \tau_B$), where non-Markovianity is $\mathcal{O}(1)$ rather than suppressed. Candidate physical realizations — strongly-driven superconducting qubits with two-level-system baths, biological electron transport with vibrational coupling, spin-boson dynamics at strong coupling — are the systems where the classical-memory simulability commitment is most stringently tested, and where emerging experimental and theoretical tools [28, 29, 30] are converging on the classical-vs-quantum memory distinction.

Theorem-level falsifiability. The characterization theorem is mathematical — the fixed-basis representation and the hidden-memory theorem are internal (§3.4), with Barandes retained for compression and Stinespring as supporting structure — so no experiment falsifies the theorem itself; what is empirically falsifiable is the framework's physical identification of nature with its deterministic finite ontology. The classical-memory test above is therefore the primary falsifiability route.

Laboratory tests of the characterization theorem may be possible in tabletop systems where the hidden sector capacity is tunable.

The posit ledger. The two-axiom presentation of §1.2 names the foundation's primitives; the full argument consumes more inputs than two, and the complete list is collected here so it is countable in one place. The framework's substantive posits, beyond Axiom 1 (tokened differentiation, indubitable), are: (i) Axiom 2 — recurrence — owned as a posit with a non-derivability countermodel; (ii) finiteness of the *total* state space S — Lemma 1 delivers finiteness of the observer's distinguishable states, and the extension to all of S is A1 of the companion substratum construction — there scoped as the gauge choice of the minimal finite representative, the deep-sector cardinality being unobservable (deep-sector enlargement, developed there) — consumed here by the recurrence step of §2.3 and by §4.6, whose conclusions accordingly hold on the finite representative and, being boundary-only in content, on every member of the gauge class (the transfer stated and proved as a corollary in the companion substratum construction); (iii) the record-preservation and eventual-image arguments that ground determinism (Lemma 3) close the hidden-sector margin using one *empirical* input — the observed unitarity of quantum dynamics — and one structural principle, so bijectivity is part-empirical rather than a pure lemma (a reader who declines the recovery may instead adopt reversibility as the standard information-conservation postulate of unitary quantum mechanics and classical Hamiltonian flow [42, 43], at no cost to the downstream results — see the remark following Lemma 3); (iv) the measure selection is a maximal-entropy principle ("the maximal-entropy invariant measure compatible with the observer's records") — a standard but substantive epistemic postulate, not derived; (v) the hidden-sector mixing hypothesis (ETH-motivated) conditioning the C2-necessity direction, for a system class (emergent descriptions of linear-mod-q automata) on which the standard ETH evidence base is silent — this is the same open dynamical question as the chaotic-mixing suppression invoked to motivate laboratory setting/preparation decorrelation (§3.3) and the typicality selection in (iv), one conditional wearing three hats; (vi) spatial isotropy, entering as empirical input E4 of the reconstruction; and (vii) linearity of the substratum update (A5), equivalent to amplitude-scale gauge invariance and owned at that more primitive location. Seven items, of unequal weight and mostly independently motivated. Part I of the companion methodology paper gives the complementary view: an eight-rung analysis of the primitive that recovers most of these as lemmas of the two-axiom base and isolates recurrence (Axiom 2) as the one irreducible non-lemma. The two accounts describe one structure — the rung analysis counts what is *recovered*, this ledger counts what is *assumed* — and agree on the substance, including that the determinism recovery closes its hidden-sector margin with an empirical input and that finiteness is delivered only per-moment. The two-axiom statement remains correct about what is *primitive*; this ledger states what is *assumed*, and the framework's claims should be read as conditional on the full list.

Status ledger (theorems and conditions). The table transcribes the standing status of each load-bearing claim of this paper's foundational argument, with its canonical site; the matter-sector flags (the chirality hypothesis H-χ' and the Casimir-chain carrier input) are carried at their sources in the companion matter paper and in the book's open-problems ledger. It introduces no new claims; a row may not state a status stronger than its cited site, and any apparent strengthening is an error in the table, not a result.

Claim	Status	Site
(ii) $\implies$ (i): P-indivisible $\implies$ nontrivial quantum representation (the representation itself universal, §3.4)	Imported ([11, 12]); accessible timescales; fixed-$\hat{H}$ under time-homogeneity. The converse is retired: (i) alone does not imply (ii) (diagonal-$\hat{H}$; slow clock; `review3_probes.py`)	§3.4
(ii) $\implies$ (iii): process $\implies$ deterministic realization	Proved constructively, per accessible horizon — finite-horizon process dilation reproducing the full multi-time law; the realization inherits (C4); C1/C3/C4 legs unconditional, C2 leg conditional on the mixing hypothesis; arbitrary-prior form via the response-table extension; Consolidation: universal exact all-time form impossible (finite-order periodicity); recurrent-family gluing and pre-recurrence window approximation open	§3.4
Fixed-$\hat{H}$ form for realized processes	Proved, uncompressed — the realization's permutation unitary, $\hat{H} = i \log U$, exact at integer horizon times for uniform, slice, and weighted priors (`review4_probes.py`, P-F(a)–(c)); the $\leq n^3$-bounded form is the external correspondence's, for the source's finite-configuration class	§3.4

Claim	Status	Site
(iii) $\implies$ (ii): realization $\implies$ accessible non-Markovianity	Proved — the readback lemma: a (C4) gap forces $I(X_{<k}; X_{k+1} \mid X_k) \geq p_0\delta^2/\ln 2$ on the window; bare-recurrence form from (C1) alone at t_R	§2.3
CP-indivisibility of $\{\Phi_t\}$	Proved for the framework's diagonal-preserving (permutation-dilation) channels; fails for general CPTP families (Hadamard boundary)	§3.2
C1 necessity	Proved — any realization of a non-Markovian process has hidden influence at some accessible step (none anywhere $\implies$ Markov); a non-permutation one-step T is the sufficient first-step witness, not an equivalence	§3.4, §1.3
C2 necessity	Conditional on the hidden-sector mixing hypothesis; the quantitative bound $\varepsilon \geq \delta/(k-1)$ is proved, its constant saturated in the certification suite (`c2_mixing_probes.py`); ETH is physical motivation, not a premise	§3.4, §1.3
C3 necessity	Proved — data-processing bound in per-process form, $m \geq 2^{I^*}$	§3.4, §1.3

Claim	Status	Site
C4 necessity	Immediate given observed non-Markovianity (any realization of a non-Markov process reads differing past-conditionals); separation certified — (C1)–(C3) alone admit zero-backflow realizations (the coin)	§1.3, §2.3

4.6 Structural Exclusion of Equilibrium-Phase Observers

An immediate corollary of (C1)–(C4) is a structural selection rule on where in a finite bijective (S, φ) observers can exist. The rule replaces a postulate often imported from elsewhere — the "past hypothesis" of low initial entropy — with a theorem conditional on the effective-mixing hypothesis (EM) below; under (EM) it is derived from the framework's own definitions, and without (EM) it is a conjectural application.

Setup. Let V be a partition with hidden sector $B = S \setminus V$. Define the coupling-graph neighborhood $\Sigma_{\text{loc}}(V)$ as the subset of S reached from V by at most $\tau_S/\Delta t$ steps of φ, where Δt is the update step. The exact Koopman operator of φ — and of its restriction to any invariant subsystem — is a permutation operator: unitary, with modulus-one spectrum and exact recurrence. It has no dissipative gap; exponential mixing cannot hold for it. Mixing, where it holds, is a property of a *compression*: let $\mathcal{K}_\Sigma := P_\Sigma \, \mathcal{U}_\varphi \, P_\Sigma$ denote the coarse-grained effective channel on the observable algebra of a local region Σ. **Effective-mixing hypothesis (EM).** *The local effective channel* $\mathcal{K}_{\Sigma_{loc}(V)}$ *mixes, with mixing time* $\tau_{mix}(\Sigma_{loc}(V))$ *set by its (dissipative) spectral gap.* (EM) is an explicit physical hypothesis — motivated by local thermalization phenomenology, not derived from φ — with the same status as the conditional-mixing curve of (C2).

Theorem (structural observer selection — conditional on (EM)). Under the effective-mixing hypothesis, for any partition V of bounded coupling-graph diameter,

$$\tau_B(V) \le \tau_{\text{mix}}(\Sigma_{\text{loc}}(V)) + O(\varepsilon),$$

where ε absorbs typical-set and finite-size corrections. In particular, if $\Sigma_{\text{loc}}(V)$ is in its local equilibrium macrostate in the sense of canonical typicality [31], $\tau_B(V)$ is bounded by the local thermal decorrelation time.

Proof. Correlations between a V-localized observable f and a B-localized observable g at time t require information to propagate from V to B at time 0 and back to influence the V-correlated component of g at time t. By the bounded coupling degree of φ, this round trip stays within $\Sigma_{\mathrm{loc}}(V)$ for $t \leq \tau_S$. Under (EM), the dissipative gap of the effective channel $\mathcal{K}_{\Sigma_{\mathrm{loc}}}$ — which on an equilibrium macrostate sets the local thermal decorrelation rate — bounds the decay of such coarse-grained correlations exponentially at rate $1/\tau_{\mathrm{mix}}$; canonical typicality [31] supplies the local-equilibrium *state*, not the temporal mixing, which is exactly what (EM) posits. Hence $\tau_B(V) \leq \tau_{\mathrm{mix}}(\Sigma_{\mathrm{loc}}) + O(\varepsilon)$. $\square$

Corollary (anti-Boltzmann-brain). A Boltzmann brain is by construction a local fluctuation within a neighborhood Σ_{loc} that is itself in local thermal equilibrium — that is what "fluctuation from equilibrium" means. The exclusion is the conditional chain: (EM) together with $\tau_{\mathrm{mix}}(\Sigma_{\mathrm{BB}}) \ll \tau_S$ implies the candidate partition fails (C2), and hence — by the fast-bath erasure lemma of §2.3, which suppresses the order-k readback gap as $e^{-(k-1)\tau_S/\tau_B}$ — fails (C4), which is the condition the characterization actually requires. For a numeric illustration under a further generic-thermal estimate — (GT): the region's effective-channel correlation time is set by the thermal scale, $\sim \hbar/(k_B T_{\mathrm{local}}) \cdot O(1)$, an estimate that fails for protected slow modes (conserved-quantity bottlenecks, activation barriers, glassy or critical slowing) — the bound is at most $\sim 10^{-12}$ s for temperatures warm enough to support structured matter. The illustration rides (GT); the exclusion itself rides only the chain. Any finite-information observer has τ_S bounded below by its own internal processing time — for biological or synthetic BBs of any currently-envisioned type, $\tau_S \gtrsim 10^{-6}$ s. The slow form (C2-slow), $\tau_S \ll \tau_B$, therefore fails by at least six orders of magnitude for any BB whose local environment satisfies (GT) — and because (EM) posits genuine mixing rather than mere speed, the persistence form (C2-structural) fails with it: records are scrambled before readback. The chain closes in the memory currency: $\neg$(C2-structural) $\Rightarrow \neg$(C4) by the fast-bath erasure lemma $\Rightarrow M_t = 0$ on accessible windows (§3.4) — Boltzmann-brain observers see no accessible memory, so they are not observers in the sense the framework's definitions require; the fixed-basis quantum representation they trivially possess ($S \iff D \iff Q_{\mathrm{fb}}$) carries no discriminating content. The exclusion is structural, not measure-theoretic — it follows from the incompatibility of (C2) with local equilibrium, not from arguments about the ratio of BBs to "ordinary" observers in an anthropic sample.

Corollary (past-hypothesis replacement, with its rate condition). The theorem above bounds $\tau_B(V) \leq \tau_{\mathrm{mix}}(\Sigma_{\mathrm{loc}}(V)) + O(\varepsilon)$, while (C2)-slow requires $\tau_S \ll \tau_B$; those do not compose. Contradicting (C2) additionally requires $\tau_{\mathrm{mix}} \ll \tau_S$ — the condition the Boltzmann-brain corollary states explicitly. A slowly mixing equilibrium with $\tau_S \ll \tau_{\mathrm{mix}}$ is entirely compatible with $\tau_S \ll \tau_B \leq \tau_{\mathrm{mix}}$, so the valid form is that a SUFFICIENTLY FAST-MIXING equilibrium supports no (C2)/(C4) observer. This matters for the de Sitter route, where $\tau_{\mathrm{thermal}} \sim \hbar/(k_B T_{\mathrm{GH}}) = 2\pi/H$ is Hubble-order and hence enormously longer than laboratory τ_S: a de Sitter thermal state does not erase observer memory on accessible timescales. Under $\tau_{\mathrm{mix}} \ll \tau_S$: For (S, φ) in its typical orbit phase — equivalently, in its equilibrium macrostate — canonical typicality [31] implies every local subsystem is also in its local equilibrium. By the theorem — under

(EM) for each region's effective channel — $\tau_B(V)$ is bounded by the local τ_{mix} for every bounded-diameter partition V. No partition supports (C2). That every relevant equilibrium region satisfies (EM) at the required rate is itself a physical hypothesis, to be established model by model; the theorem converts it into the selection rule and does not supply it. The framework's observers therefore exist only in the non-equilibrium phase of (S, φ). The low-entropy initial condition commonly imported as a "past hypothesis" to select observer-compatible configurations is replaced here by a structural selection rule: observers exist where (C1)–(C4) are satisfiable, and (C1)–(C4) are not satisfiable in the equilibrium phase.

Scope. The theorem uses bounded coupling, finiteness of S, the canonical-typicality result [31], and the effective-mixing hypothesis (EM) — the last an additional conditional physical premise, established model by model; it does not follow from isotropy or bounded-coupling regularity. Three scope notes. (i) *Locality of V.* The theorem assumes V has bounded coupling-graph diameter. Adversarially non-local partitions — cherry-picking distant subsystems that happen to be correlated — are excluded by the bounded coupling degree of φ and the finite response time of any physical observer. (ii) *Ergodicity.* The gap the argument needs belongs to the coarse-grained effective channel $\mathcal{K}_\Sigma$, not to φ itself: the exact Koopman operator of a finite permutation has modulus-one spectrum and cannot mix, so no spectral gap on typical orbits of φ is available or assumed. A gap for $\mathcal{K}_\Sigma$ is exactly the content of (EM) and does not follow from statistical isotropy or bounded-coupling regularity; pathological non-ergodic φ (many short orbits) is excluded by the structural assumptions already in force, but that exclusion is a necessary, not a sufficient, condition for (EM). (iii) *The cosmological realization.* The de Sitter horizon is thermal in the static-patch sense at Hawking temperature, but the globally expanding system is not in equilibrium with respect to any static-observer partition. The horizon's expansion sets $\tau_B \sim H^{-1}$ structurally, not thermally, which is the OI-native mechanism making human-timescale laboratory observation compatible with (C2).

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5. CONCLUSION

The paper's central result is stronger than a finite-accuracy analogy and narrower than an unqualified claim that C1–C4 uniquely generate operational quantum mechanics. At finite observable horizon the stochastic laws, finite reversible deterministic realizations, and fixed-basis unitary/Born representations are exactly the same class:

$$S \iff D \iff Q_{\text{fb}}.$$

Thus the reduced law of a finite embedded observer has an exact quantum representation, and every finite quantum record has a finite reversible realization. The representation statement is universal; it is not the selector. The content of observation incompleteness lies in history readback. C4 forces conditional memory, readout-relevant hidden predictive structure in every faithful completion, and — on a fixed finite reversible physical representative — global indivisibility somewhere in the recurrence cycle. C1 and C3 are necessary diagnostics of that mediation and its capacity, while C2-structural is the persistence condition and C2-slow one sufficient physical mechanism.

The spatial theory adds an exact causal-cone result. Because the coupling graph is defined by one-step dynamical dependence, k updates can carry influence no farther than graph distance k. Bell locality then follows when settings are genuine local interventions on a common setting-independent pre-setting ensemble **and** the Bell-relevant response graph is pointwise separated before readout; by §3.3 the resulting deterministic completion has $|S_{\mathrm{CHSH}}| \leq 2$. Reproducing quantum correlations therefore requires ontic parameter dependence, which is the option §3.3 takes and which the specific SM/GR target implements through preparation-indexed state-dependent adjacency. The superdeterministic alternative is not adopted: the manuscript proves only the model-independent necessary lower bound $I_{\mathrm{ont}} \geq 0.0309$ bits, while Hall–Branciard's optimized causal benchmark is about 0.080 bits. The PR-box construction remains the correct negative control for dropping the locality/intervention structure, not a counterexample to unitary representability.

The reverse direction is equally concrete. Every finite stochastic law has a finite reversible process dilation, and the operational gluing theorem shows that for any fixed finite dimension, finite instrument menu, finite horizon, and $\varepsilon > 0$, one finite reversible OI realization covers the whole corresponding quantum experimental envelope to accuracy ε. A single fixed finite carrier cannot equal the full continuum operational theory at all precisions, but the framework does not require an ontically infinite carrier; the continuum is the mathematical idealization of its finite-resolution physical description.

Accordingly, the phrase *equivalence of quantum mechanics and embedded observation* has a precise theorem-level meaning: **exact bidirectional equivalence of finite observable stochastic laws and their quantum representations, with history readback selecting the indivisible memory-bearing sector.** One boundary remains explicit rather than hidden in the slogan. Full equality with the standard local operational quantum theory requires a common compatible representation of the complete coherent instrument/composite hierarchy. Bell compatibility is a separate spatial-composite condition: the adopted deterministic branch keeps measurement independence but requires ontic parameter dependence with operational no-signaling. The existing classical action-labelled comb construction already covers finite menus of classical controls, so this residual is specifically the coherent/noncommuting and compositional layer. It is the remaining operational-lifting problem, not a defect in the exact stochastic–quantum equivalence proved here.

Code and Data Availability

The verification code for §3.2 is under `papers/oi_lattice_code/coherence/`: `coherence_enumeration.py` (the (C1) counterexample, the exhaustive small-system enumeration behind the counts quoted in the Remark on the failure of (C1), and the Lemma 1 boundary bounds) `coherence_theorem_tests.py` (the Weyl transfer rule and normal form of Lemma 2, the separability threshold exhaustively at $s = 2, 3$, and both directions of Corollary 1 — a negative partial transpose wherever $w < \kappa$, an explicit measure-and-prepare form wherever $w = \kappa$ — for $q = 2$ and $q = 3$), and `eb_pindivisibility_tests.py` (the strict one-way containment of complete dephasing in entanglement breaking, and an explicit reversible dilation whose one-step channel is entanglement-breaking while its visible process is P-indivisible). All are deterministic, depend only on NumPy and SciPy, and exit non-zero on failure; no result in this paper rests on sampling or on a random seed.

The repository is archived on Zenodo with concept DOI 10.5281/zenodo.19060318, which resolves to the latest version; per-release version DOIs are minted at release time, and a reader reproducing a specific claim should cite the version DOI of the release carrying it rather than the concept DOI. Paths above are relative to the repository root, which is also the root of the archived snapshot. The code is also available on GitHub at https://github.com/amaybaum/incompleteness and is released under the MIT License.

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